



PROJECT SAFETY MANUAL & SAFETY PLAN COMPLIANCE DOCUMENT

Hillwood

DTWR - Sierra Air Cargo Consolidation
Romulus, Michigan

10/6/2022 – Revision 0

Statement of Safety Program

It is a fundamental value of The Christman|Brinker Company that safety always is a primary consideration. It is our policy and a top priority to do all in our power to provide a safe workplace for all workers and to mandate the use of good safety practices. Adherence to established safety standards and procedures is required of everyone at all times on the DTWR - Sierra Air Cargo Consolidation Project site.

There is no phase of the operation that has greater importance than accident prevention and the preservation of human resources.

This Project Specific Safety Program has been developed in support of these concepts to serve as a reference in assisting all project trade contractors in satisfying their responsibility “to provide a safe, healthful workplace, free of recognized hazards.”

Leigh Woodbury

Digitally signed by Leigh Woodbury
DN: C=US,
E=leigh.woodbury@christmanco.com,
O=The Christman Company, CN=Leigh
Woodbury
Date: 2022.10.26 09:19:01-04'00'

The Christman|Brinker Company
Leigh Woodbury Jr.
Senior Project Superintendent

Steve Busen

The Christman|Brinker Company
Steve Busen
Senior Project Manager

Paul J. Yambor

Digitally signed by Paul J. Yambor
DN: C=US,
E=paul.yambor@christmanco.com,
O=The Christman Company, CN=Paul
J. Yambor
Date: 2022.10.25 09:02:38-04'00'

The Christman|Brinker Company
Paul Yambor
Project Executive

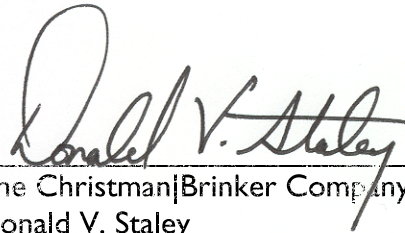
Troy E. Moulton

Digitally signed by Troy E. Moulton
DN: C=US,
E=troy.moulton@christmanco.com,
O=Christman, OU=Operations,
CN=Troy E. Moulton
Date: 2022.10.25 07:15:44-04'00'

The Christman|Brinker Company
Troy Moulton
Vice President – Industrial/Power



The Christman|Brinker Company
John Holmstrom
Senior Vice President/Principal



The Christman|Brinker Company
Donald V. Staley
Senior Vice President/Corporate
Safety Director

Statement of Safety Compliance Document

As in accordance with the FAA AC 150/5270-2G, the Christman|Brinker team has prepared a Safety Plan Compliance Document (SPCD) within our Project Safety Manual. This SPCD covers all information not discussed or disclosed in the original project Construction Safety Phasing Plan (CSPP) as well as all items requested per Wayne County Airport Authority Safety Standards dated 12/13/2021.

The Christman|Brinker team has read and understands the CSPP dated **09/16/2022** in full and agrees to comply with all requirements for both safety and security procedures as detailed within. Upon the event that any information or procedure within the CSPP or throughout the Christman|Brinker Safety Manual is changed, The Christman|Brinker team will submit to Hillwood, before being forwarded on to the WCAA & FAA for review and approval.

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DN: C=US,
E=leigh.woodbury@christmanco.com,
O=The Christman Company, CN=Leigh
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Table of Contents

- ◆ **PROJECT SAFETY MANUAL**
 - I. Introduction
 - II. Implementation Responsibility
 - III. Subcontractor Safety Programs
 - IV. Discipline Policy
 - V. Weekly Safety Meetings
 - VI. Safety Inspections
 - VII. Drug/ Substance Abuse Testing Program
 - VIII. Hazardous Material Control Program
 - IX. Existing Hazardous Material
 - X. Personal Protective Equipment and Clothing
 - XI. Emergency Telephone Numbers
 - XII. First Aid
 - XIII. Injury/ Accident Reporting and Investigation
 - XIV. Assured Grounding Program
 - XV. Jobsite Specific Safety Rules
 - XVI. Housekeeping
 - XVII. Ancillary Safety Procedures
 - XVIII. Smoking
 - XIX. Crane Inspections
 - XX. Visitor Policy
 - XXI. Fire Prevention and Protection
- ◆ **ATTACHMENT I - DRUG SCREENING LIST**
- ◆ **ATTACHMENT II - ACCIDENT INVESTIGATION REPORT**
- ◆ **ATTACHMENT III - LOCKOUT/ TAGOUT PROCEDURES**
- ◆ **ATTACHMENT IV - CONFINED SPACE ENTRY PROCEDURE**
- ◆ **ATTACHMENT V - EMERGENCY CONTACT NUMBERS**
- ◆ **ATTACHMENT VI - EMERGENCY EVACUATION PLAN**
- ◆ **ATTACHMENT VII - EMERGENCY EVACUATION MAP**
- ◆ **ATTACHMENT VIII – WAYNE COUNTY AIRPORT AUTHORITY SAFETY STANDARDS**

DTWR - Sierra Air Cargo Consolidation – Project #222031

- ◆ ATTACHMENT IX – SITE LOGISTIC PLAN
- ◆ ATTACHMENT X – JOB HAZARD ANALYSIS (JHA)
- ◆ ATTACHMENT XI –WAYNE COUNTY – LOSS MANAGEMENT MONTHLY REVIEW
- ◆ ATTACHMENT XII – WAYNE COUNTY – CONTRACTOR EMPLOYEE FORM
- ◆ ATTACHMENT XIII – WAYNE COUNTY – CONTRACTOR INSURANCE TOOL KIT
- ◆ ATTACHMENT XIV – WAYNE COUNTY – REQUEST FOR MARKING UTILITIES
- ◆ ATTACHMENT XV – WAYNE COUNTY – HOT WORK PERMIT
- ◆ ATTACHMENT XVI – WAYNE COUNTY – UTILITY FIELD AWARENESS FORM
- ◆ ATTACHMENT XVII – WAYNE COUNTY – UTILITY INTERRUPTION FORM
- ◆ ATTACHMENT XVIII – WAYNE COUNTY – UTILITY SYSTEM CUSTODY FORM
- ◆ ATTACHMENT XIX – DAILY CONSTRUCTION CHECKLIST
- ◆ ATTACHMENT XX – CONTRACTOR INCIDENT/ACCIDENT DOCUMENTATION
- ◆ ATTACHMENT XXI – SUPPLEMENTAL UTILITY STRIKE FORM

I. Introduction

This Project Safety Manual provides an administrative structure within which The Christman|Brinker Company and subcontractors present on the Project site shall provide for the safety and health of their employees and other individuals affected by their activities and for the protection of property. This Manual does not relieve subcontractors of any of their traditional or specific legal responsibilities with respect to occupational safety and health or the protection of property. Instead, this Manual provides for coordination among the various subcontractors' individual programs, monitoring of subcontractors' conformance with their individual programs, and the requirements of this Manual, initiation of corrective actions when nonconformances are identified, and documentation of safety related programs, meetings, and incidents. This Project Safety Manual for **DTWR - Sierra Air Cargo Consolidation** has been developed to assist all subcontractors working at the Project site in satisfying their responsibility to provide a safe and healthful work place, free of recognized hazards, for their employees. **This Safety Manual Issue No. 01 Revision 0 has been developed based on current summation of specifications, contract drawings, Wayne County Airport Authority Safety Standards dated 12/13/2021, Construction Safety Plan (CSPP) dated 09/16/2022, state requirements, and all OSHA Standards.**

DTWR - Sierra Air Cargo Consolidation is located on the Southeast side of Detroit Metropolitan Airport (DTW) between Middlebelt Rd and Superior Rd. This project consists of extensive site work including hauling off PFAS contaminated soil, multiple phases of AOA perimeter & jobsite fence adjustments, relocation of AOA Gate #27, installation of the RVR Road, underground storm detention system and AOA paving, seven new plane parking gates, striping and taxiway light installation along the new taxiway paths. Additionally, we will be making improvements to Superior Road & making right-a-way adjustments to Middlebelt Road.

There are six buildings that will make up this distribution center. First, is Building AB which will be a 2-story slab-on-grade Administration and Sort Facility sitting at 357,900 sqft. The Sort Facility is the primary building of the project, inclusive of a sort warehouse that connects directly to overhead coiling doors for pickup & drop-off directly with the FedEx airplanes & trucks. The AB Building office serves as the primary office facility, with a secured employee entrance, offices, conference rooms, equipment & maintenance shops, storage facilities etc. Building C will be the Generator Building housing multiple generators, switchgears, and our fuel storage tank. Building D is the Ground Service Equipment and Truck Wash Building, and our second largest building to Building AB. Building D includes multiple service and wash bays, a five-ton bridge crane included at each individual bay. Additionally, this building includes an upstairs office space with a mezzanine totaling the building sq. footage at 25,647. Building E will be the De-Icing & Glycol Storage Building that will include a de-icing shop facility. Additionally, this building will house our four additional glycol storage tanks that sit around the perimeter of the facility. Building F and Building G will be the ACMX & Sort Canopy facilities. These sort facilities will include multiple storage decks & tug paths for managing materials between the different buildings, but also between the AB Sort Facility & the FedEx planes.

Prior to accessing site, the following processes are required to be followed:

- **Badging with WCAA**
 - All employees that will require access to the AOA (airside) areas of the project site will need to be badged by Airport Security. This is inclusive of a background check and fingerprinting. Each subcontractor is required to keep a list of all of their badged employees and corresponding badge numbers at all time.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- If subcontractors require any escorting privileges or driving privileges on the AOA, additional badging measures and training will be required to be taken with Airport Security prior to start of work.
- **Mandatory Drug Testing**
 - Ref. Section VII – Drug/Substance Abuse Testing Program for more information. All employees will be required to be drug tested prior to start of work.
- **Wayne County – Contractor Employee Form Submission**
 - Ref. Attachment XII. Compliance Document requiring signature for all personnel that will be entering the site. Outlines compliance and understanding of Wayne County Airport Authority Contractor Safety Guidelines.
- **Wayne County – Contractor Insurance Tool Kit Submission**
 - Ref. Attachment XIII. Insurance Certificates will be collected by the Christman|Brinker team. Upon request for the subcontractor's insurance certificate, the contractor insurance tool kit form will also be required to be filled out and returned prior to start of work.
- **Christman|Brinker Project Safety Manual/SPCD Signed**
 - All subcontractors are required to sign the bottom of this document in agreement & confirmation the project safety manual has been reviewed in full and understood prior to work.
- **Subcontractor Safety Plan Submission**
 - Ref. Section III. Final subcontractor safety plan submission is required prior to start of work.
- **Subcontractor Project JHA Submission**
 - Ref. Section III. All trades are to complete a Jobsite Hazards Analysis (JHA) form every day for all work activities. Each crew performing specific daily job functions shall complete a JHA relating to their tasks.
- **Subcontractor Project SDS & Chemical Inventory List Submission**
 - Ref. Section III. All safety data sheets (SDS) and chemical inventory list is to be provided by each subcontractor prior to start of work. The Christman|Brinker Company will maintain a master file of all jobsite SDS' for every subcontractor in the on-site Project office. This information will be made available to any employee upon request.
- **Subcontractor Emergency Contact Information**
 - Reference Section XI. Each contractor shall furnish The Christman|Brinker Company with the emergency contact phone numbers for all key project personnel (e.g. jobsite supervisors, project managers, foreman, etc.).
- **Christman|Brinker Site Specific Safety Orientation Video**
 - All employees, subcontractors, trades people, and visitors are required to watch The Christman|Brinker Company Safety Orientation Video prior to performing work at the site. This orientation program will highlight specific safety concerns to consider while working on the project. The video may be viewed at the site office. It is the responsibility of each foreman to insure all new workers, visitors (vendors), and office personnel on the site receive the Orientation prior to entering the work site. Upon completion of the orientation, personnel will receive an orientation hardhat sticker. All hardhat stickers are to be placed on the hardhats of personnel prior to entering the site, along with verification that names are correctly labeled on both the front and back of all hardhats.

The prime responsibility for safety, supervision, and inspection of airfield construction projects lies with the airport owner/operator. Aviation safety is the prime consideration at airports, especially during construction. The primary goal is to provide the highest possible level of safety, security, and efficiency to all the airport users; from the airlines and tenants to the construction contractors and ultimately the traveling public.

II. Implementation Responsibility

Each and every person working at the Project site shall be responsible for being knowledgeable of the contents of this Manual and comply with every requirement contained herein. Below are the following safety documents pertaining to this project that must be strictly adhered to. **Where the documents contain overlapping sections, the document with the more stringent requirement shall be followed.** The following documents are required to be read and thoroughly reviewed prior to safety orientation. Upon completion of orientation, all personnel will be required to sign the Wayne County Airport Authority Contractor Employee Review Form confirming compliance and understanding of the safety requirements outlined in both this Christman|Brinker Project Safety Manual/SPCD, but also the following project documents:

- **Wayne Country Airport Authority Safety Standards (SP-10).** Prior to mobilization, the Contractor shall complete Contractor Employee Review and Contractor Safety Guidelines.
- **Construction Safety and Phasing Plan (CSPP) dated 9/16/2022.** This document shall be kept on site and updated for every new employee who will work on the project. This CSPP is reviewed and approved by the FAA. Christman|Brinker has read and understood all requirements detailed within the CSPP.

III. Subcontractor Safety Programs

- Each subcontractor is required to submit a copy of its written safety program to The Christman|Brinker Company **prior to commencing work at the Project site**. The Contractor shall maintain a Safety Program, for the purpose of safety, security, orientation, education, training, enforcement, and distribution. Each program shall meet the requirements of the codes and regulations of OSHA, local, and other authorities having jurisdiction over the work to be performed and shall include disciplinary procedures and safety orientation training procedures applicable to subcontractor personnel.

Each subcontractor safety program shall include, as a minimum, the following requirements:

- All trades are to complete a Jobsite Hazards Analysis (JHA) form every day for all work activities. Each crew performing specific daily job functions shall complete a JHA relating to their tasks.
- Conduct weekly safety (Tool Box) meetings.
- Monitor jobsite safety issues daily.
- Conduct fire inspections.
- Review project specific requirements with employees.
- Conduct tool and equipment inspections.
- Review all personal protective equipment with employees.
- Address fall protection measures.
- Address Right-To-Know regulations, including Safety Data Sheets (SDS')
- Provide hazardous material awareness training.
- On-Site Staff OSHA 10hr Certifications
- A site specific fall protection program is required to be included in the subcontractor safety program submission if that contractor or employee will be utilizing any passive or active fall protection systems.

IV. Discipline Policy

All Subcontractors entering onto this Project are required to comply with all aspects of the project safety program requirements. Those that do not adhere to this program will be subject to this discipline policy. The Project will utilize a three-step process for minor infractions; repeat or major offenses of the safety program are subject to immediate removal from the Project.

- Any re-entry employment programs for safety violations will be initiated only by The Christman|Brinker Company Superintendent and Project Manager.
- Verbal warning - The verbal warning **shall be considered given** at the signing of the subcontract and signed acknowledgement of the project safety plan. This warning is to be communicated by the Subcontractor to **all** Subcontractor personnel prior to entering this Project.
- Written warning - The second warning will be a written warning, documented, with a copy sent to the employee's Supervisor and Corporate Office.
- A written warning, at the discretion of The Christman|Brinker Company Superintendent, can result in either removal from the Project or re-orientation of the safety program. The re-orientation will be attended by both the violator and his/her Foreman. This re-orientation may consist of:
 - Video training in the related area of the violation
 - Written verification by the employee that he/she is capable of identifying the hazards related to the violation.
 - The completion of a written Job Hazard Analysis (see attachment X) by the employee and his Foreman for the task in which the violation occurred.
- In addition, any Foreman that has had his/her crew members attend three (3) re-orientation sessions for any reason is subject to removal from the Project.
- The third infraction of the same safety program issue will result in removal from the Project.
- Any Foreman, General Foreman, or Superintendent who forces or directs any employee to violate a safety procedure, will be removed from the Project immediately.
- Any violation deemed threatening to life, limb or major property damage will be grounds for immediate removal from the Project.

Subcontractors will comply with all provisions of The Christman|Brinker Company's safety program. In the event of Subcontractor's noncompliance, the Subcontract may be canceled, terminated, or suspended, in whole or in part, and The Christman|Brinker Company may complete the work at the cost of the Subcontractor in accordance with the Subcontract Agreement.

If a contractor or subcontractor interferes with or makes more difficult or expensive the Airport Authority's operations or required compliance with any law, statute, code, ordinance, or regulation, action will be required. WCAA will notify the primary contractor, and they shall respond within 48hrs, making whatever changes are necessary to remedy the situation.

V. Weekly Safety Meetings

The Christman|Brinker Company will hold weekly construction coordination/safety meetings with all on-site subcontractors. All active, on-site subcontractors shall be represented at these meetings. At the beginning of each meeting current safety topics, issues, concerns and problems will be discussed.

Additionally, Christman will also be holding weekly OAC meetings with project stakeholders in which a portion of the meeting will be dedicated to safety.

Prior to start of work, the Christman|Brinker team will be holding a safety phase planning meeting with Hillwood and WCAA prior to mobilization to the AOA. During the meeting, the approved Safety Manual/SPCD and CSPP contents will be discussed and reviewed.

Each active, on-site subcontractor shall conduct weekly toolbox meetings. Every worker active on site shall attend. Each subcontractor shall furnish the following documentation to The Christman|Brinker Company after each toolbox meeting: date, topic(s) discussed, names of attendees, subcontractor's name, and any safety concerns noted.

The Christman|Brinker team will be required to submit monthly safety updates to the Hillwood & WCAA teams through the Wayne County – Loss Management Monthly Review form (Ref. attachment XI). This form is to be filled out monthly, and include details identifying & summarizing total man hours, first aid cases, near miss incidents, medical cases, and property damage incidents.

VI. Safety Inspections

The Christman|Brinker Company's Project Superintendent will conduct informal inspections of conditions, practices, and equipment **on a daily basis**. The Christman|Brinker Company's Safety Manager shall schedule visits to the Project site, unannounced, to inspect the work in progress. All safety discrepancies noted shall be reviewed with the Project Superintendent responsible to ensure that corrective measures are implemented.

Additional safety inspections will be implemented following new construction activities, during weather changes, immediately after an incident or accident, and following any unusual condition at the airport.

Prior to daily start of work, The Christman|Brinker team will fill out the Wayne County – Daily Construction Checklist (ref. attachment XIX). Daily Construction Checklist reviews all construction activities in relation to the AOA, any NAVAID's or additional required safety requirements prior to the start of work each day. Each daily checklist is required to be signed by a WCAA inspector.

In addition to the Daily Construction Checklist – the Christman|Brinker team will be required to call Airport Operations prior to start of work every day. Contact information is **734-942-3685**.

Supervisors of all active on-site subcontractors are responsible for conducting daily inspections of conditions, practices, and equipment relating to their work areas/activities. Safety discrepancies discovered, along with the corrective measures taken, shall be reviewed with The Christman|Brinker Company Project Superintendent.

When working in an AOA area, runway or taxiway – upon reopening the space to airfield personnel and traffic, all areas must be thoroughly inspected for no construction debris or standing water and ensure all airfield lighting, signage and pavement markings are all in place.

VII. Drug/ Substance Abuse Testing Program

- It is Christman|Brinker's desire to provide a drug-free, healthy, and safe workplace. To promote this goal, employees are required to adhere to the following Drug and Alcohol Policy ("Policy"). Notwithstanding anything to the contrary herein, as this Policy relates to subcontractors, this Policy is subordinate to any union agreements.
- Random and Post Accident Drug Testing Program must be in place. The Project is using MUST for this. Christman|Brinker is registered with MUST, Project/Site #816. Must Safety Program (Management and Unions Serving Together (MUST)) is a one stop source for all drug testing and applicable safety awareness training. All tradespersons must be assigned to this MUST project # by their company before arriving on this site. Link to MUST website is below:

(1) [Login \(mustbsafe.com\)](https://mustbsafe.com)

- **Pre-Employment Testing.** Christman|Brinker and Subcontractors conduct drug testing of all applicants for employment.
- **Random Testing.** Christman|Brinker may randomly test employees or require subcontractors to test their employees for compliance with this policy. As used in the Policy, "random testing" means a method of selection of employees and/or subcontractors for testing, performed by an outside third party. Furthermore, Christman|Brinker and its Subcontractors, has no discretion to waive the selection of an employee and/or subcontractors selected by this random selection method.
- **Scheduled Periodic Testing.** Christman|Brinker reserves the right to conduct or require subcontractors to conduct periodic testing on a regularly scheduled basis for designated departments, classifications, work groups or construction sites.
- **Post-Accident Testing For Drugs and Alcohol.** Any employee injured at the Project site and needing offsite medical attention will be tested for alcohol and drugs. When an employee or subcontractor has been involved in a construction incident that has caused property damage or has been injured, such employee or subcontractor will be required to follow Christman|Brinker drug testing procedure. Within 8 hours of the construction incident, drug testing must be performed by a designated testing lab. If the injured party was hurt by the action of another person who was not injured, this person must also submit to substance abuse testing. Additionally, any employee causing or contributing to a major incident not resulting in injury will be tested for alcohol and drugs. An employee who tests positive for illegal substances or alcohol will not be allowed to return to the project site. Any employee who refuses to submit to substance abuse testing will not be allowed to work at the project site.
- **It is the responsibility of the employer to make sure that the testing is performed at the time of the incident.** It is also the responsibility of the employer to cover any costs incurred due to the testing. Results of the testing will be considered confidential. **An employee that is tested late (not within 8 hours of the incident) will be considered as "not tested" and will not be allowed to return to work on the project site.**
- **Substances Covered By Drug/Alcohol Testing.** Employees and subcontractors will be tested for their use of commonly-abused controlled substances, which include: Amphetamines, Barbiturates, Benzodiazepines, Opiates, Cannabinoids, Cocaine, Methodone, Methaqualone, Phencyclidine (PCP), Propoxyphene, and chemical derivatives of these substances.

Employees and subcontractors must advise testing lab employees of all prescription drugs taken in the past month before the test, and to be prepared to show proof of such prescription to testing lab personnel.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- **Testing Methods and Procedure.** All testing will be conducted by a licensed independent medical laboratory, which will follow testing standards established by the State and/or federal government. Testing will be conducted on a urine sample provided by the employee and subcontractors to the testing laboratory under procedures established by the laboratory to insure privacy of the individual, while protection against tampering/alteration of the test results.

Employees (only) will be engaged at work for the time spent in taking any tests, and will be compensated for such time at their regular rate.

Christman|Brinker will pay for the cost of the testing of Christman|Brinker employees. Subcontractors will pay for the cost of the testing of their own employees. This includes the confirmation of any positive test result by gas chromatography. The testing lab will retain samples in accordance with State law, so that an employee and/or subcontractor may request a retest of the sample at his/her own expense if the individual disagrees with the test result.

- **Refusal to Undergo Testing.** Employees and subcontractors who refuse to submit to a test are subject to immediate discharge. Zero Tolerance
- **Positive Test.** If an individual tests positive on an initial screening test the following will happen:
Employee: The employee will be temporarily suspended while the confirmation test is being conducted. On receipt of the confirmation test, the employee will be subject to disciplinary action, up to an included discharge. Discipline selected by Christman|Brinker will depend upon a variety of factors, including the prior work record of the employee; the length of employment; the prior accident and attendance recorded of the employee, the circumstances which led to the testing; and the proposals by the employee to address the problem.
Subcontractor: Will be temporarily suspended from the jobsite while the confirmation test is being conducted. If the additional test is also positive, the individual will no longer be able to conduct work on the jobsite. Additionally, the subcontract's contract may be cancelled. Further ramifications will be decided on a case by case basis.
- **Right to Explain Test Results.** Employees and subcontractors have a right to obtain copies of all test results from the testing laboratory. When the individual disagrees with the test results, the individual may request that the testing laboratory repeat the test. Such repeat test shall be at the expense of the individual, unless the repeat test overturns the original report of the lab, in which case Christman|Brinker will reimburse the employee (only) for the costs incurred for the retest.
- **Confidentiality Requirements.** All records for employees and subcontractors concerning test results will be kept confidential. Information shall not be released to any third parties without the individual's specific written consent. Testing laboratories may conduct testing only for substances stated in this policy provided to the individual, and may not conduct general testing related to the medical conditions of the individual which are unrelated to drug usage.
- **Retesting.** Employees (only) may request a retest of their positive test results, within three (3) working days after notification by Testing Lab of such positive test result. This retest is at the expense of the individual, unless the original test result is called into question by the retest.

Where the employee believes that the positive test result was affected by taking of lawful or prescribed substances, the individual may be suspended without pay pending receipt of confirming information to substantiate the claims of the individual. Normally, the individual will

DTWR - Sierra Air Cargo Consolidation – Project #222031

be provided no more than three (3) business days in which to provide this additional information.

Once Christman|Brinker has determined whether or not there is evidence to indicate that the test results are incorrect, Christman|Brinker will advise the individual of its decision.

Notes:

- Visitors and delivery personnel are not included.
- The drug-screening test shall include testing for cannabinoids, depressants, dissociative anesthetics, hallucinogens, opioids and morphine derivatives, and stimulants. **Refer to Attachment I.**

VIII. Hazardous Material Control Program

Each subcontractor is responsible for the safe storage, use and disposal of any hazardous material required to perform their scope of work. Each subcontractor using such materials shall provide The Christman|Brinker Company with a copy of their Hazardous Material Control Program before any of his material is delivered to the Project site. Each program shall, at a minimum, include the following:

- Identification and classification of hazardous materials
- Safety and health precautions
- Training in safe storage, handling, and use
- Transporting, labeling, packaging, documenting, and disposal requirements
- Safety Data Sheets (SDS) for all chemicals

Each subcontractor is to submit to The Christman|Brinker Company a complete list of hazardous chemicals that may be used at the Project site along with a copy of a Safety Data Sheet (SDS) for each. Copies of any subsequent SDS amendments, additions, or corrections shall also be promptly submitted. Additionally, a list of the chemicals that are on site is to be submitted to The Christman|Brinker Company.

The Christman|Brinker Company will maintain a master file of all jobsite SDS' for every subcontractor in the on-site Project office. This information will be made available to any employee upon request.

Any hazardous materials that are to be brought to site are required to be labeled & identified per the U.S. Department of Transportation (DOT) system. All wastes are to be properly stored, labeled, and disposed of in accordance with WCAA project specifications, as well as Michigan Department of Environmental Quality (MDEQ) regulations.

Upon the event that there is a hazardous material or chemical spill – subcontractors and the Christman|Brinker team are to immediately report to Airport Authority Operations and EGLE. Any transports or handling of hazardous material, or waste sites for recycling and disposing of materials, are required to be pre-approved by the WCAA. All bill of ladings and manifests are required to be submitted to WCAA for record.

It is the responsibility of each subcontractor to provide proper and adequate employee training as required under this Act.

Compliance with the storage, use and disposal requirements as outlined on labels and SDS's is mandatory for all users; full understanding of these requirements is expected of all users and suppliers of hazardous materials.

Christman|Brinker will submit a forthcoming Storm Water Pollution Prevention Plan (SWPPP) and a Spill Prevention, Control and Countermeasures Plan (SPCCP) to WCAA for review. These subsequent plans will include any additional necessary measures in regards to handling hazardous material on site.

IX. Existing Hazardous Material

The Christman|Brinker Company Site Manager and / or Superintendent and Owner/Client shall review all areas of the Project site for existence of hazardous materials, such as asbestos and lead. If asbestos exists in areas that involve construction work, the Owner/Client shall arrange to remove the asbestos before any construction work takes place in each area.

If lead-based paint exists in areas, which involve construction work, each Subcontractor working in the areas shall be responsible for complying with the requirements of the OSHA standard for lead exposure. The recognition of possible contaminated areas is key to avoiding exposure. If you discover an area that you suspect lead may be present, you are to bring it to the attention of The Christman|Brinker Company Project Superintendent. Proper communication regarding the hazardous materials is to be conveyed to all people that may possibly be exposed to the hazard.

It is confirmed that this project has 165,000 cubic yards of PFAS contaminated soil that needs to be removed from the project site prior to start of work. Please review the Environmental Construction Management Plans for further details.

Remember - If a suspect hazardous material is discovered notify The Christman|Brinker Company's Project Superintendent immediately.

X. Personal Protective Equipment and Clothing

- Each subcontractor shall supply each of their workers with the proper safety equipment, take any necessary precautions to maintain the equipment according to current regulations and specifications and accept responsibility to assure that proper safety equipment is used when required. All workers shall be required to wear appropriate personal protective equipment and clothing where there is exposure to hazardous conditions and where needed to reduce hazards to the worker. Hard hats and safety glasses are to be worn 100% of the time while working on this project. There are NO exceptions to this requirement.

The following list represents the minimum requirements:

- **Outer Garments**
 - Suitable outerwear is defined as clothing designed to protect from expected hazards. Outer garments should be well fitting. Short sleeve garments may be worn where no hazards to exposure exist. High Visibility Shirts or Vests must be worn at all times. For daytime and nighttime work within the right-of-way, apparel shall be labeled as meeting the requirements of ANSI 107-1999 standard performance for Class 2 risk exposure. For the same condition described above but the activity is flagging traffic, a Type 3 is to be used. Contractors performing site work, site utility work, foundation work, slab-on-grade work, and building framing and enclosure work are to wear reflective safety vests 100% of the time.
 - Tank tops or “muscle shirts” are not permitted. Full pants are to be worn. Neckties should not be worn when working around moving machinery. All employees are to avoid loose sleeves, frills or trims that could present hazards. Racist or profane slogans on outer garments are not permitted.
- **Gloves**
 - Hands are to be protected as dictated by working conditions. Gloves, mittens or hand leathers will be of the type consistent with the hazards present.
- **Hardhats**
 - Hardhats must be worn on site at all times. Acceptable hardhats must meet the requirements of ANSI Standard Z89.1 – Type I Class E. All hard hats shall have employees name and respective company on the front and back.
- **Eye and Face Protection**
 - Safety glasses are always required. Acceptable safety glasses must meet the requirements of the most current ANSI Standard Z87.1.
 - Employees shall be required to wear proper eye and face protection during operations where a hazard or risk of injury exists from flying objects or particles, harmful contacts, exposures such as glare, liquids, injurious radiation, electrical flash, or a combination of these hazards.
 - Contact lenses or US food and drug administration standard hardened lenses shall not be considered as eye protection.
 - An employee who needs corrective lenses in spectacles where eye protection is required shall be protected by one of the following:
 - Spectacles whose protective lenses provide optical correction.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- Goggles that can be worn over the corrective lenses without disturbing the adjustment of the spectacles.
 - Goggles that incorporate corrective lenses mounted behind the protective lenses.
- Face and eye protection equipment shall be of proper size to fit the employee and protect against the intrusion of foreign objects. Face and eye protection shall be kept clean and in good repair. Equipment with structural or optical defects shall not be used.
- Face shields and goggles shall be used, as required, to provide eye and face protection when safety glasses are insufficient.
- Employees shall wear welding helmets with filter lenses when engaged in welding, cutting, or burning operations or the employee shall wear safety glasses with side shields or goggles under the shield when the shield is raised and is exposed to flying objects. Manufactured welding shields shall be placed around the work area to prevent flash exposure to other employees.
- Jewelry
 - Bracelets, necklaces, rings and earrings are not to be worn during operations where they present hazards to the wearer.
- Foot Protection
 - Work boots or approved safety shoes are to be worn at all times on the project. Special safety footwear for protection against exposure to electrical shock, chemicals, acids, caustics, and hot or molten materials may be required and it is the responsibility of the employer to provide their employees with the appropriate “special footwear” if it is deemed necessary. If an employee is using a Jack Hammer toe guards are required.
- Personal Protective Equipment
 - The use of personal protective gear, such as gloves, respirators, coveralls, safety glasses, ear protection, and the like, will be required as conditions demand and shall be reviewed at the time of employee orientation or when assigning duties. The contractor is required to supply all required PPE to their employees.
- Respirators
 - Any Subcontractor responsible to perform work that requires respiratory protection shall have a written respiratory program in place. A copy of the program shall be submitted to The Christman|Brinker Company’s Superintendent before commencement of any work activities that require the use of respirators. At a minimum, each respiratory protection program shall provide each employee with training relative to the need for use, fitting, fit testing, cleaning and storage of respirators.
- Fall Protection
 - **Policy:** Fall protection shall be worn and used, 100% of the time, by all persons when there is exposure to a fall greater than six (6) feet unless other provisions such as guardrails, safety nets, or fall restraints have been provided. This includes, but is not limited to, steel erection (including connecting, bolting-up, decking, welding or any other steel erection activity), precast erection, roofing activities and masonry work including overhand laying operations.
 - **Exceptions:** The safe use of ladders up to twenty-four (24) feet in length.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- **Unacceptable means of compliance: Safety** monitors.
- **Example of an acceptable means of compliance:** A warning barrier fifteen (15) feet from the potential fall hazard. The exception to this will be for roofers performing roofing activities. They will be allowed to move closer to the edge and establish their warning barrier at six (6) feet. Anyone required to cross the fifteen (15) foot or six (6) foot barrier line must comply with the fall protection policy. The barrier shall consist of ropes, chains or wires supported by stanchion posts erected as follows:
 - The system shall be flagged at no more than six (6) foot intervals with a high-visibility material.
 - The system shall be rigged and supported in such a way that its lowest point (including sag) is no less than thirty-four (34) inches from the walking/ working surface and its highest point is no more than thirty-nine (39) inches from the walking / working surface.
 - After the system is erected, the stanchions shall be capable of resisting, without tipping over, a force of at least sixteen (16) pounds applied horizontally against the stanchion, thirty (30) inches above the walking / working surface, perpendicular to the warning line, and in the direction of the floor, roof or platform edge.
 - The system line shall have a minimum tensile strength of 500 pounds and after being attached to the stanchions, shall be capable of supporting without breaking, the loads applied to the stanchions noted in the above item.
 - The line shall be attached at each stanchion in such a way that pulling on one section of the line between stanchions will not result in slack being taken up in adjacent sections before the stanchion tips over.

XI. Emergency Telephone Numbers

Emergency telephone numbers for the ambulance, hospital, fire department, and police shall be posted in each jobsite office at each phone. Each contractor shall furnish The Christman|Brinker Company with the emergency contact phone numbers for all key project personnel (e.g. jobsite supervisors, project managers, foreman, etc.). Contact numbers shall include mobile, home, and other phone numbers available and necessary to access key personnel in the event of a project emergency. Distribution of these numbers will be limited and considered confidential.

Christman|Brinker prime contacts for any on site emergencies are as follows. Please note, the following should be contacted in this order in case of an emergency:

Leigh Woodbury (24/7 Contact)
Senior Project Superintendent
Leigh.woodbury@christmanco.com
Cell – (810) 588-7771

Steve Busen
Senior Project Manager
Steve.busen@christmanco.com
Cell – (248) 514-9417

Kyle Jones
Project Superintendent
Kyle.jones@christmanco.com
Cell – (313) 949-2842

Wayne Compton
Project Superintendent
Wayne.compton@christmanco.com
Cell – (517) 896-7896

Caitlin Spiegel
Project Manager
Caitlin.spiegel@christmanco.com
Cell – (520) 820-0706

Amanda Korte
Project Manager
Amanda.korte@christmanco.com
Cell – (636) 236-5056

XII. First Aid

Each subcontractor shall provide The Christman|Brinker Company with the name or names of each designated first-aid certified worker. Each subcontractor is required to have at least one (1) certified worker on site during periods when the subcontractor is active.

The Christman|Brinker Company will have a first aid kit on site at the Project office. Each subcontractor shall also provide a first aid kit on site.

XIII. Injury/ Accident Reporting and Investigation

Any worker incurring an injury while working at the DTWR - Sierra Air Cargo Consolidation site shall report the injury to his supervisor immediately. **All injuries shall be verbally reported to The Christman|Brinker Company the day of the incident/accident, followed up in writing on the proper forms, within twenty-four (24) hours.** Each incident/accident shall be documented using an accident investigation report similar to The Christman|Brinker Company report (refer to Attachment II). In the event of an accident, the following steps shall be taken:

- Make sure that accident victim receives appropriate medical attention. This may involve calling paramedics, arranging transportation to a hospital, or sending victim to a clinic depending on severity of injury.
- Notify The Christman|Brinker Company's Superintendent immediately regardless of the seriousness of the incident or accident. This includes property damage.
- Perform a thorough investigation of all facts and circumstances surrounding the accident. This should include:
 - Photographs of the accident site and any other pertinent areas or conditions, which document the circumstances existing at the time of the accident.
 - Determine who, if anyone witnessed the accident. Obtain names, addresses and phone numbers and get written statements from them as to what they saw and any other facts they may be aware of regarding the accident and its cause.
 - Interview the accident victim to hear his/her explanation of how the accident occurred.
 - Fully complete an accident investigation report and submit a copy to The Christman|Brinker Company's Superintendent.
 - All accidents resulting in property damage shall be reported immediately to The Christman|Brinker Company Project Superintendent in order to allow for documentation of the extent of damage and determination of any required corrective action.
 - After the occurrence of an accident or damage on the jobsite, the Christman|Brinker team will fill out and submit the Wayne County – Contractor Incident/Accident Documentation (see attachment XX). This is required for any injury, property damage, environmental release, FAA incursion, FAA surface incident, unsafe act, or utility strike. If the event does in fact involve a utility strike, the Christman|Brinker team will also submit the Wayne County – Supplemental Utility Strike Form (see attachment XXI), with the Contractor Incident/Accident Documentation.
- The WCAA has the authority to monitor Contractor's and subcontractor's operations for safety performance, workmanship, protection of operations, work progress, housekeeping and compliance to design specifications. WCAA has the right to investigate an accident or incident.
- If any incident occurs on the AOA related to construction or driving, the incident is required to be reported immediately to the DTW or YIP Airfield Operations Units. Contact information is (734) 942-3685. Airport Authority DTW, YIP Airfield Operations, and Airport Authority Public Safety have a right to investigate all incidents and issue citations.

XIV. Assured Grounding Program

- This program may be used to ensure that the scheduled inspection and safety condition of construction electrical equipment grounding conductors is satisfactory. The program applies to all extension cords and receptacles not part of the permanent building or structural wiring. The Contractor shall comply with the National Electric Code (NEC) requirements regarding ground fault circuit interrupters for construction field tools and equipment.
 - Inspection Procedure
 - The inspection procedure for all extension cords, attachment caps, plug/receptacle sets and equipment powered by a plug-in cord will include visual inspection at the beginning of each shift for signs of damage, defects, missing pins or insufficient insulation. Defective equipment will be removed from service until repaired and / or replaced.
 - Test Procedure
 - The test procedure for all extension cords and receptacles that are not permanent wiring includes testing for electrical conduction and correct attachment to the equipment ground conductor. The frequency of these tests will be as follows:
 - Prior to the first use.
 - Before equipment is returned to service following any repairs.
 - Before equipment is used following an incident which may have resulted in equipment damage.
 - At intervals not exceeding three months. Quarters and tape color should be coded following inspection. Winter (January through March) will be coded in white; Spring (April through June) will be coded in green; Summer (July through September) will be coded in red; and Fall (October through December) will be coded in brown.
 - Testing for each quarter is expected to occur between 15 days prior to and 15 days after the beginning of that quarter.
 - Each subcontractor is responsible for assigning a properly qualified person to coordinate their Assured Grounding Program and ensure compliance with this procedure and any other applicable codes and regulations.
 - Other forms of testing will be accepted as long as the record documentation of the testing can be provided upon request.

XV. Jobsite Specific Safety Rules

Violation of any of the following general jobsite safety rules are considered to be extremely serious and a detriment to the safety and welfare of all workers. Violation of these safety rules will be cause for disciplinary action up to and including permanent removal from the project.

- This project will take place in an airfield secured area until the Temporary AOA Perimeter Fencing is installed and inspected WCAA. The Wayne Co. Airport Authority (WCAA) and the Transportation Safety Administration (TSA) have very strict safety and security policies. All construction personnel working on this site must obtain and display airport-issued identification badges until temporary fencing is installed. Obtaining such badges is a two-part process. The first part consists of background checks. The second part consists of SIDA training and non-movement driver's training.
- Only applicants who clear the initial background check will be allowed to proceed to part two for fingerprinting & actual badging. ID badges will be issued upon successful completion of both reviews. In addition, all contractor foreman, all gate attendants, and all flags' persons are required to be trained and familiarized with the safety and security requirements of this project.
- Per AC 150/5370-2F "Operational Safety on Airports During Construction" all training will be provided by the Owner/WCAA in regards to airfield operations. This training is required prior to receiving your personnel badge.
- **Vehicles & Equipment on AOA property**
 - All non-working personnel (delivery drivers, contractor's visitors, etc.) requiring access to any portion of the project site must be escorted by construction personnel who have received and are displaying an ID badge with escort privileges.
 - The contractor shall verify and assure that every employee who operates any mobile equipment on Airport Authority properties shall have a current valid driver's license.
 - Onsite vehicles are to have no more passengers than seats available on the vehicle. Personnel carts, golf carts or similar vehicles, where permitted, shall only have as many riders as seats available. Failure to follow this requirement will lead to the removal of personnel from the project.
 - No vehicles will be allowed onsite in an active AOA area without an active WCAA badge with driver privileges, and DTW driving sticker permit displayed on the dashboard of the vehicle. Without these provisions, all vehicles, equipment, and pedestrians will need to be escorted by a WCAA approved & badged person who is trained and certified to drive on AOA property.
 - When driving on AOA property, aircrafts always have the right-of-way and all Contractor vehicles, equipment, and employees shall give way to aircraft at all times and on all surfaces.
 - No vehicles or equipment will be permitted on site without operational head lights, tail lights, and brake lights. The use of high beams is not permitted on airport property.
 - All contractor vehicles shall display in full view, company logos, affixed to each side of the vehicle. Logos shall be no less than 12" x 12" and can be magnetic, printed or painted on, but must be commercially made. Any vehicle that does not have a company logo installed on both sides of the vehicle will not be allowed onto AOA property.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- All construction vehicles are required to display an operational beacon on the top of the vehicle. Vehicles without a beacon will not be allowed onto AOA property.
- All construction vehicles and equipment shall have a 3' x 3' orange and white checkered flag located on the uppermost portion of the vehicle/motorized equipment.
- If a vehicle is escorting another vehicle or a delivery, an escorting flag is required to be displayed. The escorting flag required is a 12" x 18" yellow flag fixed to the top of the escorting vehicle.
- Employees shall not operate any equipment or vehicles for more than 16 hours consecutively.
- Vehicles and equipment that are deemed a potential hazard by Hillwood or WCAA shall be removed from the job site and airport property. Vehicles and equipment that leak any automotive fluid especially oil, hydraulic fluid, transmission fluid, gear oil, gasoline, diesel, etc. shall be removed to the staging area and shall not be allowed to operate on any paved surface. If the vehicle cannot be repaired, the vehicle shall be removed from the airport.
- **Escorting & Flagging on AOA Property:**
 - In order to access airport or AOA property without a badge, all personnel, equipment, deliveries, and vehicles are required to be escorted by personnel that is fully badge & trained, equipped with an escorting privileges on their badge.
 - Escorts (SUV/Pickup Escort) can only escort a maximum of 3 hauling or delivery trucks at a time. Haul trucks are not permitted to escort other haul trucks – a separate escort must be used.
 - Escort vehicles are also required to have "Escort Vehicle" signs fixed to both front doors of the vehicle. Signs shall be no less than 12" x 12" and can be magnetic, printed, or painted on; but must be commercially made.
 - All construction vehicles requiring an escort shall display a 12" x 18" yellow flag fixed to their vehicle.
 - For all flagging services, flaggers must be trained by WCAA staff and equipped with approved hazard communication devices (red/green flag, radios etc.).
 - For escorts contracted through WCAA – The contractor or subcontractor is responsible for the cost of each flagger/escort at the hourly rate as identified by WCAA.
 - Contractor employees designated as escorts, are required to be badged by airport security, go through Security Identification Display Area (SIDA) training program, and pass the computerized test. Office hours for credentials are Monday thru Friday from 8:30 am until 4:00 pm. The office can be contacted at (734) 942-3606. Upon successfully completing the security requirements, those employees who have been selected to drive on the AOA must contact the WCAA Airfield Operations office to coordinate training on the AOA with a WCAA Airfield Operations agent. Training is performed Monday – Friday at 9:30 am and 12:00 pm. The WCAA Airfield Operations Construction Office can be reached at (734) 942-3823.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- **Project Fencing, Barricades, and Perimeter Requirements:**

- This Project includes the removal and installation of the permanent AOA perimeter fence within the project limits. Temporary AOA fences and additional personnel will be utilized to ensure that there are no disruptions to aircraft operations or security.
- For more information on fencing materials for both the permanent and temporary fence installations – ref. specification section F-162 – Chain Link Fence. All fences will be chain link fence, equipped with barbed wire and wildlife deterrent fence skirt.
- All barricades, lighting, and warning devices used to delineate any construction or hazardous areas are to be supplied by Christman|Brinker.
- No trenching is allowed under AOA fence without prior approval and coordination by the Airport Authority Security Division.
- All barricades and fencing installed onsite is required to be secured in place against any movement or jet blast. All barricades are to be reviewed for damages, displacement, security, and to ensure that hazard lighting is installed and operational at the completion of each workday.
- No caution or danger tape is allowed on site (for more information ref. section for Foreign Object Debris (FOD)). In lieu of tape, other barricade materials such as wood, steel cables, or safety chain may be used. These items are to be properly secured at all times to prevent from becoming airborne or displaced by potential jet blast. The Christman|Brinker company suggests using yellow safety chain in place of caution tape, and red safety chain in lieu of danger tape. Additional safety signage will be required on all caution or danger barricades.
- No project fencing, perimeter protection, guardrails, barricades, or weather protection is ever to be modified or removed without direction from the Christman|Brinker team and prior approvals.
- The exterior perimeter fence is required to have a drivable lane around the entire linear footage of fencing. This is for applicable inspections by WCAA & Airport security at any time.
- Marking and lighting shall be in accordance with the requirements contained in the Barricade Details Checklist (Ref. attachment #8 in SP-10).

- **Project Phasing:**

- The project is currently divided into multiple different phases as outlined in the project schedule and project phasing plan. This phasing is due to ensuring that construction activities do not affect any airfield operations or impact any taxiways or runways at any time. The primary phasing events are those directly affecting the AOA perimeter & construction activities within the airfield. All project phasing is detailed throughout the project baseline schedule and within the Kimley Horn Construction Safety Phasing Plan (CSPP).
- At this point there is no anticipation of phasing changes. If the need arises to deviate from the current project phasing, a written request will be sent to the WCAA detailing the reasoning behind the requested change. Project phasing will not be altered until approval from the WCAA is received.

- **Protection of NAVAIDS:**

- NAVAIDS are a physical device on the ground that aircraft can detect and fly to. There are a variety of different NAVAIDS on airport property, and each have a specific use

DTWR - Sierra Air Cargo Consolidation – Project #222031

to guide the pilot to the designated flight path. All NAVAIDS that are within a construction zone are required to be protected and flagged prior to start of work. Other electronic signals or radios are to be monitored and controlled to protect any frequencies from conflicting with DTW NAVAIDS. All navigation equipment and NAVID critical areas that are to remain active inside the limits of construction during construction will be delineated with orange safety fence. If required for operational purposes, wooden posts will be utilized to secure the orange safety fence in lieu of the metal posts.

- All NAVAIDS will be protected in place unless otherwise specified in the construction documents. Any damage to navigation equipment will be reported immediately to the WCAA. Storage of construction equipment or materials will not be permitted near any navigation equipment. Prior to commencement of construction and prior to construction within any subsequent phases of construction, The Christman|Brinker team will coordinate with WCAA Airfield Operations and the Construction Administration Team to ensure that the FAA has the proper notification to evaluate any effects construction activities may have on the navigational equipment.

- **Contractor Access:**

- Project access points and haul routes are shown in the contract documents and detailed in the CSPP. Any deviation from the routes shown in the CSPP will not be allowed without prior written approval from the Airport Authority.
- Contractors will be able to use WCAA contract security guards to man the AOA security check point (s) for access. The only access point onto WCAA property will be through AOA Gate #27.
- Christman|Brinker employees and on-site subcontractors will all be required to obtain a photo ID access badge through WCAA Security Office. The number of these badges that are tied to our DTWR project will be monitored and tracked via the Christman|Brinker team. Additionally, each subcontractor will be required to maintain a log of all badge holders at each company's site office or with their respective supervision. All badge expiration dates, issue dates, or returned dates are required to be closely monitored throughout the course of the project.
- Any individual requiring AOA Drivers Training will be noted on their badge application and will be required to go through airfield drivers training throughout the course of their badging procurement. Personnel will not be allowed to drive in the AOA until their badge has been issued, drivers training is completed, and after receiving an AOA vehicle permit to display on the dashboard.
- Flaggers are required to be supplied by Subcontractor's for the AOA crossing points. All subcontractor personnel that will be flagging are required to complete the WCAA flagger training prior to being allowed to work in the AOA. Flaggers are required to always have a two-way radio with them. The Christman|Brinker team will coordinate supplying Airfield Operations with a two-way radio to facilitate contact with the flaggers. If Airfield Operations cannot reach subcontractor flaggers, Christman|Brinker team supervisors can be reached.
- Christman|Brinker team will maintain internal communication radios for all on-site personnel. No communication via these radios will be allowed outside of jobsite use only.
- All fencing that is within the construction area will be maintained per all applicable FAA and WCAA requirements and be kept clear to allow for visual inspection.

DTWR - Sierra Air Cargo Consolidation – Project #222031

WCAA has a requirement for a minimum of a 10' clear driving path to be available along the entire perimeter of the construction fence.

- If any work is required outside of the proposed and submitted construction area a request for access to that area must be submitted in writing to WCAA for approval a minimum of 72 hours in advance – excluding weekends.
- Deliveries are to be strictly controlled by Christman|Brinker. All deliveries are required to either be driven by badged personnel with driver privileges, or to be escorted. Ref. escorting section for further information on required escorting provisions.
- Parking will be made available for the workers onsite after AOA fencing is installed. Prior to that, only permitted vehicles will be onsite.

• **Contractor Staging Area & Material Laydown Spaces**

- Access to the Contractor staging area, batch plant, and Project Site will be Superior Road. The Contractor shall provide signage to/from the Contractor staging area. The signage plan shall be submitted and approved by Hillwood, Airport Staff, and any other regulatory agencies. The signage plan shall also meet the requirements of all federal, state, and local requirements.

• **Wildlife Management:**

- The project site is placed on an existing wetland space under WCAA property. The Christman|Brinker team's goal to prevent impacts to any surrounding wildlife and existing environmental features that is not completely necessary for construction.
- In order to prevent not only FOD, but also attract wildlife – all trash cans and dumpsters are required to be completely covered at all times. For dumpsters, this would include either a dumpster roller cover or secured tarp cover. For trash cans around site, a secured manufactured lid will be required to be placed with each trash can.
- All standing water or pond water that has collected on the jobsite must be pumped out and removed immediately. If any water is caused from a subcontractors excavations, tire ruts, or low graded areas, subcontractors will be notified immediately and requested the water be removed promptly.
- The project boundaries include existing vegetation that must be maintained throughout construction. To prevent the attraction of wildlife, all grass over 10" within the construction area will need to be cut and maintained. This included any grass or vegetation around project fence and perimeter.
- If any existing wildlife habitats are discovered that will be required to be disrupted, the Christman|Brinker team will be required to immediately notify WCAA Airfield Operations prior to any interaction with the existing wildlife. The designed wetlands are excluded.

• **Underground Utilities:**

- Christman|Brinker and its subcontractors will utilize the MISS DIG System for location of all public utilities. In addition, WCAA Maintenance and if needed, a private utility locating service will also be utilized. The WCAA Request for Marking Utilities Form (ref. Attachment XIV) will be filled out at least two weeks prior to excavation

DTWR - Sierra Air Cargo Consolidation – Project #222031

being performed. Prior to performing the field locates, Christman|Brinker will schedule a Utility Locate Meeting with the project team to facilitate communication.

- All underground utilities that are found will be located with GPS and noted on a project drawing to ensure that the information is preserved and readily available. Utilities will be marked in the field with paint and or stakes. Color coding / labeling of the locate marks will be done per SP-10 3.3D. Upon completion of utility markings, the WCAA Utility Field Acknowledgement Form (ref. Attachment XVI) will be filled out noting all utilities that were marked or could not be located and including all marking documentation. Upon survey being conducted, underground work is only allotted for 21 days thereafter.
- If any utility is going to be tapped into or shutdown, a WCAA Utility Interruption Form (ref. Attachment XVII) and a Utility System Custody Form (ref. Attachment XVIII) is required to be submitted at least (5) days prior to shutdown.
- Prior to commencing any excavation, fence installation, saw cutting or pavement removals Christman|Brinker and our subcontractors will review all applicable drawings with WCAA personnel to ensure that all known utilities have been clearly marked and that there is no questions regarding the locates. Areas that conflict with work or are of special concern will be discussed with WCAA personnel. The Excavation checklist will be filled out and signed by the foreman that is supervising the crew performing the excavation work prior to the start of work.
- All trenches / excavations will be inspected and monitored by a trained, excavation competent person. All trenches / excavations that are in excess of 5 feet in depth will be properly sloped or a protective system used. Excavations / trenches that are left open during non-work hours will be properly marked and barricaded as required.
- Excavation within 5 feet of a known utility will be monitored using a hand probe, shovels or a soft dig vacuum until the utility is physically located. Existing utilities that are to be exposed will be properly supported. As each situation will be unique, details for any required utility support will be submitted at that time.
- All abandoned or removed utilities will be noted on the project as-built drawings for future reference.

- **Special Conditions:**

- During non-working hours and/or periods of weather conditions that are approaching a 1,000-foot ceiling or 3 mile visibility (as indicated by Airfield Operations) all men, equipment and materials will be pulled back to a location that is a minimum of 250-feet from the centerline of any active runway. At no time will equipment be operated or remain within 400-feet of an active centerline if its height would exceed 25-feet in height above the active centerline elevation.
- If questionable sky conditions are noted, Christman|Brinker will call 734-941-5363 to check current sky conditions at DTW.

- **Runway & Taxiway Visual Aids:**

- Runways that are closed to aircraft operations will have portable lighted runway closure markers installed over each end of the runway. Closure markers will be placed over the runway designations. Additionally temporary closed runway markers will be placed along the length of the closed runway.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- All AOA pavements closed to aircraft operations will be delineated with lighted low profile barricades and taxiway ending markers at the intersections. Any guidance signs for the runways and taxiways that are closed will either be removed or covered, airfield lighting for the closed runways and taxiways will either be de-energized or disconnected. Any centerline markings that lead into the construction area will be removed.
- The permanent and temporary lighting and signage will be done in accordance with AC 150/5340-30F and 150/5340-18F. Permanent and temporary pavement markings will be done in accordance with AC 150/5340-1K
- **Markings and Signs for Access Routes:**
 - Access routes to and from the construction area will be as shown on the operational plans. Traffic Control Plan will be developed by Christman|Brinker per the current MUTCD standards. A Traffic Control Plan will be developed and updated as required as a separate document.
- **Hazard Marking and Lighting:**
 - Open trenches and excavations will be marked utilizing orange safety fence and low profile barricades with red obstruction lights. Open trenches and excavations will be minimized to limit the hazard. Excavations or trenches that will remain open for an extended period of time or are adjacent to paved areas that carry vehicle traffic will be protected by concrete barrier walls equipped with flashing warning lights.
 - Heavy equipment and vehicles will be marked in accordance with FAA Circular AC 150/5210-5D. The concrete batch plant located in the staging area will have a 36" x 36", white/orange checkered flag installed at its tallest point. In addition to the flag a red obstruction light will be installed on the highest point.
 - Material stockpiles will be kept away from any active taxiway surface. If material stockpiles are deemed a hazard to vehicle traffic they will be delineated with low profile barricades.
- **Protection of Runway and Taxiway Safety Areas:**
 - Limitations for work, personnel, equipment and materials in the safety areas as designated in the project CSPP will be strictly adhered to. All personnel work on the site will be briefed on the requirements in the CSPP.

If during periods of inclement weather Airfield Operations determines that work is to be suspended, all employees and equipment will be removed to either the project staging area or as directed by Airfield Operations.

Work of any type will not be performed in the safety areas listed in the CSPP until all requirements for the closure or restriction of the adjacent movement area(s) have been completed and is acceptable to WCAA Airfield Operations and the FAA.
- **General Construction Jobsite Safety Rules:**
 - Report all injuries to the field office. Get first aid promptly.
 - Keep your mind on your work at all times. No horseplay on the job.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- If any part of your body should come in contact with an acid or a caustic substance, rush to the nearest water available and flush over the affected part. Secure medical aid as soon as possible.
- Watch where you are walking. Don't run at anytime on the jobsite.
- The unauthorized use, sale, possession or distribution of alcohol, narcotics, drugs or controlled substances while working or working while under the influence of these substances is prohibited.
- Do not distract the attention of fellow workers – to do so may cause injury.
- Sanitation facilities are provided for your use. Use them.
- Good housekeeping is a must. Keep your work area free from rubbish and debris. Keep materials, tools, and equipment in a neat and orderly fashion.
- Do not use a compressor to blow dust or dirt from your clothes, hair, face, or hands.
- Use a respirator when working on operations where dust or fumes are present.
- Never work aloft if you are afraid to do so, are subject to dizzy spells, or if you are apt to be nervous or sick.
- In case of injury to a fellow worker, go for help. Never move an injured person unless it is absolutely necessary – further injury may result. Keep the injured as comfortable as possible and utilize jobsite first aid supplies until medics arrive.
- Know where firefighting equipment is located and learn how to use it.
- Learn to lift correctly – with the legs, not the back. If a load is too heavy, GET HELP.
- Riding material hoists, crane ball, and sideboards or with your legs dangling over the ends or sides of trucks is not permitted.
- Do not use power tools and equipment until you have been properly instructed in their use and operation. Be sure that all guards are in place. Do not remove, displace, damage, or destroy any safety device or safeguard furnished or provided for use on the job, nor interfere with the use thereof.
- All Contractor equipment brought onsite for use on or during the construction project shall be kept in a safe operating condition. Worn or damaged equipment shall be repaired, replaced or taken out of service (locked out) and removed from the job site.
- Rope off or barricade danger areas. Do not enter an area, which has been roped off or barricaded.
- If you must work around backhoes, cranes, trucks, and dozers, make sure operators are aware of your presence. All mobile equipment must have backup alarms.
- Never oil, lubricate or fuel equipment while it is running or in motion.
- Trenches 4 ft. in depth must have a ladder and over 5 ft. must be shored or sloped as required. Keep out of trenches or cuts that have not been properly sloped or shored. Inspect trenches and banks daily. Backfill material must not be stored closer than two feet from top of bank.
- Use the “four and one” rule when using a ladder – one foot of base for every four feet of height. Always secure the bottom of the ladder with cleats and/or safety feet. Lash off the top of the ladder to avoid shifting. Ladders must extend three feet above a landing for proper use. Defective ladders must be properly tagged and removed from service.
- Erect scaffolds according to manufacturers' recommendations.
- Use only extension cords of the three prong type and protected with a GFCI.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- The use of fall protection with safety lines when working from unprotected high places is mandatory. Always keep your lines as tight as possible.
- Never throw anything “overboard”. Someone may be passing below.
- Open fires are prohibited.
- Floor perimeter and floor openings shall be properly guarded at all times. Perimeter cabling must be flagged. All covers for floor openings must be secured and marked, “Hole Cover Do Not Remove”. A “hole” means a gap or void 2 inches or more in its least dimension, in a floor, roof, or other walking/working surface. Note – this includes mechanical and electrical sleeves.
- All personnel operating a manlift or forklift must be certified.
- No music radios, MP3 players or personal listening devices are allowed on site. No headphones of any kind will be allowed on site or are permitted during construction activities.

XVI. Housekeeping

A clean work place is a safe work place. Every subcontractor and worker shall be responsible to adhere to the following housekeeping guidelines:

- Contractors must control their operations in a manner that minimizes the migration of construction materials, waste dirt, dust, debris rocks or other matter. Such material must be prevented from entering the airfield or limiting visibility or causing Foreign Object Debris (FOD) to park or operating aircraft.
 - Pick up and dispose of trash and debris on a daily basis. A sufficient number of trash receptacles shall be provided and emptied with regularity.
 - **Work areas are to be kept clean, orderly and floors are to be swept daily.**
 - Leads, hoses, and extension cords shall be hung up with a nonconductive material, off all floors, stairways, and walkways.
 - Trash such as drinking cups, cans, and scraps from lunch are not to be thrown down, but disposed of properly in marked containers.
 - Materials, equipment, concrete forms, pipe, etc., are to be orderly stacked so as not to obstruct walkways, doors, stairways, ladders, other points of access or egress, etc.
 - Oil, grease, and other such liquid spills shall be cleaned up at the time of spill and are not to be left unattended.
 - Participation in mandatory cleanup programs if the project requires it. At the very least, all subcontractors are responsible for their identifiable debris to be picked up and put into a trash container. Subcontractors must take away all pallets, boxes and dunnage to dumpsters provided by Christman|Brinker. Christman|Brinker laborers will perform unidentifiable clean-up and general sweeping along with hauling/emptying mobile trash containers and replacing them where found.
- **Foreign Object Debris (FOD)**
 - Foreign Object Debris (FOD) is any trash, soil, material, debris etc. that is loose/accessible on the jobsite and has potential to become airborne and can damage airplane or personnel. This project includes working on an active airport facility, which means planes are consistently being operated around the jobsite. It is to be assumed by the Christman|Brinker team and all subcontractors that the jobsite work area will be subject to potential jet blast and high wind velocities at any time. These airplanes have engines that can very easily suck in any loose items into their engines, or from the back side have a very strong jet blast that can easily blow items into the air. It's our main priority to prevent any impact to airport operations or aircrafts – thus additional precautions are required to prevent any FOD on the project. This includes but is not limited to having lids placed on all trash cans and dumpsters at all times, having multiple vacuum type mechanical sweeper and water truck to clean all taxiway and apron pavement of dirt, stones, and loose debris where construction traffic operates, and collectively working to identify and remove all FOD on a daily basis but also through weekly safety inspections.
 - No debris shall be allowed to remain on the roadways or airport paved surfaces. Active runways, taxiways, and aprons shall be kept free of debris at all times. If any FOD substances are found on any active roadways, runways, taxiways or aprons it must be swept up and cleaned immediately.
 - Other pavements shall be cleaned by the Contractor daily, and as required, using power vacuum sweepers to keep all access and construction areas clear of soils, clods, or other debris.

DTWR - Sierra Air Cargo Consolidation – Project #222031

- All stockpiled material and supplies shall be constrained in a manner to prevent movement resulting in an aircraft blast or high wind conditions. All stockpiles shall not exceed 15' in height and cannot be placed within 6' of an AOA fence. All materials cannot be stored within 500 feet of an aircraft turning or any movement areas.
- No chemicals shall be used for removal of FOD.
- The Contractor shall have available on-site at all times a method of periodic spraying of any stockpile or exposed areas to limit dust. All waste receptacles shall be covered when not in use to prevent potential FOD from being blown out by wind or jet blast and also to prevent wildlife attractants.
- All airfield personnel, construction workers, and vehicle drivers will monitor construction sites and all paved areas within construction limits for potential FOD. Personnel are to make all efforts to keep areas open to aircraft operations clear of debris and FOD. If hazardous conditions exist, notify WCAA Airfield Operations immediately at (734) 942-3823.

XVII. Ancillary Safety Procedures

- **Lockout/Tagout:** a procedure for cutting off all sources of energy or flow of medium to a piece of equipment or system. Refer to Attachment III.
- **Confined Space Entry:** a procedure to assure that workers who perform work in a confined space is protected. Refer to Attachment IV.
- **Powder-Actuated Tool Operator:** each powder-actuated tool operator will be certified in accordance with OSHA requirements. Operators will carry their certification card on their person while operating powder-actuated tools.
- **Hole Covers:** Any contractor / trade that creates or exposes a hole or opening in a floor or wall is responsible to immediately protect the opening in accordance with OSHA regulations. At a minimum the hole is to be protected by barricades or a secured cover that is sufficient in strength to support any potential load that may be imposed. Additionally, anyone who removes a hole cover is responsible to protect the area while the exposure is present and to re-secure the cover or protection once the work is completed for that shift.
- **Scaffolding:** The use of makeshift, defective or inadequate scaffolding, rigging, or staging is prohibited.
- **Utility Installation:** Potable water line installation will require coordination with the Airport Authority. In addition, a permit application may have to be completed and submitted to the Airport Authority. The Contractor shall follow Detroit Water and Sewer Department guidelines for the installation and storage of potable water lines. Installation, storage, and testing results shall be coordinated with the Airport Authority.
- **Underground Utilities:** This project includes modifications to storm and sanitary systems and water services. All above grade and below grade utilities must be maintained during this work. Contractors are advised to coordinate with Christman|Brinker and Wayne Co. Airport Authority (WCAA) before beginning any work in the vicinity of existing utilities. Contact WCAA 2 weeks prior to starting any work onsite for site wide utility sweep/marketing. Use Attachment #3 in the WCAA Safety Standards Utility Marking Request Forms. Each contractor is also responsible for contacting MissDig or other responsible agencies for identification before beginning work.
- **Excavation Work:** Prior to any Subcontractor excavating, it is imperative that Miss Dig be contacted. Arrangements must be made 5 days prior to starting work by contacting (800) 482-7171. There are known underground utilities in this area, including power, gas, cable and water. All known utilities are to be located by hand dig prior to using any mechanical equipment for excavation. There will be deep excavations that have steep slopes, limited access and are barricaded or have handrails around them. It is important that the safety features of the excavations are not compromised. If a handrail or barricade must be temporarily removed, Construction Manager must be notified prior to doing so. No handrails or barricades are to be removed and the leading edge left open; immediate replacement of handrails is mandatory.

XVIII. Tobacco Use Protocol

Smoking and the use of all other tobacco products are strictly prohibited on the project site. Violation of this policy will be cause for disciplinary action up to and including permanent removal from the project.

XIX. Crane Inspections

- Each crane used at the Project site shall have a current annual crane inspection report, which documents that the crane has been inspected within the current year in accordance with established OSHA standards. The Owner has specific controlling requirements for the use of cranes and other equipment with booms. Contractors are to strictly adhere to these requirements.
- All cranes used on this project must be operated by a Certified Crane Operator.
- All equipment exceeding 25 feet in height must be obstruction marked and lighted at night, and when not in use must be lowered to its stored height.
- Christman|Brinker is required to submit a critical lift plan to WCAA that is prepared and approved by a professional engineer that is registered with the State of Michigan, for when a crane lift exceeds 70-91% the crane capacity and/or requires two cranes for one lift. The critical lift plan will cover: the description of the lift, crane position, lift height, load radius, boom and angle, size and weight of load, percent of cranes capacity, personnel involved, rigging plan, communication methods, ground conditions, inspection procedures, procedures for hoisting (if applicable), weight, crane & rigging capacities, inspections and wind speeds.

For any crane or hoisting operations, it is imperative to perform the required crane inspections before any hoisting work takes places. The crane inspections must be passed to let any hoisting work start. Please review and follow the crane and hoisting guidelines:

- The operator shall perform a visual inspection of the crane each day before use.
- If the crane requires emergency inspection of repair, the operator will inform the superintendent who shall arrange for maintenance.
- A daily inspection shall be documented as indicated on Daily Crane Inspection Record.
- Each crane shall have an annual inspection by a qualified person. The inspection shall be documented in writing and be copied to the yard resource manager, the safety director, and be maintained on the project site with the crane. All deficiencies shall be corrected prior to putting crane back into service.
- Following any major repair, overhaul, modification, or reconfiguration, all cranes must be re-certified before being returned to service.

It is imperative to perform safe operation of the equipment as well as safely operating or conducting operations around them. Be sure to coordinate your work when near the operation of the crane and inform your employees of the inherent dangers while working in the area. Additionally, please review and follow the crane and hoisting guidelines:

- All employees who rig loads to be hoisted by cranes must be “qualified” by industry definition. Qualified is defined as: (1) experience in that job task and (2) has attended training on inspection and use of rigging devices, such as nylon slings, cable chokers, etc. (3) attaches tag lines to loads.
- It is the policy of The Company that all crane operators for tower cranes or cranes of 150 ton or more in capacity should be certified operators. Exceptions may be made by the Christman|Brinker Safety Director, if proof exists that the operator has more than five years’ experience in a similar crane. The Project Superintendent shall obtain and review this information from each trade contractor prior to lifts being allowed on the project site.
- A pre-hoist checklist is to be submitted by each trade contractor prior to performing any crane hoist that involves hoisting of materials more than 8 feet above the ground that is considered a critical pick (i.e. – greater than 80% of the crane capacity).

DTWR - Sierra Air Cargo Consolidation – Project #222031

- All lifts shall be planned and review by all personnel involved in rigging, hoisting or receiving. YOUR Operator and YOUR Riggers shall be in communication with each other by the use of a designated radio communication channel provided by you.
- All riggers will instruct the operator to sound an audible horn (minimum 80 decibels) prior to lifting and/or swinging into an intended work/receiving area if the operator observes workers in the intended swing path.
- All workers should look overhead when the horn is sounded. Observe the intended path of travel and ensure they are not standing in the fly zone and in harms way.
- All employees will be instructed on this procedure in the new hire orientation.
- Critical picks will be preceded by a meeting attended by crane operator, supervisors and riggers.
- Riggers / Crane Operators – Christman requires the use of qualified Riggers / Certified Crane Operators (CCOs) on all cranes on its jobsites. Qualified riggers / CCOs on all cranes is a requirement for each project thereby ensuring minimum rigger/operator qualifications and operating standards are meant to protect the safety of all workers and the public.
- Prior to performing any hoisting with crane, submit all your documentation to the Construction Manager and get approval.
- All cranes and cable (wire rope), rigged hoisting equipment shall have a current annual inspection by an accredited agency prior to working at this site. Contractors also are required to maintain a current annual inspection for the duration of the work. An accredited agency is a third party, which is recognized by the Department of Labor, Occupation Safety and Health Agency. A copy of the annual certification must be submitted to the respective general contractor immediately upon arrival on site. Rigging must be inspected and have required capacity tags.

XX. Visitor Policy

All visitors to the Project site must report to The Christman|Brinker Company Project Office upon arrival. All visitors shall comply with the policies and procedures of this Project Safety Manual while on site. Visitors are required to watch The Christman|Brinker Company safety orientation video and sign the visitor release form. Visitors must wear a hardhat, Hi-Visibility shirt or safety vest, and proper footwear at all times while on site. Other PPE such as safety glasses are to be worn when the conditions warrant such protection. The contractor requesting approval of the visitor is responsible to issue the proper PPE that may be required.

No visitors are allowed onto the jobsite or AOA property without a WCAA approved or badged escort. All escorts need to be arranged a minimum of 24 hours in advanced and inputted into WCAA Escort Required Name Check Website at the link below:

[DTW Escort Required Name Check \(metroairport.com\)](https://metroairport.com)

If an escort is needed for more than one day, the name of the personnel being escorted is required to be inputted into the WCAA Escort Required Name Check Website on a daily basis.

XXI. Fire Prevention and Protection

A. Fire Prevention

1. Housekeeping

Proper Housekeeping practices by all subcontractors will help to establish and maintain free and unobstructed egress from the project site. Housekeeping of the project is to be maintained by all subcontractors performing work on the site. It is each subcontractor's responsibility to report areas of the jobsite that present a likelihood of fire to The Christman|Brinker Company superintendent. Depending on the type of construction and the participation of the subcontractors in daily housekeeping efforts, The Christman|Brinker Company may create and enforce a mandatory cleanup policy that will be adhered to by all subcontractors.

2. Storage of Flammable and Combustible Liquids

A. General Requirements

1. Only approved containers and portable tanks will be used for storage and handling of flammable and combustible liquids.
2. Approved metal safety cans will be used for handling and use of flammable liquids in quantities greater than one gallon. Plastic containers are prohibited. Flammable liquid materials may be used and handled in original shipping containers.
3. For storage, use and handling of flammable liquids in quantities of one gallon or less, only the original container or approved metal safety cans may be used.
4. Flammable or combustible liquids are not to be stored in areas used for exits, stairways or other high-traffic areas.
5. No storage of flammable or combustible materials is to occur on any roof.

B. Indoor Storage

1. No flammable materials may be stored inside.
2. Materials, which will react with water and create a fire hazard, shall not be stored in the same room with flammable or combustible liquids.

C. Outdoor Storage

1. All storage areas are to be created as to prevent the contamination of surrounding areas in the event of a spill. Proper consideration is to be given to the runoff of rainwater vs. the control of spilled or leaking materials.
2. Containers less than 60 gallons in size are to be placed in areas not to exceed a total of 1,100 gallons. Containers are to be separated by a five feet clearance and will not be nearer than 20 feet to a structure.
3. Portable tanks will not be nearer than 20 feet from any structure.

D. Gas and Oxygen Bottles

1. Full and/or partially used compressed gas cylinders must be attached to approved carts and turned off after each use.

DTWR - Sierra Air Cargo Consolidation – Project #222031

2. All compressed gas cylinders regardless of content amount (full, partially full or empty), shall be properly stored and secured in the area designated by The Christman|Brinker Company.
3. Valve protecting caps shall always be in place when cylinders are not connected for use.
4. When moving a cylinder, be sure to remove all gauges prior to the move. Be sure to use a rack when lifting a cylinder to and from upper elevations. Never move a cylinder by rolling it or by dropping it down to lower elevations.
5. Oxygen cylinders must be stored 20 feet away from fuel cylinders or separated by a solid 1- hour divider. Storage is considered as not being used within the next shift of work or 24-hour period, which is ever shorter. Oil and grease must be kept away from oxygen cylinders.
6. Compressed gas cylinders shall be legibly marked with the chemical or trade name of the gas. Marking shall be by stenciling, stamping, or labeling and shall not be readily removable.
7. All L.P.G. not connected for use must be properly stored in the designated area. **NO EXCEPTIONS!**

E. Temporary Heat

1. Temporary heaters must be installed and serviced by authorized technicians only.
2. Housekeeping in the area of the heater is extremely important and all subcontractors are responsible to insure that no materials are stored near the heater or in such a fashion that they may create a hazardous condition.
3. All temporary fuel burning heaters are to have functioning safety valves that will eliminate fuel flow in the event of flame outage.
4. If a heater is to operate for an extended period of time it must have a daily scheduled inspection to insure that it is operating correctly and that the area is clear of possible obstructions.
5. The Christman|Brinker Company superintendent may assign maintenance, operation, and inspection responsibilities to a subcontractor.
6. All subcontractors are responsible to notify The Christman|Brinker Company immediately in the event of a malfunctioning heater or potential for a hazardous condition.

F. Hot Work Operations and Permit Procedures

Many of Christman|Brinker projects are within existing operating facilities. Clients may have established procedures at their facilities to ensure all safety precautions have been taken prior to initiating operations involving Hot Work. Additionally, Hot Work permits may be issued by Client operations personnel to control the work within operating areas and ensure that operator/maintenance/construction personnel are aware of client required safe work practices. It is the responsibility of all personnel entering client controlled operating areas to obtain the proper Client authorization and appropriate work permits. The Christman|Brinker Company shall be informed of any intended Hot Work prior to initiating. The Christman|Brinker Company will coordinate our Hot Work permitting program with any other existing program being implemented by the Client. Regardless if the Client has a Hot Work procedure, NO Hot Work shall take place without first coordinating with The Christman|Brinker Company. Wayne County – Hot Work Permit (ref. Attachment XV) is also required to be filled out, submitted, and signed

DTWR - Sierra Air Cargo Consolidation – Project #222031

upon evaluation of any possible fire hazards near any construction work, building materials, equipment and near any personal or public property. WCAA will require and issue the Hot Work Permit.

The purpose of this procedure is to eliminate the potential for injury to personnel and damage to equipment and property due to fire, sparks, heat or explosion resulting from cutting, welding and similar hot work.

A "Hot Work Permit" will be required by The Christman|Brinker Company when using any equipment, tools or apparatus capable of generating heat, sparks or flames. The Christman|Brinker Company, through designation of a Permit Authorizing Individual (PAI), will ensure adherence to this Hot Work permit procedure. Failure to comply with this Hot Work Permit procedure may result in immediate and permanent removal from the project. The discipline policy as described in section IV of the project specific safety manual will be enforced.

Upon issuance of the proper work permit by Christman|Brinker or the Client's representative, the PAI shall completely perform the required actions indicated on the permit, check all boxes on the permit, and sign off on it. The PAI shall print their name and sign the work permit verifying he has reviewed the permit and understands the limitations and requirements within.

Upon receiving the permit, it is the responsibility of the PAI to review the permit with his employees/crew explaining the precautions and limitations of the proposed task and assure that the permit system is implemented correctly.

The following conditions apply to all hot work permits:

- The contractor will be responsible for providing all the required training, materials, personnel and protective equipment to conduct all hot work.
- Contractor personnel who perform fire watch duties must be trained to perform the task and trained to use a fire extinguisher.
- A permit is not valid for more than one shift. In any case, the duration of the Hot Work Permit shall not exceed 12 hours. A new permit is to be issued if the job/task duration exceeds 12 hours.
- Prior to starting operations, the working areas and any levels below must be reviewed to assure that flammable material and equipment has been removed from harm's way.
- Adjacent contractors have been informed of the impending hot work.
- Contractor supplied firefighting equipment must be in place and readily available for worker use.
- A restricted zone must be established outside of the working area and at any areas levels accessible below.
- A fire watch must be established and maintained for a designated period after hot work has been completed. The duration of the fire watch is dependent upon the type of hot work being performed and the working area. Fire watch can range from one (1) to two (2) hours.
- Hot work permits must be prominently displayed within the hot work area and available for review. Workers assigned to fire watch duty must be readily identifiable and within the area. Fire watch must be maintained through break and lunch periods.

Occasionally Special conditions may apply to hot work permits, based on the environment of

DTWR - Sierra Air Cargo Consolidation – Project #222031

the work area and the type of operations being performed. Special conditions to the permit may include the following:

- Fume monitoring and control measures.
- Flame retardant ground and wall coverings.
- Extended fire watch
- Dosing or spray down of work areas after hot work completion.
- Use of smoke and carbon monoxide detectors.
- Oxygen and gas monitoring.

Hot work permits are to be collected by the PAI at the expiration of the permit, conclusion of task or end of shift. After the permit has been closed out they are to be submitted to The Christman|Brinker Company superintendent or site safety manager for review.

Copies of the permit will be retained by The Christman|Brinker Company for the duration of the project.

B. Fire Protection

- A means of egress shall be established and maintained. Each subcontractor is responsible to notify The Christman|Brinker Company if the means of egress is not obvious or if it is obstructed.
- The rules for evacuation will be posted on the project bulletin board area adjacent to the other OSHA required postings. During establishment of AOA fencing, the emergency meeting point will be inside the sliding gate at entrance of Superior Rd off Middlebelt Rd. If the location of the work is such that posting of the rules is not feasible, a copy of the rules and a map of the evacuation route (if applicable) will be given to each subcontractor working on the project. It is the responsibility of each subcontractor to make their employees aware of any such posted or otherwise conveyed rules, maps or procedures for the safe and orderly evacuation of the project.
- Emergency phone numbers shall be posted next to each phone in the jobsite trailer or office.
- Firefighting equipment shall be provided by the subcontractor for protection of their employees. In some cases, The Christman|Brinker Company may provide “common area” protection. Each subcontractor is to coordinate their efforts to comply with this rule with The Christman|Brinker Company superintendent. Defective equipment shall be immediately replaced and brought to the attention of The Christman|Brinker Company superintendent.
- Fire walls and exit stairways required for the completed buildings are given construction priority. Fire doors with automatic closing devices shall be hung on openings as soon as practicable.
- Retain existing fire separations in buildings undergoing alterations or demolition until operations necessitate their removal.

C. Employee Emergency Action Plan

DTWR - Sierra Air Cargo Consolidation – Project #222031

The Christman|Brinker Company's project superintendent shall be the person in command in the event of a fire or other emergency. It shall be his/her responsibility to oversee that employees are evacuated and the emergency response team is notified. The superintendent shall designate another employee as second in command. Fires and other emergencies shall be reported to the superintendent immediately. All employees are required to familiarize themselves with the evacuation plan for the site and with the designated contacts for emergency response team.

In the event of a fire or other emergency, employees shall evacuate the building through the nearest exit after determining it is safe to do so. Upon exit from the unsafe area, the employee should contact his superintendent and coordinate efforts to call the emergency response team (911). Each foreman or superintendent is responsible for a head count of their employees after evacuation. The Christman|Brinker Company superintendent is to be notified of any missing employees.

If critical operations must be conducted or continued within or near an unsafe area during a fire or other emergency and the superintendent deems that limited activity may continue in the area, the superintendent shall establish a plan to monitor those employees in the unsafe area as well as a way to quickly evacuate them should the conditions become worse. If monitoring and quick evacuation are not possible, the operations shall be abandoned until conditions improve.

All employees are responsible for the joint welfare of all other employees. During evacuation, ambulatory employees trained in first-aid/CPR are responsible to assess the conditions of non-ambulatory employees and assist in their evacuation. Employees who cannot be moved must be reported to the superintendent upon evacuation.

ATTACHMENT I - DRUG SCREENING LIST

Drug Screening List

- A. Any employee injured at the project site and needing offsite medical attention will be tested for alcohol and drugs. If the injured party was hurt by the action of another person who was not injured, this person must also submit to substance abuse testing. An employee who tests positive for illegal substances or alcohol will not be allowed to return to the project site. Any employee who refuses to submit to substance abuse testing after an injury/accident which involves offsite medical attention will not be allowed to work at the project site.

Notes:

- a. All staff/trades working at the project site shall be subject to the provisions of this program. Visitors and delivery personnel are not included.
- b. Substance abuse testing shall include the drugs listed in c (below) and alcohol.
- c. The drug-screening test shall include testing for amphetamines, barbiturates, benzodiazepines, cannabinoids, cocaine, methadone, methaqualone, opiates, phencyclidine, and propoxyphene. Refer to the following listing, commonly referred as a Panel 10 Drug Screening:

DRUG GROUP	COMMON NAME	DRUG OR METABOLITE DETECTED
Amphetamines	Speed, Eve, Crystal	Amphetamines, Methamphetamines
Barbiturates	Secobarbital, Pentobarbital, Phenobarbital, Butabarbital	Diverse
Benzodiazepine	Valium, Librium	Oxazepam
Cannabinoids	Marijuana, Pot, Smoke, Weed, Mary Jane, Ganja	Delta-9-THC-9-COOH
Cocaine	Coke, Blow, Crack, Tool	Benzoylcegonine
Methadone	Methadone, Met	Methadone
Methaqualone	Quaalude, Ludes	Methaqualone
Opiates	Morphine, Codeine, Heroin, Horse	Codeine, Morphine
Phencyclidine	PCP, Angel Dust, Acid, Ecstasy	PCP
Propoxyphene	Darvon	Propoxyphene

ATTACHMENT II - ACCIDENT INVESTIGATION REPORT

Accident Investigation Report

Employee Name	_____	Date of Birth	_____
Home Street Address	_____	Male or Female	_____
City, State, Zip	_____	Date Hired	_____
Home Phone	_____	Date of Injury or illness	_____
Emergency Phone	_____	Time Employee Began Work	_____
Case # from Log	_____	Time of Event	_____

If a subcontractor on job, please fill out the following information:

Employer Name	_____	Position	_____
Work Comp Insurance	_____	Policy #	_____

Project Name & Location _____ TCC Project # _____

Did injured person die? _____ Date of death _____ Did injury result in lost time? _____ # Days _____

Drug test given? _____ Did injured seek medical attention offsite? _____ Name of Physician _____

Name and address of facility _____

Was person treated in an emergency room? _____ Was person hospitalized overnight as an in-patient? _____

What was injured doing just before incident occurred? (Describe activity as well as tools or equipment used. Be specific.)

Describe the injury. (Be specific: cut, fracture, burn, strain, body part affected.)

Describe events causing injury. (who, what, where, when, why, how)

Identify unsafe acts, conditions, contributory factors, environmental conditions.

Witnesses to accident. (name, address, phone number)

Describe steps taken to prevent future accidents of this nature.

Severity Potential: _____ Major _____ Serious _____ Minor _____ Have similar accidents occurred before? _____

Superintendent Signature & Date

Injured Signature & Date

Upon completion email to: incidentreporting@christmanco.com

ATTACHMENT III - LOCKOUT/ TAGOUT PROCEDURES

Lockout/Tagout Procedure

Requirement

Whenever service, maintenance, or inspection is performed on machines, equipment, electrical circuits, or systems, it must be done with the machine, equipment, electrical circuit, or system stopped and isolated from all sources of energy or flow of medium. The energy isolation device(s) or isolation valves for that machine, equipment, electrical circuits, or systems must be locked out and tagged out in accordance with a documented procedure. Employees involved in the energy control program must be given training. When subcontractor employees are performing work at the Project site, they must coordinate with The Christman Company and any other employer to ensure that no employees are endangered. When a group of employees is performing a service, maintenance, or inspection activity, each employee must be afforded protection equivalent to the utilization of individual lockout/tagout.

Lockout/Tagout Procedure

This procedure establishes the minimum requirements for the lockout/tagout of energy isolating devices and flow isolation valves whenever maintenance, servicing, or inspection is done on machines, equipment, electrical circuits, or systems. It shall be used to ensure that the machine, equipment, electrical circuit, or system is isolated from all potentially hazardous energy sources or flows, and locked out before employees perform any servicing, maintenance, or inspection where the unexpected energization or startup of the machine, equipment, electrical circuit, or system or release of stored energy/flow could cause injury.

All subcontractors are required to comply with the restrictions and limitations imposed upon them during the use of this procedure. All employees are required to perform the lockout in accordance with this procedure. All employees, upon observing a machine, piece of equipment, or system which is locked out to perform servicing, maintenance, or inspection, shall not attempt to start, energize, or use that machine, equipment, or system. Failure to follow this procedure will result in disciplinary action.

Sequence of Lockout

1. Notify all affected employees that servicing, maintenance, or inspection is required on a machine, equipment, electrical circuit, or system and that the machine, equipment, electrical circuit, or system must be shut down and locked out to perform the servicing, maintenance, or inspection.
2. The employee shall refer to any and all sources to identify the type and magnitude of the energy that the machine, equipment, or electrical circuit utilizes, shall understand the hazards of the energy, and shall know the methods to control energy. The type and source of medium which is flowing in a system to be worked on shall be identified and system isolation valves located. All flows shall be stopped and the lines to be worked on purged prior to the commencement of work.

3. If the machine, equipment, or electrical circuit is operating, shut it down by normal stopping procedure (depress stop button, open switch, close valve, etc.).
4. Deactivate the energy isolation device(s) so that the machine, equipment, or electrical circuit is isolated from all energy sources. Close isolation valves or shut down equipment which is used to provide flow to the system.
5. Lock out the energy isolation device(s) with assigned individual lock(s). Tagout isolation valves and/or equipment.
6. Stored or residual energy (such as that in capacitors, springs, elevated machine members, rotating flywheels, hydraulic systems, and air, gas, steam, or water pressure, etc.) must be dissipated or restrained by methods such as grounding, repositioning, blocking, bleeding down, etc.
7. Ensure that the equipment is disconnected from the energy source(s) by first checking that no personnel are exposed, then verifying the isolation of the equipment by operating the push button or other normal operating control(s) or by testing to make certain the equipment will not operate.

CAUTION: Return operating control(s) to neutral or “off” position after verifying the isolation of the equipment.

8. The machine, equipment, electrical circuit, or system is now locked and/or tagged out. The employee(s) that installed a lock shall apply tag(s) identifying who locked the piece out, the date, and the time.

Restoring Equipment/System to Service

When the servicing, maintenance, or inspection is complete and the machine, equipment, electrical circuit, or system is ready to return to normal operating condition, the following steps shall be taken:

1. Check the machine, equipment, or electrical circuit and the immediate area around the machine, equipment, or electrical circuit to ensure that nonessential items have been removed and the machine, equipment, or electrical circuit components are operationally intact.
2. Check the work area to ensure that all employees have been safely positioned or removed from the area.
3. Verify that the controls are in neutral.
4. Remove the lockout/tagout devices and re-energize the machine, equipment, or electrical circuit.
5. Notify affected employees that the servicing, maintenance, or inspection is complete and the machine, equipment, or electrical circuit is ready to use.

ATTACHMENT IV - CONFINED SPACE ENTRY PROCEDURE

CONFINED SPACE ENTRY PROCEDURE

I. Definitions

A. Confined Space

Confined space means any space having a limited means of egress which is subject to the accumulation of toxic or flammable contaminants or has an oxygen-deficient atmosphere. Confined spaces include, but are not limited to: storage tanks, tunnels, pipelines and open top spaces such as pits, tubs, vaults, caissons and vessels.

B. Hazardous Atmosphere

Hazardous atmosphere means an atmosphere that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue (escape unaided from a permit space), and injury or illness from one or more of the following causes:

1. Flammable gas, vapor, or mist in excess of 10% of its lower flammable limit (LFL).
2. Airborne combustible dust at a concentration that meets or exceeds its LFL.
Note: This concentration may be approximated as a condition in which the dust obscures vision at a distance of five (5) feet or less.
3. Atmospheric oxygen concentration below 19.5% or above 23%.
4. Atmospheric concentration of any substance for which a dose or a permissible exposure limit is published as subpart "G", Occupational Health and Environmental Control, or in subpart "Z", Toxic and Hazardous Substances which could result in employee exposure in excess of its dose or Permissible Exposure Limit.
5. Any other atmospheric condition that is immediately dangerous to life or health.

II. Owner/Client's General Requirements

- A. The owner/client shall evaluate the work place to determine if any spaces are permit-required workspaces.
- B. If the work place contains permit spaces, danger signs shall be posted at all points of entry to each space. Example: Danger, Permit Required Confined Space, Do Not Enter.
- C. Identify the elements, hazards, and previous experience with each space that make it a permit space.
- D. Identify any implemented precautions and procedures used by the owner/client to protect his employees in or near each space.

III. Subcontractor Requirements

- A. Each subcontractor shall obtain any available information regarding permit space hazards and entry operations from the owner/client and coordinate employee entry operations in or near permit spaces.
- B. Each subcontractor who performs any work in permit required work spaces shall be responsible to follow the requirements of this confined space entry procedure.

Each subcontractor shall provide training for each employee involved with work located in permit required work spaces.

- C. Subcontractors must provide documentation that training has been given. Said documentation shall contain each employee's name, ID of trainer, and dates of training. Subcontractor employee duties relative to this procedure are as follows:

1. Entry Supervisors Duties

- a. Identify the need to implement this procedure.
- b. Responsible to follow the procedure requirements.
- c. Complete safe entry checklist.
- d. Complete a confined space entry permit for each 8-hour period.
- e. Knows, understands and insures training of the authorized entrants and attendants.
- f. Checks appropriate permit information, assuring all tests have been made and recorded. Knows the hazards that may be faced during entry, including mode information, signs or symptoms, and exposure consequences.
- g. Terminates the permit if required.
- h. Verifies rescue services are available.
- i. Removes unauthorized individuals from the permit space.

2. Attendant Duties

- a. Knows the hazards that may be faced during entry, including mode information, signs or symptoms, and exposure consequences.
- b. Is aware of possible behavioral effects of hazard exposure of entrants.
- c. Continuously maintains a count and identification of who is in the permit space. Verifies that all entrants have exited the confined space at the end of the work shift.
- d. Remains outside the space until properly relieved by another attendant.
- e. Monitors activities inside and outside the space to determine if it is safe for entrants.
- f. If the attendant detects behavioral effects of hazard exposure, he/she must be knowledgeable to:
 - i. Summon rescue and other emergency services.
 - ii. Warn authorized entrants.
 - iii. Perform non-entry rescues.
 - iv. Perform no duties that might interfere with the primary duty to protect authorized entrants.

3. Authorized Entrant Duties

- a. Know the hazards that may be encountered.
- b. Know how to properly use protective and emergency equipment required.
- c. Know how to communicate with attendant.

IV. Procedures

A. General

All employees required to enter into a confined space shall be instructed in:

1. The nature of the hazard involved.
2. The precautions required.
3. The use of personal protective equipment and emergency equipment required.
No employee shall be permitted to enter a confined space unless the following steps have been completed:
 - a. A tested ventilation system is in place to ensure the removal of harmful gases, vapors, smoke, fumes, mists or dusts from within the confined space.
 - b. The employee has been provided with and is wearing appropriate respiratory protective equipment.
 - c. Where presence of toxic gas is suspect, appropriate gas tests shall be performed immediately prior to entry to determine the presence of any harmful gases, fumes, mists, vapors, smoke, dusts or an oxygen deficiency. Re-testing shall be done at least every four (4) hour intervals. During the presence of employees in the confined space, to ensure harmful concentrations of gases, vapors, smoke fumes, mists or dusts or any oxygen deficiency has not developed, gas test results shall be taken and logged.

B. Environmental Testing

Before entry into any confined space suspected of having mixtures or concentrations of flammable and/or toxic air contaminants or oxygen deficiencies, appropriate tests of the atmosphere shall be made by a qualified person to assure that, explosive limits are less than 10% LEL, that toxic gas levels (TLVs) as specified by established health data sheets are not exceeded and that oxygen concentrations are greater than 19.5% but less than 23% of the total air mixture.

A confined space found to have or suspected of having oxygen deficiency or exceeding flammable or toxic limits shall be:

1. Reported promptly to The Christman Company Supervisor on site.
2. Posted with the appropriate warning signs.
3. Ventilated or exhausted.
4. Re-checked by a qualified person prior to entry.

During inert gas welding, portable and/or fixed oxygen analyzers with visual/audible, alarms shall be provided in areas where oxygen deficient atmospheric conditions may

develop. Continuous oxygen monitors, equipped with alarms shall be provided in areas designated.

All monitoring and testing equipment shall be maintained in accordance with the manufacturer's specifications.

C. Personal Protective Equipment

Work and/or rescue equipment shall be immediately available at all times and shall be selected with the potential hazards or possible contingencies determining the equipment which may be required during the work operations.

Appropriate eye, face, and ear protection and the designated protective clothing shall be worn by employees exposed to physical hazards.

Where air sampling has determined that flammable or toxic limits have been exceeded or an oxygen deficiency exists and accepted engineering control measures such as general and local ventilation are not feasible, air supplying respiratory protection shall be worn.

Employees entering a confined space where a harmful or potentially harmful atmosphere exists, shall wear an approved safety harness attached to a life line which is attended at all times by another worker stationed outside the confined space entrance. The outside worker shall be equipped to be capable of immediately rescuing the worker in the confined space. A mechanical device shall be available to retrieve personnel from vertical spaces more than 5 feet deep.

When an employee enters a confined space from the top, the employee shall wear a safety harness, which will keep the individual in a vertical position if a rescue is required.

Where the work being performed requires more than one worker to enter the confined space, provisions shall be made to prevent safety lines or air hoses from becoming entangled.

Where air sampling has determined flammable or toxic limits have been exceeded or an oxygen deficiency exists, and acceptable engineering controls such as general or local ventilation is not feasible, appropriate respiratory protection for the identified hazards shall be worn.

D. Ventilation/ Exhaust

When ventilation is used as a control method in maintaining acceptable concentrations of flammable and toxic contaminants, this section will be used as a guide in the installation and operation of ventilation systems.

1. Ventilation/exhaust will be constructed, maintained and operated to ensure the required protection, by maintaining a volume and velocity of exhaust air sufficient to gather dusts, fumes, vapors or gases from the confined or enclosed space and to convey them to suitable points of safe disposal.
2. Periodic air sampling for flammable and toxic materials and oxygen deficiencies shall be performed before and during employee work assignments in the confined or enclosed space to ensure a safe environment is maintained.

E. Lighting/ Electrical

1. Lighting will be provided in areas where there is insufficient natural light.
2. For work areas that may contain concentrations that exceed safe limits of flammable or combustible materials, appropriate lighting systems such as explosion-proof fixtures, switches or equipment shall be used. All electrical equipment is to conform to article 500 of the National Electrical Code.
3. Emergency lighting shall be provided at all points of access and egress. When this is not practical, explosion-proof flashlights shall be provided to persons required to enter confined or enclosed spaces, which are subject to blackout.
4. In areas of high moisture, portable electric lighting shall be operated at a maximum of 12 volts.

F. Communications

Constant communications shall be maintained with all personnel in confined spaces by personnel outside those areas. This shall be accomplished by utilizing one or more of the following methods:

1. Visual surveillance
2. Voice communication
3. Telephone communication
4. Two-way radio communication

Note: Proper method(s) will vary depending on whether an explosive atmosphere exists.

G. Fire Protection

Fire protection shall be assured in confined spaces at all times. This shall be accomplished by adherence to the following:

1. Safe access and egress will be maintained at all times when work is being performed in a confined space. Access ladders, floors and components that are constructed out of combustible materials shall be protected, covered or wrapped with a flame-retardant material.
2. Flammable liquids (i.e. acetone, alcohol, etc.) must be stored in approved (UL or FM) flammable liquid containers or dispensers. The amount of such flammable

liquid(s) in a confined space shall not exceed the amount necessary to perform the work each day.

3. Properly rated fire extinguishers shall be immediately available. In instances where extreme fire potential exists, a charged fire hose shall be readily available for immediate use.
4. Cylinders containing oxygen, acetylene or other fuel gases shall not be taken into confined spaces.
5. All rags, brushes, wipes, gloves, etc. shall be stored in metal containers with lids.
6. A person shall be stationed during all welding, burning and heating operations to monitor for fires and to ensure that, during and after the work has ceased or at the end of a work shift, there are no fire conditions present. A thorough check of welding and cutting areas is to be made at the end of each welding operation to assure NO fires have been started.
7. All flammable gas equipment, hoses, torches, etc. shall be free of defects and are to be inspected prior to each use.
8. To eliminate the risk of fire in confined spaces because of gas escaping through leaking or improperly closed torch valves, the gas supply to the torch shall be shut off whenever the torch is left unattended for long periods of time. Fuel gas and oxygen hoses shall be immediately removed from confined spaces before they are disconnected from the torch or other gas-consuming device, and at the end of each work shift.

H. Lockout Procedure

All lines, pipes or other conveyance of flammable and/or toxic materials into a confined space shall be positively locked out and tagged in accordance with the Project Lockout/Tagout Procedure.

I. Recordkeeping Requirements

1. A safe entry checklist (example attached) shall be completed and reviewed with The Christman Company Supervisor on Site.
2. A properly executed confined space work permit (example attached) is to be obtained before work may begin in a confined or enclosed space. All entrances into the confined space shall be posted with a copy of the permit.
3. A file of sampling results will be kept for each confined space and will be maintained by The Christman Company Supervisor on Site.

J. Rescue

Whenever work is to be performed in a confined space, emergency rescue must be anticipated and a rescue plan developed. In the event of an emergency, The Christman Company Supervisor on Site will coordinate and work with emergency medical and rescue personnel. Rescue plans will encompass all foreseeable contingencies that might occur when working in confined spaces. The Christman Company Supervisor on Site

will be responsible for training employees in emergency procedures such as rescue and escape.

DO NOT ENTER A CONFINED SPACE UNTIL YOU HAVE CONSIDERED EVERY QUESTION AND HAVE DETERMINED THE SPACE TO BE SAFE.

Description of confined space _____
Subcontractor involved _____ Date _____
Subcontractor Supervisor _____

YES NO
[] [] Is entry necessary?

TESTING

YES NO
[] [] Are the instruments used in atmospheric testing properly calibrated?
[] [] Was the atmosphere in the confined space tested?
[] [] Was oxygen at least 19.5% - not more than 23%?
[] [] Were toxic, flammable, or oxygen-displacing gases/vapors present?
- Hydrogen sulfide - Carbon dioxide
- Carbon monoxide - Other (list)

- Methane

MONITORING

YES NO
[] [] Will the atmosphere in the space be monitored while work is going on?
[] [] Continuously?
[] [] Periodically? (If yes, give interval:)

REMEMBER – ATMOSPHERIC CHANGES MAY OCCUR DUE TO THE WORK PROCEDURE OR THE PRODUCT STORED.

CLEANING

YES NO
[] [] Has the space been cleaned before entry is made?
[] [] Was the space steamed?
[] [] If so, was it allowed to cool?

VENTILATION

YES NO
[] [] Has the space been ventilated before entry?
[] [] Will ventilation be continued during entry?
[] [] Is the air intake for the ventilation system located in an area that is free of combustible dusts and vapors and toxic substances?
[] [] If atmosphere was found unacceptable and then ventilated, was it retested before entry?

ISOLATION

YES NO
[] [] Has the space been isolated from other systems?
[] [] Has electrical equipment been locked out?
[] [] Have disconnects been used where possible?
[] [] Has mechanical equipment been blocked, chocked, and disengaged where necessary?

[] [] Have lines under pressure been blanked and bled?

CLOTHING/ EQUIPMENT

YES NO

[] [] Is special clothing required (boots, chemical suits, glasses, etc.)? (If so, specify:)

[] [] Is special equipment required (e.g. rescue equipment, respirators, communications equipment, etc.)? (If so, specify:)

[] [] Are special tools required (e.g. spark-proof)? (If so, specify:)

RESPIRATORY PROTECTION

YES NO

[] [] Are MSHA/NIOSH-approved respirators of the type required available at the work site?

[] [] Is respiratory protection required (e.g. air purifying, supplied air, self-contained breathing apparatus, etc.)? (If so, specify type:)

TRAINING

YES NO

[] [] Have you been trained in proper use of a respirator?

[] [] Have you received first aid/CPR training?

[] [] Have you been trained in confined space entry and do you know what to look for?

STANDBY/RESCUE

YES NO

[] [] Will there be a standby person on the outside in constant visual or auditory communication with the person on the inside?

[] [] Will the standby person be able to see and/or hear the person inside at all times?

[] [] Has the standby person(s) been trained in rescue procedures?

[] [] Will safety lines and harness be required to remove a person?

[] [] Are rescue procedures available and can they be followed in the event of an emergency?

[] [] Are you familiar with emergency rescue procedures?

[] [] Do you know who to notify and what to do in the event of an emergency?

PERMIT

(This permit is an authorization in writing that states that the space has been tested by a qualified person, that the space is safe for entry; what precautions, equipment, etc. are required; and what work is to be done.)

YES NO

[] [] Has a confined space entry permit been issued?

[] [] Does the permit include a list of emergency telephone numbers?

REVIEWED: _____ DATE: _____

The Christman Company

Confined Space Entry Permit
(Permit valid for 8 hours only)

Site Location and Description	Date:
	Time: AM/PM
Purpose of Entry	Expires: AM/PM
Name of Employer (Subcontractor)	

SUPERVISOR(S) IN CHARGE OF CREWS	TYPE OF CREW	PHONE NUMBER

Italic print areas denote minimum requirements to be completed and reviewed prior to entry

REQUIREMENTS COMPLETED	DATE	TIME	REQUIREMENTS COMPLETED	DATE	TIME
<i>Lockout/De-Energize/Tryout</i>			<i>Full Body Harness with "D" Ring</i>		
<i>Line(s) Broken/Capped/Blanked</i>			<i>Emergency Escape Retrieval</i>		
<i>Purge/Flush/Vent</i>			<i>Life Lines</i>		
<i>Ventilation</i>			Fire Extinguishers		
<i>Secure Area (Post and Flag)</i>			Lighting (Explosive Proof)		
<i>Breathing Apparatus</i>			Protective Clothing		
<i>Resuscitator/Inhalator</i>			Respirator(s) Air Purifying		
<i>Standby/Safety Personnel</i>			Burning and Welding Permit		

Note: Items that do not apply, enter N/A in the box.

***Record continuous monitoring results every 2 hours.**

CONTINUOUS MONITORING TEST(S) RESULTS:	PERMISSIBLE ENTRY LEVEL	TIME:	TIME:	TIME:	TIME:	TIME:	TIME:
Percent of Oxygen (O ₂)	19.5% - 23.5%						
Lower Exposure Limit (LEL)	Under 10%						
Carbon Monoxide (CO ₂)	+35 PPM						
Hydrogen Sulfide (H ₂ S)	+10 PPM						
Other							

NOTE: "+" denotes 8-hour time weighted average. Employee can work in area 8 hours (longer with appropriate respiratory protection).

ATTENDANT/GAS TESTERS	INSTRUMENT USED	MODEL AND/OR TYPE	SERIAL OR UNIT NO.

Attendant Standby Person is Required for all Confined Space Work

NAME OF CONFINED SPACE ENTRANTS(S)	NAME OF CONFINED SPACE ENTRANTS(S)

Supervisor Authorizing Entry: _____ Emergency Phone Number: _____

Reviewed by: _____

The Christman Company

Issue Date: 0/00/2014

ATTACHMENT V - EMERGENCY CONTACT NUMBERS

IN CASE OF EMERGENCY CALL 9-1-1

NON-EMERGENCY CONTACTS

Emergency	Responder	Location	Phone
Fire	Romulus Fire Department	28777 Eureka Rd, Romulus, MI 48174	734-941-8585
Police	Romulus Police Department	11165 Olive Street Romulus, MI 48174	734-942-6879
Ambulance	Rapid Response EMS	29045 Airport Dr, Romulus, MI 48174	313-817-6000
Hospital	Beaumont Taylor	10000 Telegraph Rd, Taylor, MI 48180	313-295-5005
WCAA Fire	Fire Station #1	W Service Dr, Detroit, MI 48242	734-942-3603

Directions to Hospital:

- Head north on Middlebelt Rd toward Northline Rd
- Turn right onto Goddard Rd
- Turn left onto Beech Daly Rd
- Turn right onto Wick Rd
- Turn right onto US-24 S/Telegraph Rd
- Turn right onto Haig St
- Road names changes to Winston St
- Turn right onto Oakwood Heritage Hospital
- Turn left into destination

Report to the hospital registration/information desk or the emergency room in the event of an emergency.

ADDITIONAL PHONE CONTACTS

DTE (Electricity)	800-477-4747
DTE (Gas)	800-477-4747
Romulus Water Department Utilities (Water & Sewer)	734-942-7560
Wayne (County Offices)	313-224-6262

NON-EMERGENCY CONTACTS

The Christman Company

On-Site Contact(s):	Leigh Woodbury, Sr. Superintendent	810-588-7771
	Kyle Jones, Superintendent	313-949-2842

Off-Site Contact(s):	Don Staley, Vice President/Safety Director	517-819-9166
	John Holmstrom, Senior Vice President	517-712-2042

DTW Sierra BTS/Hillwood

On-Site Contact(s):	DTW Sierra BTS/Hillwood	847-737-0273
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Date: May 8, 2014

ATTACHMENT VI - EMERGENCY EVACUATION PLAN

EVACUATION PLAN

EVACUATE

Fire
Chemical Spills (interior)
Earthquake
Bomb Threat
Violence
Bio Terrorism

NON-EVACUATE

Weather Conditions (Tornado)
Chemical Spills (exterior)
Civil Disorder

Designated Horn Signals

Stop Work Order- Repeated blasts of the Air Horn
Evacuation Order- Repeated two (2) pulse blast of the Air Horn
All Clear Order- Repeated three (3) pulse blast of the Air Horn

Non- Evacuate Plan

Repeated blasts from the air horn, work is stopped. Workers are to gather at their immediate location of work and report to their foreman for a head count.

Evacuation Plan

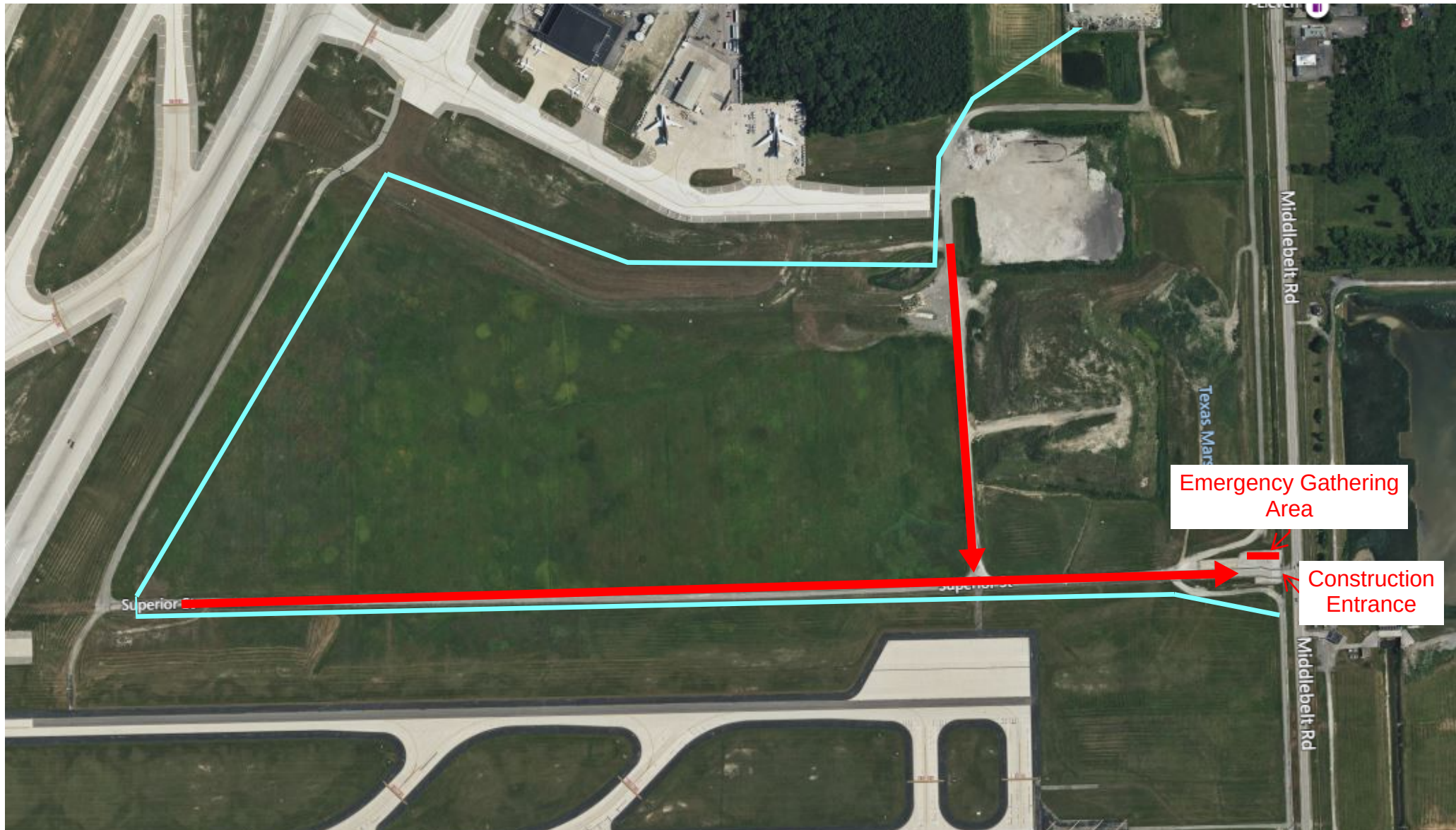
Upon enclosure of structure, emergency escape routes shall be posted for the exit ways. The procedure to be used for employee accountability after an emergency evacuation shall be as follows: Upon sounding of repeated two (2) pulse Air Horn blasts, all employees must immediately exit the building/job site and report to their foreman, and assemble at Gate #27 entrance near the orientation trailer off of Middlebelt Rd.

All Clear

Repeated three (3) pulse blasts from the air horn signals that everything is all clear. Work may resume and/or workers may re-enter the building.

ATTACHMENT VII - EMERGENCY EVACUATION MAP

DTWR - FedEx Expansion Project Emergency Egress Drawing



Egress Route: →

Temporary/Permanent AOA Fence: —

ATTACHMENT VIII – WAYNE COUNTY AIRPORT AUTHORITY SAFETY STANDARDS



Wayne County Airport Authority

Safety Standards

Table of Contents

Section 1.0	General Requirements
Section 2.0	General Airfield Safety
Section 3.0	Specific Airfield Safety
Section 4.0	Underground Work – Excavation, Shoring, Trenching, Saw Cutting, and Drilling
Section 5.0	Fall Protection
Section 6.0	Fire Prevention and Hot Work
Section 7.0	Cranes/Derricks/Lifting Equipment/Rigging
Section 8.0	Hazardous Energy Control
Section 9.0	Confined Space Entries
Section 10.0	Hazardous Materials/Environmental
Section 11.0	Roofing Operations
Section 12.0	Removal of Existing Equipment
Section 13.0	Vehicle Operations on the Airport Operations Area (AOA)
Section 14.0	AOA Contractor Escorts and Flagging
Section 15.0	Special Construction Rules on the AOA
Section 16.0	Job Hazard Analysis
Section 17.0	Housekeeping
Section 18.0	Site Sanitation
Section 19.0	Personal Protective Equipment
Section 20.0	Compressed Gasses
Section 21.0	Ladders
Section 22.0	Material Handling/Ergonomics
Section 23.0	Hand Powered Tools
Section 24.0	Portable Electric Tools
Section 25.0	Heaters
Section 26.0	Barrier Tape Identification
Section 27.0	General Electrical Rules
Section 28.0	Disposal Chutes
Section 29.0	Non-AOA Motor Vehicles and Mechanized Equipment
Section 30.0	Silica Exposure Plan
Section 31.0	Investigation, Documentation. And Reporting of Incident and Accidents

Attachment #1	– Loss Management Monthly Summary
Attachment #2	– Contractor Employee Review
Attachment #3	– Insurance Submittal Tool Kit
Attachment #4	– Request for Marking
Attachment #5	– Hot Work Permit
Attachment #6	– Utility Field Acknowledgement Form
Attachment #7	– Utility Interruption Form
Attachment #8	– Utility Custody Form
Attachment #9	– Barricades Details
Attachment #10	– Daily Construction Checklist – AOA Construction
Attachment #11	– Incident Accident Documentation Template
Attachment #12	– Supplemental Utility Strike Incident Form

1.0 General Requirements

This document is specifically intended for the use of Construction Managers, Vendors, Contractors and Subcontractors who perform work on or at (Airport Authority) Detroit Metropolitan Airport (DTW) or Willow Run Airports (YIP) and facilities. It is made available to amplify your Safety Program. All or parts of the contents will apply to your contract depending upon the type of contract and the sequence in which the phases are conducted. ***BE ADVISED you will still be subjected to a Construction and Alteration Permit for work to be performed (if C/A permit project).***

The following information is provided as an example in which to determine what items to include in the safety program for work performed and does not constitute the requirements of a complete, comprehensive safety policy or Safety Program. It shall be the Contractors responsibility to provide a safe and healthful work environment for their workers on-site during the performance of the contract or work performed from a purchase order or work order. This shall include the Contractor and all its Subcontractors, all trades and supervision, suppliers, Airport Authority personnel, visitors and additional parties having access to the designated project work areas at the Airport Authority's facility. The safety goal of Airport Authority construction projects is to achieve zero fatalities, zero permanent disabilities, and zero lost time accidents.

The generalized overviews presented in this document are statements of expectations that the Contractor will be measured against. Failure to meet these requirements may be grounds for the removal of the individual employee from the worksite and also could lead to grounds for termination of the Contract by the Airport Authority.

The Contractor must not interfere with or make more difficult or expensive Airport Authority's compliance with any law, statute, code, ordinance or regulation. Airport Authority will notify the Contractor, orally or in writing, and the Contractor shall within forty-eight hours of receiving Airport Authority's notification make whatever changes are necessary to remedy the situation, including, without limitation, changes in the work schedule, installation of safety devices. Airport Authority's exercise of its rights under this guideline will not be grounds for an increase in the Contract Sum under the Contract.

The Airport Authority has the right to monitor (Contractor shall still be responsible for assuring safe work practices) the Contractors' operations for safety performance, workmanship, protection of operations, work progress, housekeeping, and compliance to design specifications. It is a general practice that the Airport Authority will work through the Contractor's supervision and not directly with the employee. The Airport Authority has the right to participate with and investigate any accident or incident.

Contractor shall develop and implement a system for assessing appropriate requirements applicable to its employees, including removal from Airport Authority property, for violation of safety laws statutes, codes, ordinances and regulations, safety requirements specified by Job Hazard Analysis; safety requirements specified by the Contract, including, without limitation, any other condition that presents a safety hazard to the employee or others. Contractor shall prepare and maintain a detailed written report of each instance where it has assessed discipline for a safety related infraction including identification of the employee, the nature of the infraction and their discipline assessed.

Prior to beginning any work, the Contractor shall conclusively demonstrate to the Airport Authority that the Contractor has in place and actively maintains a comprehensive random and post accident drug and alcohol testing program that meets or exceeds the standards for post accident and random testing in accordance with Department of Transportation (DOT) Drug and Alcohol Testing - 49 Code of Federal Regulations (CFR) Part 40 (hereinafter "Drug Testing"). Written proof of Drug Testing shall be submitted to the Airport Authority simultaneously with the required Payment and Performance Bonds

and Insurance. The Drug Testing requirement shall apply to all subcontractors of the Contractor and Insurance. The Drug Testing requirement shall apply to all subcontractors of the Contractor as well as the Contractor.

All persons entering the project area designated as the construction site shall strictly follow Michigan OSHA, EGLE, FAA, DOT, and TSA regulations.

These established safety requirements shall govern Contractors and all persons within the designated construction site and are outlined to avoid infractions of common accepted safety practices. Safety Program – The Contractor shall submit its Safety Program to the Airport Authority and obtain approval prior to issuance of the Notice to Proceed. The Safety Program includes, but is not limited to the following:

- Contractor's Corporate Safety Policy and/or
- Contractor's Site Specific Safety Plan;
- Construction Safety and Phasing Plan (CSPP) – The Contractor shall abide by the CSPP, approved by the FAA and provided by the Airport Authority (for all Airfield, PFC, and AIP funded projects only).
- Safety Plan Compliance Document (SPCD) – The SPCD details how the contractor will comply with the CSPP. The Contractor shall prepare the SPCD and obtain approval by the Airport Authority prior to issuance of the Notice to Proceed (for all Airfield, PFC, and AIP funded projects only).

Regular progress meetings will be conducted during construction. Part of the meeting will be dedicated to safety. During these meetings, the Contractor shall submit to WCAA on a monthly basis the following safety information (Attachment #1):

- Estimated man-hours worked from the previous period;
- Number of near misses from the previous period;
- Number of accidents from the previous periods;
- Number of recordable injuries from the previous period; and
- Summarization of any accident that took place from the previous period.

The Contractor shall carry out its operations in a manner that will cause a minimum of interference with air traffic, and shall be required to cooperate with the Federal Aviation Administration (FAA), Wayne County Airport Authority (Airport Authority) the airlines, and other contractors working in the area. All work shall be completed in accordance with the Contract Documents including the Safety Program and FAA Advisory Circular 150/5370-2F, Operational Safety on Airports During Construction or current edition as of bid date.

If the Contractor uses resources from any union halls for staffing, those individuals shall have a current OSHA 10 hour certification as well as go thru safety orientation with the Contractor. Copies of any names of individuals who are used from the local halls shall be submitted with a copy of the certification to the Airport Authority or the Airport Authority's representative one week prior to allowing the individual onsite.

The Contractor shall supply, place, maintain, move and store the items listed herein, as appropriate, to facilitate construction and protect air traffic. The Contractor shall maintain an adequate extra supply of these items on site.

The following information is provided as an example in which to determine what items to include in the safety program for work performed and does not constitute the requirements of a complete, comprehensive

safety policy or Safety Program. It shall be the Contractors responsibility to provide a safe and healthful work environment for their workers on-site during the performance of the contract or work performed from a purchase order or work order. This shall include the Contractor and all its Subcontractors, all trades and supervision, suppliers, Airport Authority personnel, visitors and additional parties having access to the designated project work areas at the Airport Authority's facility. The safety goal of Airport Authority construction projects is to achieve zero fatalities, zero permanent disabilities, and zero lost time accidents.

These established safety requirements shall govern Contractors and all persons within the designated construction site and are outlined to avoid infractions of common accepted safety practices. These safety requirements shall not be construed as complete and any requirements of the guidelines in conflict with Michigan OSHA and FAA shall be superseded by Michigan OSHA or FAA regulations.

Tools, equipment or materials shall not be left or placed on beams, overhead walkways, or places where they may fall, causing injury.

To prevent possible explosive or incendiary devices from being hidden in areas close to Airport Facilities, equipment, aircraft, or vehicles, no containers (tool boxes, storage containers, materials trailers shall be left unsecured or unattended in public areas.

The Contractor shall not bring any tool through passenger screening. No tools shall be left unsecured in public areas. Any temporary doors that lead to construction areas that are accessible by the public shall be equipped with a push button cipher lock (installed at the contractor's expense). The access number to the door shall be provided to Airport Authority designees.

The Contractor is not permitted to carry liquids (except drinking water), gels, or aerosols into sterile areas except for those liquids, gels, or aerosol necessary for operational or medical needs, all of which shall be subject to inspection.

Eye protection shall be worn at all times.

While working on the airfield hearing protection is mandatory. It is up to each contractor and subcontractor to determine if a hearing conservation program is necessary. The Contractor will comply with industry standards for hearing protection of personnel and visitors as appropriate.

Upon request the Airport Authority will submit any historical noise surveys to the Contractor.

Contractor shall verify and assure that every employee who operates any mobile equipment on Airport Authority properties shall have a current valid driver's license.

The Contractor shall comply with the National Electric Code (NEC) requirements regarding ground fault circuit interrupters for construction field tools and equipment.

In matters concerning interpretation of the foregoing requirements, the decision of the Airport Authority will be final and binding.

Onsite vehicles are to have no more passengers than seats available on the vehicle. Personnel carts, golf carts or similar vehicles, where permitted, shall only have as many riders as seats available. Failure to follow this requirement will lead to removal of personnel from the project.

The Contractor shall maintain a Safety Program, for the purpose of safety, security, orientation, education, training, enforcement, and distribution.

If a fire line or any type of fire suppression service is going to be taken out of service, the Contractor must coordinate with the Airport Authority's Public Safety Fire Marshal or a designated representative from the Airport Authority's Fire Division at least three days in advance. In addition, the contractor shall complete an impairment notification to the insurance carrier. Impairment kits can be obtained by contacting the Airport Authority building operator, Airport Authority Maintenance Division, or Airport Authority Risk Management Unit.

If a security system (cameras, gates, lifts, doors, etc) is going to be taken out of service, the Contractor shall coordinate with the Airport Authority Security Division at least three days in advance. Pending certain security threat levels, permission may not be granted to take down a security system.

Potable water line installation will require coordination with the Airport Authority. In addition, a permit application may have to be completed and submitted to the Airport Authority.

The Contractor shall follow Detroit Water and Sewer Department guidelines for the installation and storage of potable water lines. Installation, storage, and testing results shall be coordinated with the Airport Authority.

Employees shall not operate any equipment or vehicles more than 16 hours consecutively.

Any individual failing to follow these safety requirements will be directed by the Contractor to immediately abate the unsafe act, behavior, or equipment.

All Contractor equipment brought onsite for use on or during the construction project shall be kept in a safe operating condition. Worn or damaged equipment shall be repaired, replaced or taken out of service (locked out) and removed from the job site.

Contractor shall keep its work area in a clean and safe condition.

The use of makeshift, defective or inadequate scaffolding, rigging, or staging is prohibited.

The Contractor shall provide barriers, railings or coverings for all areas including, but not limited to, elevated work platforms, holes, excavations, roof openings, along roof edges, manholes, and/or unfinished work causing floor obstructions. Outside excavations, barriers, and safety signs shall be adequately illuminated during darkness.

No Contractor is permitted to use any powered industrial moving equipment, trucks, tools, or ladders owned or rented by, the Airport Authority unless it is an emergency and the request form is executed.

The Contractor shall provide physical barriers along the perimeter of its work site and place signs identifying the area as a construction site. In some cases where the general public or open airfield is to be protected, additional and/or specialty barriers might be required and will need to be determined by Airport Authority.

All individuals in the designated construction areas are required to wear hardhats, safety glasses, protective footwear, and any other protective clothing or gear as required by safety codes and regulations or as deemed necessary by the Contractor.

Contractors shall not perform work overhead of any member of the general public. The Contractor shall use physical barriers to prevent access by non-construction personnel to areas with overhead construction work. If the Contractor cannot restrict access to the overhead work area, work will cease until a method of restricting access is developed and implemented.

Welding screens and/or curtains will be used in areas where cutting or welding operations are being performed where non-construction personnel may be exposed to weld flash or sparks. Refer to Airport Authority Hot Work permit requirements.

Non-Airport Operations Areas (Non-AOA) road closures shall be coordinated with Airport Authority Facilities, Design, and Construction Division. If road closures are required at YIP, then YIP Management shall be involved as well. Coordination shall take place at a minimum of five days prior to closure. Public roadways will be closed following the Michigan Uniform Traffic Code procedures.

Prior to mobilization the Contractor shall complete Contractor Employee Review, Contractor Safety Guidelines (Attachment 2). This document shall be kept on site and updated for every new employee who will work on the project.

Regular progress meetings will be conducted during construction. Part of the meeting will be dedicated to safety.

During the project the Contractor shall complete and keep up to date Attachment #3 – Insurance Submittal Tool Kit. The purpose of this kit is to assure the Prime Contractor assure any subcontractors maintains the appropriate insurance and is current. This document should be made available upon request.

2.0 General Airfield Safety

This section is instead to be used to assist those Contractors who have access to the airfield. Warning lights shall meet the requirements of FAA Advisory Circular 150/5370-2G, or current edition as of bid date, Operational Safety on Airport during Construction. The Contractor's vehicles shall meet the requirements of FAA Advisory Circular 150/5210-5D or current edition as of bid date, Painting, Marking, and Lighting of Vehicles Used on an Airport.

Low profile barricades shall be in accordance with the details in the Contract Documents and meet the requirements of FAA Advisory Circular 150/5370-2G, or current edition as of bid date. The barricades shall be furnished, maintained and relocated during each phase by the Contractor. At the completion of the Contract (based on the contract language), the entire quantity of barricades shall be delivered to the Airport Authority complete and in good working order and shall become property of the Airport Authority. Barricades shall be as detailed and installed per the Drawings along the affected pavement edge or access to a closed runway, taxiway or apron.

Safety fence shall be furnished and installed at the locations as indicated on the Contract Documents and/or directed by the Designer.

The power vacuum sweepers shall be Tymco, Model HSP-600 or Elgin Model Crosswind or an approved equal. Depending on the size of the project and construction activity on aircraft operations area, a broom type sweeper may be allowed by the Designer. The Contractor shall provide a minimum of one sweeper per taxiway intersection at all times.

Taxiway ending marker shall be furnished and installed at the locations as indicated on the Contract Documents and/or directed by the Designer. Taxiway ending marker shall meet the requirements of FAA Advisory Circular 150/5345-44K, Type L-858C, or current edition as of bid date.

Airfield Rescue and Firefighter (ARFF) roads cannot be taken out of service unless approved by the WCAA.

Equipment that is used for material handling on the airfield shall have working headlights and brake lights. Any equipment that exits the AOA onto public roadways is subject to MDOT regulations.

Contractor to coordinate jet fuel line work with Airport Authority Operations prior to disturbance

The Contractor shall not pull any associated airfield lighting electrical cables through existing

If working on the AOA, the Contractor will be required to prepare an FAA Safety Plan Compliance Document (SPCD) that is a part of the Safety Program.

Airport Operations Area (AOA) zipper road closure shall be coordinated with the DTW or YIP Airfield Operation Units. Coordination shall take place at a minimum of five days prior to closure.

The Contractor shall submit an FAA Form 7460-1 at least 90 days prior to any crane erections. All construction involving cranes shall further be coordinated at least 72 hours in advance, excluding weekends, with the applicable DTW and YIP Airfield Operation Division or Airport Authority Facilities, Design, and Construction Division. This does not include the time required for airspacing. The following information and actions are required:

1. Location of the Crane.
2. Maximum extendable height.
3. Hours of operation.

4. The top of each crane boom shall be marked by a 3' x 3' orange and white checkered flag — each box being 1' square.
5. Each crane shall be lowered at night and during periods of poor visibility as directed by Airport Authority Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division. In the event the crane is approved to remain extended during the hours from sunset to sunrise, the highest point of the crane boom will be lit with a red obstruction light in accordance with AC 70/7460-1, and the Michigan Tall Structures Act.

3.0 Specific Airfield Safety

Taxiway ending marker shall be furnished and installed at the locations as indicated on the Contract Documents and/or directed by the Designer. Taxiway ending marker shall meet the requirements of FAA Advisory Circular 150/5345-44K, Type L-858C, or current edition as of bid date.

Runway Closure Marker (lighted X) shall meet the latest edition of FAA AC 150/5345-55A, or current edition as of bid date, Lighted Visual Aid to Indicated Temporary Runway Closure. Runway Closure Marker shall be approved in accordance with the details shown on the Contract Documents and as approved by the Designer. At the completion of the project the Runway Closure Markers and all associated equipment shall become the property of the Airport Authority. The following are requirements for the Runway Closure marker:

- A. Be a portable, towable unit that can be quickly removed from the runway. Includes a two inch ball hitch on the trailer tongue and second hitch mounted on the rear of the trailer.
- B. Consist of clear incandescent lamps or transmit a white color, arranged in the shape of a letter "X" with arms crossed at an appropriate angle to make the "X" discernible. The arms shall be painted aviation yellow on all sides so that the unit will be clearly visible when it is in position.
- C. Be energized by a portable power supply, water-cooled diesel engine or by outlets on the airfield as available.
- D. Be controlled so that the lighted signal will flash all lights simultaneously at an approximate rate of 2.5 seconds "on" (+/- 20%) and 2.5 seconds "off" (+/- 20%).
- E. Provide the following daytime and nighttime visual reference during Visual Flight Rule (VFR) conditions when placed on centerline and within 250 feet of the runway end:
- F. Visible to the pilot at a range of at least 5 nautical miles.
- G. Recognizable as a letter "X" from a range of at least 1-1/2 nautical miles.
- H. Provide lamp dimming capability for nighttime operations.
- I. Produce a signal that provides a horizontal coverage to at least 15 degrees on each side of the runway centerline, and a vertical coverage from 0 degrees to 10 degrees above horizontal, both day and night, at a range of 1-1/2 nautical miles.
- J. Adjustable aiming and leveling to allow tilting to an optimum angle of three degrees from vertical.
- K. Withstand a minimum wind speed of at least 40 mph without affecting aiming or operation.
- L. Include an illuminated failure indicator that is visible from the back (runway side) of the unit.
- M. Include an operations placard in a conspicuous location that instructs operators to visually check the operation of the device every two hours.
- N. One person set up in less than 5 minutes.

- O. Diesel portable power with adapter to run directly from electrical outlets.
- P. Trailer hitch options including tandem towing for on-airport operations.
- Q. Ability to provide up to 120 hours of continuous operation.
- R. Fail safe protection to ensure that the unit stays on as continuous light if the flasher unit should fail.
- S. Dimensioning and lights arrangement shall follow FAA recommendations of AC 150/5345-55A, or current edition as of bid date.

The power vacuum sweepers shall that meet requirements of dust control that will assure air emissions will comply with MIOSHA standards. Depending on the size of the project and construction activity on aircraft operations area, a broom type sweeper may be allowed by the Designer. If a broom type sweeper is used it must be used water or a vacuum system for dust control. The Contractor shall provide a minimum of one sweeper per taxiway intersection at all times.

In compliance with FAA AC 150/5370-2G, or current edition as of bid date, the Contractor shall prepare a Safety Plan Compliance Document (SPCD). This document shall include a general statement by the Contractor that he/she has read and will abide by the CSPP. Any details not identifiable for the CSPP should be included within the SPCD. The SPCD is similar to the CSPP but shall not contain duplicate information. The contractor must submit the SPCD to the Airport Authority for approval prior to the issuance of the NTP. The SPCD shall include but not be limited to the following Checklist as applicable to the scope of the project:

Coordination. Discuss details of proposed safety meetings with the Airport operator and with contractor employees and subcontractors.

Phasing. Discuss proposed construction schedule elements, including:

- Duration of each phase.
- Daily start and finish of construction, including “night only” construction.
- Duration of construction activities during:
 - o Normal runway operations.
 - o Closed runway operations.
 - o Modified runway “Aircraft Reference Code” usage.

Areas and operations affected by the construction activity. These areas and operations should be identified in the CSPP and should not require an entry in the SPCD.

Protection of NAVAIDS. Discuss specific methods proposed to protect operating NAVAIDS.

Contractor access. Provide the following:

- Details on how the Contractor will maintain the integrity of the Airport security fence (contract security officers, daily log of construction personnel, and other).
- Listing of individual requiring driver training (for certificated airports and as requested).
- Radio communications.
 - o Types of radios and backup capabilities.
 - o Who will be monitoring radios.
 - o Whom to contact if the Air Traffic Control Tower (ATCT) cannot reach the Contractor’s designated person by radio.
 - o Details on how the contractor will escort material delivery vehicles.

Wildlife management. Discuss the following:

- Methods and procedures to prevent wildlife attraction.
- Wildlife reporting procedures.

Foreign Object Debris (FOD) management. Discuss equipment and methods for control of FOD, including construction debris and dust.

Hazardous material (HAZMAT) management. Discuss equipment and methods for responding to hazardous spills.

Notification of construction activities. Provide the following:

- Contractor points of contact.
- Contractor emergency contact.
- Listing of tall or other requested equipment proposed for use on the airport and the time frame for submitting 7460-1 forms not previously submitted by the Airport operator.
- Batch plant details, including 7460-1 submittal.

Inspection requirements. Discuss daily (or more frequent) inspections and special inspection procedures.

Underground utilities. Discuss proposed methods of identifying and protecting underground utilities.

Penalties. Penalties should be identified in the CSPP and should not require an entry in the SPCD.

Special conditions. Discuss proposed actions for each special condition identified in the CSPP.

Runway and taxiway visual aids. Including marking, lighting, signs and visual NAVAIDs. Discuss proposed visual aids including the following:

- Equipment and methods for covering signage and airfield lights.
- Equipment and methods for temporary closure markings (paint, fabric, other).
- Types of temporary Visual Guidance Slope Indicators (VGSI).

Markings and signs for access routes. Discuss proposed methods of demarcating access routes for vehicle drivers.

Hazard marking and lighting. Discuss proposed equipment and methods for identifying excavation areas.

Protection of runway and taxiway safety areas. Including object free areas, obstacle free zones, and approach/departure surfaces. Discuss proposed methods of identifying, demarcating, and protecting airport surfaces including:

- Equipment and methods for maintaining Taxiway Safety Area standards.
- Equipment and methods for separation of construction operations from aircraft operations, including details of barricades.
- Other limitations on construction should be identified in the CSPP and should not require an entry in the SPCD.

The Safety Program, including the SPCD, shall be submitted to the Airport Authority for review.

Contractor shall be familiar with all existing and limiting conditions that will or may have a bearing on the performance of the Contract with regard to safety. Any limiting conditions shall be identified in writing.

Throughout the duration of the Contract, any practice or situation that the Designer determines to be unsafe or a hindrance to regular Airport operations shall be immediately rectified.

The following publications contain definitions/descriptions of critical Airport operating areas. The areas defined below pertain to airfield safety requirements and are referenced throughout the Contract Documents. Copies of these publications are available from the FAA at www.faa.gov. Advisory Circular 150/5370-2G, or current edition as of bid date, "Operational Safety on Airports During Construction": Sets forth guidelines to assist Airport operators in complying with FAR Part 139, "Certification and Operation/Land Airports Serving Certain Air Carriers" and with the requirements of federally funded construction projects.

FAR Part 77, "Objects Affecting Navigable Airspace," Current Edition: Establishes standards for determining obstructions to navigable airspace. Civil Airport imaginary surfaces are defined in the publication. It also sets forth requirements for notice of certain proposed construction or alteration. Notice of construction provides a basis for recommendations for identifying the construction or alteration in accordance with AC 70/7460-1, "Obstruction Marking and Lighting," or current edition as of bid date. AC 150/5300-13, "Airport Design" or current edition as of bid date: Establishes design, operational and maintenance standards for Airports. Standard terms used in the Contract Documents are defined below:

Runway Safety Area (RSA) - The defined surface surrounding the runway over which aircraft should, in dry weather, be able to cross at normal operating speeds without incurring significant damage. A safety area is graded, drained and compacted. It is free of any holes, trenches, humps or other significant surface variations or objects, other than those which must be there because of their essential aeronautical function. The safety area requires the capability of supporting maintenance, firefighting, and rescue vehicles under normal (dry) conditions.

Object Free Area (OFA) – An area on the ground centered on a runway, taxiway, or taxilane centerline provided to enhance the safety of aircraft operations by having the area free of objects, except for objects that need to be located in the OFA for air navigations or aircraft ground maneuvering purposes.

Obstacle Free Zone (OFZ) – The OFZ is the airspace below 150 feet above the established Airport elevation and along the runway and extended runway centerline that is required to be clear of all objects, except for frangible visual NAVAIDs that need to be located in the OFZ because of their function, in order to provide clearance protection for aircraft landing or taking off from the runway, and for missed approaches. The OFZ is subdivided as follows:

Runway OFZ. The airspace above a surface centered on the runway centerline.

Inner-approach OFZ. The airspace above a surface centered on the extended runway centerline. It applies to runways with an approach lighting system.

Outer-approach OFZ. The airspace above the surfaces located on the outer edges of the runway OFZ and the inner-approach OFZ. It applies to runways with approach visibility minimums lower than $\frac{3}{4}$ -statute mile.

Taxiway Safety Area (TSA) – A defined surface alongside the taxiway prepared or suitable for reducing the risk of damage to an airplane unintentionally departing the taxiway.

The work shall proceed in such a manner as to provide safe conditions for all workers and personnel. The sequence of operations shall be such that maximum protection is afforded to ensure that personnel and workers in the work area are not subject to any dangerous conditions.

Prior to commencement of construction activity, the Contractor shall notify in writing, at least 72 hours in advance, Airport Authority Operations and the Designer of its intentions to begin construction, stating the proposed time, date, and area of which construction is to occur in order for the appropriate Notice-to-Airmen (NOTAM) to be issued. During the performance of this Contract, the Airport facility shall remain in use to the maximum extent possible. The Contractor shall not allow employees, subcontractors, suppliers, or any other unauthorized persons to enter in any Airport area which may be open for aircraft use.

Should any of the following problems or hazards arise during construction, the Contractor shall immediately rectify/correct the problem or hazard to the satisfaction of the Designer and the Airport Authority: Trenches, holes, or excavations at or adjacent to any active runway or in safety areas. Unmarked/unlighted holes or excavation at any active apron, taxiway, taxilane, or related safety area. Mounds or piles of earth, construction materials, temporary structures, or other objects in the vicinity of any active taxiway, taxilane, or in a related safety, approach, or departure area.

Vehicles or equipment (whether operating or idle) on any active runway, taxiway, taxilane, or in any related safety, approach, or departure area. Vehicles, equipment, excavations, stockpiles, or other materials which could degrade or otherwise interfere with electronic signals from radios or navigational aids (NAVAIDS). Runway surfacing projects resulting in excessive lips greater than 1 inch for runways and exceeding 3 inches for edges between the old shoulder and new surfaces at runway edges and ends.

Unmarked utility, NAVAID, weather service, runway lighting, or other power or signal cables that could be damaged during construction. Objects (whether or not marked or flagged) or activities anywhere on or in the vicinity of the Airport which could be distracting, confusing, or alarming to pilots during aircraft operations.

Unflagged/unlighted low visibility items (such as tall cranes, drills, and the like) anywhere in the vicinity of active runways, or in any approach or departure area. Misleading or malfunctioning obstruction lights or unlighted/unmarked obstructions in an approach to any active runway.

Inadequate approach/departure surfaces needed to assure adequate landing/takeoff clearance over obstructions or work or storage areas. Inadequate, confusing or misleading (to user pilots) marking/lighting of runways, taxiways, or taxilanes, including displaced or relocated thresholds. Water, dirt, debris, or other transient accumulation which temporarily obscures pavement marking, pavement edges, or derogates visibility of runway/taxiway marking or lighting. Inadequate or improper methods of marking, barricading, and lighting of temporarily closed portions of Airport operations areas. Trash or other materials with foreign object damage (FOD) potential, whether on runways, taxiways, or aprons, or in related safety areas. Inadequate barricading or other marking which is placed to separate construction or maintenance areas from active aircraft operating areas.

Failure to control vehicle and human access to active aircraft operating areas. Construction/maintenance activities or materials which could hamper the response of aircraft rescue and firefighting (ARFF) equipment from reaching all aircraft or any part of the runway/taxiway system, runway approach and departure areas, and aircraft parking locations. Bird attractants on Airport, such as edibles (food scraps, etc.), miscellaneous trash, or ponded water.

4.0 Underground Work – Excavation, Shoring, Sheet Piling, Trenching, Saw Cutting, and Drilling

The Contractor shall identify any known underground interferences or discrepancies on all available drawings that can be provided by contacting the Airport Authority Project Representative. The Contractor must coordinate a meeting a Utility Review meeting prior to any underground work to discuss the project approach as it relates to this section.

Prior to commencing any excavation (on or off AOA), drilling (on or off the AOA), driving fence posts (along the AOA), trenching (on or off the AOA), saw cutting (AOA only), the Contractor shall review drawings with Airport Authority personnel to insure that all underground obstructions and utilities are identified. In addition the contractor shall contact MISS DIG, other stakeholders, and coordinate with the Airport Authority Project Representative to assign the resources for marking the utilities. The Contractor will be completely responsible for all damage to underground utilities. The Contractor shall coordinate request of marking the utilities by completing the Utility Marking Request form (Attachment #3) at least 72 hours prior to any excavations. If the subject area of interest encompasses more than 5,000 linear feet or one acre, the request shall be made at least two weeks prior to the scheduled work. Airport Authority will notify the Contractor a minimum with 24 hours after receiving notice.

After the subjected area has been reviewed by MISS DIG, the Airport Authority, and other stakeholders the Contractor shall complete the Utility Field Acknowledgment form (Attachment #5). The Contractor shall **not** proceed until the Field Acknowledgment form has been executed by the Contractor. A Field Acknowledgment Form (ticket) is valid for 21 days from the start date of the excavation on the ticket, except that a ticket is valid for 180 days from the start date if the notice indicates that the proposed excavation will not be completed within 21 days of the start date. The Contractor shall catalog the marked utilities by means approved by the Airport Authority. The Contractor could be subjected to additional fees and penalties for mismanaging and not proper planning the coordination of this portion of the work

The Contractor shall assure and has a right to assure each utility shall be swept in the following manner. In addition for any reason the Contractor is the marking they must conform to this standard as well:

- Flags can be used but shall be color coordinated as suggested below. In addition the “acronym” for that utility shall be written on one side of the flag with a permanent marker.
- Stakes can be used. The top two inches of the stake shall be painted in color as suggested below. In addition the “acronym” for that utility shall be written on one side of the stake with a permanent marker.
- Painting is only authorized on asphalt, concrete, and metal surfaces. Markings shall be color coordinated as suggested below. The acronym for the utility shall be used for each utility. A line that shows the direction of the utility shall emanate from the acronym in each direction.
- All marking of utilities shall be every 50 feet. Use of stakes and flags on the AOA could be limited subjected to the work area:

	Acronym	Color
Electrical Loops (non AOA)	Use “Elec”	Red
Airfield Electrical	Use “Elec”	Red
Natural Gas	Use “Nat Gas”	Yellow
Sanitary	Use “Sanit”	Green
Storm	Use “Storm”	Green
Combined Sewers	Use “Combo”	Green
Water (potable and fire)	Use “Water”	Blue
Irrigation	Use “IRR”	Purple

FAA Fiber	Use "FAA Fib"	Orange
WCAA Fiber	Use "Fiber"	Orange
Telephone	Use "Tele"	Orange
Jet Fuel	Use "Jet Fuel"	Yellow
Glycol	Use "Glycol"	Blue
Oil	Use "Oil"	Yellow
Gasoline	Use "Gas"	Yellow
Diesel	Use "Diesel"	Yellow
Steam	Use "Steam"	Yellow
Condensate	Use "Conden"	Green
Temporary Mark	None	Pink
Proposed Excavation/Blasting	None	White

*if abandoned still shall stake, mark, or flag but write down "aband" before the abbreviated prefix indicated above.

The individual marking, staking, or flagging shall mark the utilities in a way that coincides with the drawings that are referenced on the Request for Sweep Form.

If a utility or any underground obstruction is found it shall be reported immediately to the Designer or the Airport Authority Project Supervisor.

Contractor employees in an excavation shall be protected from cave-ins by an adequate protective system unless the excavation is:

- Made entirely of stable rock, or
- Less than 5 feet deep and determination has been made that there is no potential for a cave-in.

Excavation shall be protected using proper barricading materials which shall be installed a minimum of 6 feet back from excavation (unless in conflict with airfield requirements). Barricade material can be wood, steel cables, or chain supported at intervals so that the barricade does not sag or drop below the required height. Caution tape is not an approved barricade material. Guardrail/jersey barriers may be required and shall provide a top rail, mid rail, and toe board at proper elevations and be able to withstand a minimum 200 pound of force without collapsing.

The Contractor is responsible for documenting utility information for use during construction and preparation of as-builts.

The Contractor, as needed, will conform to the Miss Dig Facility Damage & Safety Prevention Act.

5.0 Fall Protection Control

Fall Hazard Control requires that Contractors comply with fall prevention and protection procedures in accordance with Michigan Occupational Safety and Health Administration (MIOSHA) Part 45.

Fall Protection will be required on specific mobile equipment and platforms.

Installation of fall protection systems on the ground:

- As much as practicable, fall protection such as nets, and lifelines, are installed on material and equipment on the ground.

Reducing fall hazards in the construction process:

- As much as practicable, permanent stairs, floors, decking, handrails, and walls are installed as the structures are being built. Also, material and equipment such as structural steel and tanks are erected in sections and painted on the ground to reduce fall exposures.

Safe access:

- Methods to provide safe access to work areas are included in the JHA. Safe access includes ladders, scaffold, stairways, and ramps. Climbing structural steel and equipment is not allowed.

Training:

- Before employees are allowed to use a particular fall protection method or system, they are properly trained on proper use and limitations of the system. The training is the responsibility of their supervisor.

When not protected by any other means of fall protection such as safety nets or scaffold with proper guardrails, employees shall use full body harnesses, shock absorbing lanyards with double locking snap hooks, and an adequate anchorage (fall arrest equipment). To achieve 100 percent fall protection, employees may need to use a double lanyard system and/or vertical or horizontal lifelines, retractable lifelines, or other such approved devices.

Employees shall rig fall arrest equipment so that they can neither free fall more than 6 feet nor contact any lower object. Anchorage points for fall arrest equipment shall be capable of supporting 5,000 pounds per employee and located above the employee's body harness attachment point where practicable. Anchorage points shall be independent of any anchorage being used to support or suspend scaffolds or other platforms.

When vertical lifelines are used, each employee shall be protected by a separate lifeline. The lifeline shall be properly weighted at the bottom and terminated to preclude a device such as a rope grab from falling off the line. Horizontal lifelines should be limited to two persons at one time between supports. Horizontal lifelines shall be designed, installed, and used under the supervision of a qualified person. The horizontal lifeline shall be designed to maintain a safety factor of at least two.

Prior to each use, employees shall visually inspect all fall arrest equipment for cuts, cracks, tears or abrasions, undue stretching, overall deterioration, mildew, operational defects, heat damage, or acid or other corrosion. Equipment showing any defect shall be withdrawn from service.

All fall arrest equipment subjected to impacts caused by a free fall or by testing shall be removed from service.

Employees should store all fall arrest equipment in a cool dry place not subjected to direct sunlight.

Employees shall not use fall arrest equipment until they have been properly trained in its use.

Foremen shall ensure fall protection is available and used as required for all employees they are responsible for.

Fall arrest equipment shall not be used for any other purpose such as tow ropes or hoist lines.

Proper guardrails shall be installed on open sides of all walkways and runways where the fall distance exceeds 4 feet.

Proper guardrails shall be installed on all open sided floors where the fall distance exceeds 4 feet.

All floor openings or floor holes shall be protected by guardrails or hole covers. If hole covers are used, they shall be strong enough to support the maximum intended load, secured against displacement, and properly labeled. If the cover is subject to vehicular traffic, it shall be capable of supporting at least two times the axle load of the largest vehicle expected to cross over it. When operating a scissor lift work platform, the lift shall have guardrails on all open sides and the door access chains or rails in place.

Employees operating aerial lifts shall wear a body harness and lanyard attached to the aerial lift. Employees shall not attach the lanyard to an independent structure. Employees riding in a crane suspended work platform shall wear a body harness and lanyard attached to the grab rail of the platform.

Employees working on wall forms or rebar shall wear a body harness and lanyard in addition to a positioning device when exposed to a fall in excess of 6 feet. Position devices shall be rigged to prevent a free fall greater than 24 inches.

Stairs, ladders, or ramps shall be provided for all access ways where there is a change in elevation greater than 19 inches.

When guardrails are used for fall protection, they shall consist of a top rail, intermediate rail, and toeboard. The top rail shall have a vertical height of 42 inches, the midrail shall be at 21 inches, and the toeboard 4 inches. Guardrail systems shall be constructed so that there are no openings greater than 19 inches. When wood railings are used, the post shall be of at least 2 inch by 4 inch stock spaced not to exceed 8 feet, the top rail shall be of at least 2 inch by 4 inch stock, and the intermediate rail shall be of at least 1 inch by 6 inch stock. If pipe is used, it shall be at least 1-1/2 inch nominal diameter. If structural steel is used, it shall be of 2 inch by 2 inch by 3/8 inch angles or equivalent. If wire rope is used for railings, it shall have a diameter of at least 1/2 inch and be stretched taut to allow no more than a 3 inch deflection. Guardrail systems shall be capable of supporting a force of at least 200 pounds applied within 2 inches of the top edge.

Guardrail systems shall be constructed so that when a 200 pound force is applied in a downward direction, it will not deflect to a height less than 39 inches.

If wire rope is used for top rails, it shall be flagged at no more than 6 foot intervals with high visibility material. Manila or synthetic rope shall not be used as guardrails. Employees shall not stand or sit on guardrails.

If an employee ever feels that any piece of his or her fall protection equipment is unacceptable or unsafe, he/she must contact his/her supervisor, who will immediately turn the equipment in for replacement. Harnesses, lanyards, hooks, etc., shall be visually inspected for the condition of rivets, buckles, stitching, D-rings, tabs, frayed or broken strands, cuts and abrasions, burns, rot, soundness of latching and locking mechanisms, and general appearance. Any piece of fall protection equipment that does not pass inspection will be immediately destroyed and replaced. Any piece that is subjected to loading will be immediately destroyed and replaced.

6.0 Fire Prevention

The Contractor shall provide appropriate fire extinguishers for its employees.

The Contractor shall not shut down or in any way interfere with the normal operation of the Airport Authority's sprinkler or fire control system. In all cases where the Contractor requires draining of a sprinkler system, a request shall be made with the Airport Authority Fire Division.

The Contractor shall not attach any tooling, equipment, rigging or any other device to a sprinkler pipe. Ladders shall not be placed against sprinkler pipes.

Contractor shall field verify location at sprinkler heads prior to performing cutting, burning, and/or welding operations. If in doubt, contact the Airport Authority Fire Marshall. Any damage to a sprinkler system shall be reported and repaired immediately.

Materials, trucks, ladders, racks and all other equipment shall be kept clear of aisles, exits, and firefighting equipment, sprinkler risers, fire alarm boxes, electrical lighting, power panels, and valves.

No burning, cutting, welding, or heat generating operation shall be allowed in any area on Airport Authority property without first obtaining a Wayne County Airport Authority - Hot Work permit from the Airport Authority Fire Division (Attachment #4).

Only the Airport Authority Fire Division will issue welding and burning permits. An inspection of the work area will be conducted prior to issuance of the permit. Issuance of the permit will allow heat generating operations to be performed in the areas stated on the permit only. Permits are good for the shift they are issued on. If work is going to continue on the next shift, a new permit is required unless approved by the Airport Authority Fire Division.

Oxygen and acetylene cylinders not in use, shall be properly stored and capped, and secured by a chain or rope. All flammable liquids shall be stored in approved safety containers. Airport Authority Fire Division shall give permission before any quantity in excess of 5 gallons is brought onto the Airport Authority's property. Oily rags and/or rags soaked with flammable liquids shall be stored in approved safety containers only. Paint and painter's equipment, drop sheets, cleaning materials, shall be stored in a clear area away from any heat generating operation.

Lids shall be replaced on all opened paint and/or solvent containers. All solvents are to be stored in approved safety containers.

7.0 Crane/Derricks/Lifting Equipment/Rigging

All crane operators shall be thoroughly knowledgeable in the operations of the crane, rigging equipment, and other MISOHA requirements relating to lifting.

NOTAMS may be required depending upon the location and height of the lift. Contractor shall contact Airport Authority Airfield Operations (734) 942-3685 prior to any crane activity. Lifting shall not be done over any employees. Tag lines shall be used for all lifts greater than 20 feet. No lifts shall be made when winds are sustained over 25 mph. Outrigger shall be shored so they do not damage property. Contractor will be responsible for any property damage to paved or concrete surfaces.

A prelift review shall be performed by the Contractor for every critical lift. A Critical lift shall be defined as a lift with a hoisted load that is within 15% of the maximum load limits (normal) of the equipment that is being used. Before a critical lift is performed the Contractor shall submit a Critical Lift Plan that is prepared and approved by a Professional Engineer that is registered with the State of Michigan. The Critical lift Plan will include the following:

- Description of the lift
- Crane Position
- Lift Height
- Load Radius
- Boom and angle
- Size and weight of load
- Percent of cranes capacity
- Personnel involved
- Rigging plan
- Communications methods
- Ground Conditions
- Inspection Procedures
- Procedures for hosting (if applicable)
- The critical lift plan will also document the source (i.e. weight, crane and rigging capacities, inspections, and wind speeds).

8.0 Hazardous Energy Control

Examples of energy sources that are required to be locked out and tagged regardless of who has custody are:

Any time repairs or modifications are made to electrical systems, either temporary or permanent, they shall be locked out. Locks shall be applied to the main disconnect switch whenever possible. A tag shall accompany all locks.

Electrical systems that provide electrical power to equipment, such as pumps and electrical motors, shall be locked out by the Contractor until such time that system is released. Depending upon the location of the work, the disconnects may have to be coordinated with the Airport Authority Electrical Unit or an Airport Authority Building Operator.

Electrical systems that provide electrical power to the airfield signage and lighting.

Lines, valves and similar systems that are being tested pneumatically or with Inert gases, such as oxygen, shall be tagged and locked out to prevent an accidental discharge of the pressure within the line. In addition, areas affected by the pneumatic test shall be barricaded against entry and inspected by the Contractor prior to commencement of the test.

Any steam, natural gas, refrigeration, chemical feed, glycol, water, fuel oil, diesel, and jet fuel lines that could be inadvertently activated or discharged causing a hazardous condition, shall be locked out, blanked or otherwise neutralized to prevent accidental activation and or a hazardous condition. In some cases double block and bleed maybe required.

Access to all electrical systems will be coordinated by the Airport Authority Electrical Unit. This includes access to all panels, breakers, switches, relays, substations, and all relevant switch gear. All initial desegregation and reenergizing will be coordinated through the Airport Authority Electrical Unit.

Access to all utilities and equipment not including electrical systems can be coordinated with the Airport Authority Maintenance and the Facilities and Infrastructure Divisions.

In some cases a “system” may be turned over to a Contractor or Vendor for repairs and testing. If a system is turned over to a Contractor or Vendor for testing and repairs, the lock-out rules below in this Part still apply.

To get a system energized or de-energized, the Contractor will get approval from the Airport Authority Project Representative. This will require coordination with various Airport Authority departments. If a system is de-energized, back-up utility service may be necessary (i.e. electrical generator) pending the details of the work and proposed outage.

Shutting down systems will be conducted in a manner that will minimize interruption. The Contractor shall complete the Utility Interruption Form (Attachment #6) and submit it to the Airport Authority Project Representative at least five business days in advance prior the shutdown. The Airport Authority Project Representative will coordinate with those tenants/lease holders who will be affected by the shutdown. All outages will be conducted on midnights unless authorized by the Airport Authority Project Representative.

To take overall control and custody of a system for a period of time, the contractor shall complete and execute the Airport Authority Utility System Custody Form (Attachment #7) five business days in advance prior to work with all the proper notifications

Systems that are 480 volts or more will not be turned over to Contractors, unless authorized by the Airport Authority Electrical Department.

If more than one employee is required to lockout and tag a circuit or piece of equipment, a multiple padlock device (hasp) shall be used.

Depending upon the location of the work, some disconnects may have to be coordinated with the Airport Authority Building Operator/Tenant.

This procedure establishes a lockout practice for securing machinery and equipment during periods of construction and maintenance. It is essential that all Contractors are consistent with their lockout procedure to ensure the safety of all employees. A lockout procedure is to render inoperative electrical systems, air lines, hydraulic lines, mechanical devices, pumps, conveyors, fuel, glycol, water, gasoline, jet fuel, pipelines, valves and all other such energy and stored energy systems that may accidentally be energized or discharged while employees are working on them before they are ready and released for service.

Contractors shall administer their own lockout program where not identified by this document. The Contractor shall issue all locks and applicable tags to their foreman, general foreman, superintendents and employees as necessary. The Contractor shall maintain a lock and tag log. Tags are required to have the Contractor's name, phone number, employee name, and supervisor name easily identified on each tag. Contractors shall use lock boxes in all areas except:

- Airport Lighting Vaults at Willow Run and Detroit Metro
- North Terminal Substation #1
- North Terminal Substation #2
- North Terminal Substation #3
- North Terminal Substation #4
- Wayne County Airport Authority Boiler Plant/Powerhouse (Building 611).
- Metro Energy Center
- McNamara Terminal Substation #1
- McNamara Terminal Substation #2
- McNamara Terminal Substation #3
- McNamara Terminal Substation #4
- McNamara Terminal Substation #5
- McNamara Terminal Substation #6
- McNamara Terminal Substation #7
- McNamara Terminal Substation #8
- Building Substations – Willow Run
-

Areas above (1-15) are locations that have pre-designed lock out tag out boards, podiums, and/or Airport Authority provided lock boxes that will be utilized. After equipment has been locally de-energized (i.e. regulator or switch gear) by Airport Authority Electricians, Airport Authority Electricians will lock out the device with an Airport Authority administrative lock and HASP. The Contractor will be required to place their HASP on the Airport Authority HASP at the local disconnect. Afterwards the contractor will be required to hang their personal locks and tags at the local disconnect on their HASP. The Airport Authority will lock all the administrative keys inside the podium or the Airport Authority provided lock box. During work activities the equipment will be locked out and under control of the Airport Authority Electrical Unit. If a Contractor needs to access the electrical room or vault to remove locks and tags they shall coordinate with the Airport Authority Electrical Unit.

All other areas where lock out tag outs are to be completed and not specified above in item K will be considered as remote locations. At remote locations the Contractor and the Airport Authority will follow the same procedures as listed above, but will keep the administrative keys in a lock box at a designated location.

When Airport Authority employees are to work on equipment that contractors are also working on, WCAA employees will place their personal locks and tags on the Airport Authority HASP.

All energy sources shall be locked out and a “DANGER” tag affixed to the equipment or System indicating who installed the lock, Contractor’s name, phone number (24-hour contact), and the reason the system was locked out. Each employee shall be responsible for hanging his or her own lock and tag on the proper piece of equipment before starting work. No employee or other Contractor may work on a lock and tag belonging to another employee. Contractor supervision shall be responsible for assisting employees in locating the proper piece of equipment to be locked out and tagged. Each employee involved with “lockout” shall have a lock with an individual key. No locks with duplicate or master keys shall be used. Contractors are required to identify locks by either tags, paint or die markings. Unidentified locks are subject to removal by using the “lock removal procedure.”

After locking out and tagging a circuit, an attempt to energize the equipment shall be made by depressing or turning “on” all starting stations before work begins. In no case shall work begin before circuits and equipment are tested to ensure that they are, in fact, de-energized.

Any employee who removes a tag or lock belonging to another employee or person, or overrides a tag or lock in any way, may be removed from the Airport property. Written authorization has to be obtained from the foreman, general foreman, superintendent and Project Manager of the responsible Contractor when a lock has been left on a piece of equipment and the originator is not available for removal.

When locks and tags are required, Contractors employees working on that system shall notify their appropriate supervisor. The supervisor, or his designee, shall see that appropriate locks and tags are provided. When work is completed, the appropriate supervisor is also to be notified when locks and tags are removed.

After equipment or systems are turned over to the Airport Authority, no work or Airport Authority modifications will be performed without compliance to Airport Authority’s Lockout/Tagout Program.

There may be some equipment that cannot be physically locked out using any type of device. If a situation exists where equipment cannot be locked out the contractor shall notify the Airport Authority and a Job Hazard Analysis is to be prepared.

9.0 Confined Space Entries

The Airport Authority has several “confined spaces” which will require a written permit prior to entry in accordance with the MIOSHA standard for permit required confined spaces Part 90 and Part 490.

Airport Authority’s confined space procedures can be obtained by contacting the Airport Authority Risk Management department. After reviewing the Airport Authority Confined Space Procedures, the contractor shall evaluate and follow their confined space entry guideline. It is up to each contractor to perform a pre-entry inspection and to classify the confined space. If a classification cannot be achieved the contractor is to work with the Airport Authority Environmental and Risk Management Units.

Airport Authority Fire Division will not act as “Standby” for confined space services.

Any Contractor involved in a confined space entry shall meet all State, Federal, and Airport Authority standards, relative to confined spaces, including:

- MIOSHA requirements.
- Demonstration of proof that their employees who enter, act as standby attendants, issue permits, or perform rescue team functions have been properly trained.
- Issuance and posting of their confined space entry permits by qualified permit issuers.
- Providing appropriate confined space instrumentation to measure oxygen levels, explosive atmospheres, or the presence of toxic gases.
- Providing rescue equipment.

Airport Authority personnel will not perform the above confined space entry procedures for Contractors or provide any instrumentation or equipment. However, the Airport Authority will provide the necessary equipment where Airport Authority personnel will also be entering the confined space.

When Airport Authority employees are also entering the same confined space as the Contractors, Airport Authority employees will control the confined space entry.

10.0 Hazardous Materials/Environmental

Work on Airport Authority projects may involve hazardous materials. If working in buildings with suspect materials please contact the Airport Authority Environmental unit for clarification on those suspected material.

Hazardous materials can be easily identified using the U.S. Department of Transportation (DOT) labeling and identification system. All hazardous materials arriving on site shall be properly labeled, stored, and managed as required by the Safety Data Sheet (SDS) for that material, or as directed by Airport Authority Fire, Airport Authority Environmental and Airport Authority Risk Management. If shipments are being shipped to Willow Run, then Willow Run Maintenance (734-485-6672) shall be notified separately as well as Airport Authority Environmental, Airport Authority Fire, and Airport Authority Risk Management.

All wastes shall be properly stored, labeled, managed, and disposed of in accordance with the Airport Authority project specifications and by the State of Michigan Environment, Great Lakes & Energy (EGLE) regulations, or as otherwise directed by the Airport Authority Environmental personnel.

Contractors and Subcontractors are required to have copies of all SDS's for all materials brought on site. If suspect unknown hazardous materials are identified, then the job should stop until further direction by the Airport Authority Environmental and Risk Management Departments.

If potentially hazardous waste/materials have been indicated in the bid documents and could be foreseen in a project, proposal, or work order, then the contractor is expected to have onsite the proper personal protective equipment and instruments for detection and safety.

Contractor shall provide mitigation plans for projects that include abatement or remediation.

If required, the Contractor must submit a Storm Water Pollution Prevention Plan (SWPPP) and a Spill Prevention, Control, and Countermeasures Plan (SPCCP).

Contractor to immediately report spills to Airport Authority Operations and EGLE. Reports are to conform to EGLE requirements.

Transports and handlers of hazardous materials will be pre-approved by the Airport Authority or their representative.

Waste sites for recycling and disposing of materials will be pre-approved by the Airport authority or their representative.

All bill of ladings and manifests will be submitted to the Airport Authority.

11.0 Roofing Operations

Access to the rooftops shall be coordinated with the Airport Authority Project Representative.

Bitumen kettles will not be permitted on the Airport Authority's property until the Airport Authority Fire Division has inspected them. Once approved, the Airport Authority Fire Division will designate the location of the kettle. The Contractor or Subcontractor shall provide fire extinguishing equipment suitable for the operation. Propane tank shall be properly protected.

Smoking, cutting, burning, or welding will not be permitted on any roof where roofing operations are being performed.

Where large quantities of bitumen are required for roofing operations, the material will be mechanically pumped to the roof.

Reduce the possibility of fires by keeping work areas clean. This shall be done on daily basis.

Fall protection requirements shall be adhered to including perimeter protection, harnesses with lifelines, warning line system within 6 feet of roof edge.

Access to the roof and ladders shall be cleared by Airport Authority Security and the Building operator.

12.0 Removal of Existing Equipment

This section does not pertain to large-scale demolition work. Removal of abandoned or existing equipment, hangers, piping, conduit, etc. is a major safety concern at the construction site throughout the duration of the contract. When reviewing the scope of work, a site hazard assessment will be completed by the Contractor, with all safety and environmental concerns addressed, including methods of removal. Mechanical means of removal will be used versus oxy-gas cutting, whenever possible.

Special consideration should be given for the use of mechanical shearing where removing of building structure will be involved and the structure is contaminated with lead paint. Hydraulic lines can be cut easily. Residual hydraulic oil in lines can cause dangerous flashback if cut with a torch. Even though the lines are drained, not all of the hydraulic fluid can be removed. Galvanized conduit and raceways filled with wires can be cut with the shear. This eliminates potentially hazardous vapors, gases, and fumes coming from the galvanizing and wiring.

Selection of the mechanical equipment will be the responsibility of the Contractor, but shall meet the approval of the Airport Authority. To avoid nuisance odors and possible disruptions to adjacent populated work areas and the general public, work may have to be performed during low peak periods. All rented equipment will be equipped with suitable audio and/or visual back up alarms and shall meet applicable government requirements.

13.0 Vehicle Operation on the AOA

All vehicles accessing the AOA shall be placarded with a company name and logo or some other form of identification. All vehicles shall be limited to the Airport zipper road, paved leasehold areas and/or construction areas unless specifically authorized by the DTW or YIP Airfield Operations Unit.

All construction vehicles/mechanized equipment authorized within the Movement Area or related safety areas shall be marked with a flag on a staff attached to the uppermost portion of the vehicle/motorized equipment so that the flag will be readily visible. The flag shall be at least a 3' x 3' square having a checkered pattern of international orange and white squares at least 1' on each side in accordance with FAA Advisory Circular 150/5210-5D, or current edition as of bid date.

During nighttime hours, all equipment operating on the Airport exceeding 15 feet in height shall be lit with a red obstruction light in accordance with FAA Advisory Circular 70/7460-1, or current edition as of bid date. This light is to be located on the uppermost portion of the equipment.

All construction equipment that exceeds 20 feet in height are required to be "airspaced" as determined by the FAA Form 7460-1. This will require the contractor to notify the DTW or YIP Operations Units so that may have to issues a NOTAM.

Contractor utilized bicycles, motorcycles and two-wheeled scooters are prohibited on the AOA. Vehicle(s)/equipment shall be operated in a manner that does not interfere with aircraft operations. All vehicle(s)/equipment shall yield right of way to all aircraft and emergency vehicles. Vehicles/mechanized equipment operators shall obey all traffic signs and markings.

Vehicles/equipment shall not stop or be parked so as to block a driveway, AOA access gate, fire lane or aircraft. Vehicles/equipment shall not stop or be parked in areas other than those prearranged and approved by the DTW or YIP Airfield Operation Units.

No equipment or vehicles may be parked within six feet of an AOA fence. The established speed limit on the Ramp and AOA is 15 mph. Vehicle(s)/equipment shall not be operated by individuals under the influence of any substance which impairs the ability to do so in a safe manner.

If an incident occurs on the AOA the incident shall be reported immediately to the DTW or YIP Airfield Operation Units by contacting (734) 942-3685. Airport Authority DTW and YIP Airfield Operations and Airport Authority Public Safety have the right to investigate all incidents and issue citations. The Contractor is still obligated to produce their own incident report to be submitted to the Airport Authority upon request. The Contractor is required to submit an incident report no later than 24 hours after the incident.

Vehicle/mechanized equipment operators are not permitted to move about the Airport, outside the designated construction area, at night unless the vehicle has operating head lights, tail lights and brake lights, or is under the escort of a properly lighted vehicle. Head lights shall not be set on high beam when moving about the Airport at night.

Vehicles/mechanized equipment authorized on the Movement Area (runways, taxiways, and ramps) and/or associated safety areas shall be equipped with an electrically powered, amber color, 360-degree omni-direction light, mounted on the vehicle such that it is conspicuous from any direction.

At no time shall a vehicle enter the Movement Area and/or associated safety areas unless it is authorized by the DTW or YIP Airfield Operation Units and is in continuous radio communication with the Control Tower. If a vehicle is not radio equipped to communicate with the Control Tower, an escort vehicle equipped with such a radio

shall lead or direct the movement of this vehicle while operating on the Movement Area unless the construction area is completely closed to Airport movement and cordoned off. Coordination of escorts shall be prearranged with the DTW or YIP Airfield Operation Units. Any individual authorized unescorted access to the Movement Area or associated safety areas shall have completed the driver training program administered by the DTW or YIP Airfield Operations Department.

Seat belts shall be utilized on equipment/vehicles that are designed for usage.

The Airport Authority may remove and impound, at the owner's expense, any vehicle/equipment which is disabled, abandoned, improperly parked, or represents an operational hazard

All vehicles/equipment shall be appropriately secured such that neither aircraft blast nor wind will result in their movement.

14.0 AOA Contractor Escorts

As required the Contractor shall provide an adequate number of escorts/flaggers for material deliveries along haul routes and the movements of the Contractor's vehicles/mechanized equipment and personnel within the Movement Area and Non-Movement Areas as authorized by the DTW or YIP Airfield Operation Units. To arrange for employees to become an escort the contractor shall complete the following:

For DTW Projects: those selected Non-Movement Area escort employees are required to be badged by Airport Security, go through Security Identification Display Area (SIDA) training program, and pass the computerized test. Office hours for Credentials are Monday thru Friday from 8:30 am until 4:00 pm (closed 11:30-12:30 for lunch). Be advised that in some cases training will take at least two hours. Training shall not be requested after 2:00 p.m. The office can be contacted at (734) 932-3606 or visit the Airport Authority's website at www.metroairport.com and click on the "Badging" link on the top tool bar for additional information on Badge processing. Upon successfully completing the security requirements, those employees who have been selected as escorts shall contact the Airport Authority Airfield Operations Access and Permits office to coordinate finalization of the permit and to coordinate training on the AOA with an DTW or YIP Airfield Operations agent. Training is performed at 10:00 am and 1:00 pm each day Monday thru Friday. The DTW Operations and Permits office can be reached at 734-942-3823.

Movement Area escort/flagging employees are required to complete and pass an additional eight hour class on Ground Vehicle Operations. For scheduling contact Airfield Operations Training Unit at (734) 942-3595. For YIP Projects: All employees are encouraged, but only Supervisory personnel (Foreman, Site Superintendents, Project Managers, etc.) are required to be trained and badged. These badged individuals will be responsible for escorting their employees and Subcontractors in closed and cordoned construction sites. The YIP Operations Unit, or designee, will perform any escorts in open AOA Movement Areas. Badge applicants are to complete the required badge paperwork. Employees attend a 1.5 hour Airport Security/AOA Driver's Safety Training Class at the Operations/Security Office (734) 485-6675. Training is provided at 10:00 am and 1:30 pm Tuesdays and Thursdays, by appointment only. A \$20 non-refundable badge processing fee, payable in cash or money order only, will be charged for each badge. Escort processions will be limited to a maximum of three (3) trucks or less. Those performing the escorts will be trained by WCAA Airside Operations. If requested, variance to the procession procedure can be modified only upon written approval by the Airport Authority. An approved flagger plan maybe required.

During any absence of the approved escort(s)/flagger(s) or for periods that they are unable to perform their specified duties, all work within the Movement Area and associated safety areas for projects shall stop. Additionally, all personnel and equipment shall be escorted to approved locations outside the Movement Area and related safety areas. NO contract time extension will be granted for time lost due to the absence of escort(s). Work shall resume only with the return of the approved escort(s).

The escort/flagger shall ensure that all equipment maintains proper clearances from moving aircraft.

For flaggers/escorts contracted through the Airport Authority, the Contractor shall be responsible for the cost of each required flagger/escort at the hourly rate identified on the Wayne County Airport Authority Flagging/Escorting Service Request Authorization Form (See WCAA Airside Operations). The Contractor shall also be responsible for completing and submitting this form.

Contractor provided flagging services must be approved by the Airport Authority. Flaggers will be trained by Airport Authority staff and be equipped with approved hazard communication devices (red/green flags, radios etc.). The area where the flaggers are staged must be approved and will be properly illuminated.

15.0 Special Construction Rules on the AOA

When airfield construction is being performed on the AOA the following rules will apply unless modified in writing by the DTW or YIP Airfield Operation Units. All construction activities on the AOA shall include a specific Construction Safety Phasing Plan (CSPP) and a Safety Plan Compliance Document (SPCD) as required by the FAA. The SPCD will address compliance to and details required by the CSPP and include any other topics of discussion that might be mentioned during the safety phase planning meeting.

The safety phase planning meeting shall be held prior to mobilization to the AOA. Any Airport construction and/or alteration requires the Contractor to complete and submit FAA Form 7460-1 Notice of Proposed Construction or Alteration (available from the FAA Air Traffic Division Regional Office), and www.FAA.gov at least 60 days prior to the start of the project.

The Contractor shall complete and submit FAA Form 7460-1 for all equipment and/or temporary structures, utilized during any Airport construction and/or alteration that exceeds a height of 20 feet above ground level. This includes

- Cranes;
- Derricks;
- Stockpiles of materials or equipment; and
- Earthmoving equipment.

A copy of all completed FAA Form 7460-1's and the FAA's determination(s) shall be on file with the Department of Aviation prior to commencing the erection or construction of the item(s) proposed by the Contractor. The Contractor will provide the DTW or Yip Airfield Operation Units with the FAA determination number, for internet review, or paper copy of the full determination. The Contractor shall erect and maintain fencing, barricades, signs and warning devices used to delineate the perimeter of all construction areas, as approved by the Airport Authority Airfield Operation Units.

All escorts performed within the Movement Area and/or associated safety areas, shall be provided by an authorized Escort. The Airport Authority Airfield Operation Units and the Airport Authority Security Division shall designate all access points into the AOA. All points of entry into the AOA, which are under the Contractor's control, shall be secured and/or guarded and should be coordinated with the Airport Authority Security Division or YIP Airfield Operations/Security Division. Deliveries are to be strictly controlled (by the Contractor) using personnel specifically acquainted with these rules. The Contractor shall provide properly manned escort vehicles as required to guide and escort all deliveries to the work area(s).

All work outside an approved construction area shall be submitted, in writing, 72 hours in advance, excluding weekends

Unless otherwise specified by the DTW or YIP Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division, all work outside an approved construction area shall be marked in accordance with Barricade Details Checklist included at the end of this document. All barricades, lighting and warning devices used to delineate any construction or hazardous area(s) are to be provided by the Contractor.

At no time shall personnel, vehicles or equipment be located or enter any of the following areas unless authorized by DTW or YIP Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division.

- Within 250 feet parallel to an active runway centerline (to be indicated on the CSPP and/or SPCD).
- Within 400 feet parallel to an active runway centerline without equipment and stockpile removal.

- Within 1,000 feet of the end of active runways (each end to be indicated in the CSPP and/or SPCD).
- Within 160 feet parallel to an active taxiway centerline operating with Group V aircraft without proper approval.
- Active NAVAID Critical Areas.
-

On the Movement Area and/or associated safety areas during times of inclement weather or unusual events as determined by the DTW or YIP Airfield Operation Units. During such times all work is to be suspended. All equipment shall be removed to approved staging areas

Trenches and/or Excavations Trenches and/or excavations shall not be allowed in the following areas without closure or restriction of the adjacent Movement Area:

- Within 250 feet parallel to a runway centerline.
- Within 400 feet parallel to a runway centerline, without proper trench and excavation cover.
- Within 160 feet parallel to a taxiway centerline operating with Group V aircraft without proper approval.
- Within 1,000 feet of the end of a runway.
- Active NAVAID Critical Areas.

Barricading/plating of trenches and/or excavations shall be in accordance with the requirements contained in Barricade Details Checklist. No trenching is allowed under AOA fence without prior approval and coordination by the Airport Authority Security Division. All stockpiled material(s)/supplies shall be constrained in a manner to prevent movement resulting from aircraft blast or wind conditions. Material(s)/supplies shall not be stored within 500 feet of aircraft turning areas or movement areas. Stockpiled material(s)/supplies shall not exceed 15 feet in height unless the Contractor has complied with all requirements for airspacing and secured approval from the DTW or YIP Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division. All material(s)/supplies shall be positioned so it will not obstruct the line of sight from the Control Tower to the Movement Area. Marking and lighting shall be in accordance with the requirements contained in Barricade Details Checklist.

Stockpile material may not be within 6 feet of an AOA fence. Nighttime work, not covered by the Contract Documents, requires 72 hours advanced approval, excluding weekends, by the DTW or YIP Airfield Operation Units. Lighting for nighttime work shall be shielded and positioned downward so as not to hinder the vision of the air traffic controllers in the control Tower or the pilots of moving aircraft.

Debris, waste, and loose materials shall not be allowed on the Movement Area. If debris and/or loose materials are observed to be on active portions of the Movement Area, the Contractor will be responsible for correcting the discrepancy immediately. At the direction of the DTW or YIP Airfield Operation Units, debris problems occurring during construction, NOT corrected by the Contractor in a timely manner, will be corrected by the Airport at the Contractor's expense. The Contractor is responsible for controlling dust problems resulting from construction and clean-up processes, as defined by the DTW or YIP Airfield Operation Units or WCAA Facilities, Design, and Construction Division, resulting from construction and clean up processes. The Contractor may be working in an air operations area in which a high degree of care is necessary to control debris and dust. Spilled material on active roadways, runways, taxiways and aprons will be swept up immediately. The Contractor will be aware that the construction area may be subject to jet blast and significant wind velocities. Dust control measures will be required to prevent loose material from blowing within or outside the air operations area.

If the Contractor uses or if a haul road crosses any area used by aircraft, service or emergency vehicles, a vacuum power broom and/or hand sweeping will be used to keep this area clean of debris which could damage aircraft engines or propellers. The Contractor will be liable for any damages that occur. Power brooms and sweepers shall be vacuum capable. Prior to opening any Runway, Taxiway, Ramp, Apron or Associated Safety Area, that has been

closed for construction; the Contractor shall arrange for an inspection by the DTW or YIP Airfield operations director or his/her designee.

All barricades used by the Contractor to designate an unusable or hazardous area on the AOA, shall be secured in place against movement or jet blast. The Contractor shall ensure that all barricades and hazard lighting are operational prior to departing the construction area at the end of each workday.

In accordance with FAR Part 139, The DTW or YIP Airfield Operation Units shall, prior to the release of work crews, inspect all areas to ensure that:

- Paved areas are free of surface variations in accordance with FAR Part 139.
- All unpaved safety areas are cleared and graded and have no potentially hazardous ruts, humps, depressions or other surface variations.
- All trenches or excavations within active Runway and/or Taxiway safety areas are backfilled to support the weight of an aircraft or Aircraft Rescue and Fire Fighting (ARFF) equipment.
- If the trenches, excavations or hazardous areas have been authorized to remain in place, they are to be adequately plated and marked and lighted in accordance with Barricades Detail Checklist.

Location of haul routes on the Airport site shall be approved by the DTW or YIP Airfield Operation Units or Airport Authority Facilities, Design and Construction Division. All haul routes on the Airport shall be marked, when necessary, and maintained by the Contractor. These routes shall be restored to their original condition upon completion of the construction project. Markings, if required, shall be provided by the Contractor, in accordance with specifications established by the DTW or YIP Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division, and Airport Authority Security. Construction equipment shall not be permitted to operate upon paved areas unless the equipment has pneumatic tires or special means, approved by the Airport Authority Facilities, Design, and Construction Division, provided to protect the pavement.

Construction equipment shall not exceed a height of 20 feet above the Airport surface without approval by the Airport Authority Airfield Operation Units. All construction involving cranes shall be coordinated at least 72 hours in advance, excluding weekends, with the Airport Authority Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division. This does not include the time required for airspacing. The following information is required:

- Location of the Crane.
- Maximum extendable height.
- Hours of operation.
- The top of each crane boom shall be marked by a 3' x 3' orange and white checkered flag — each box being 1' square.
- Each crane shall be lowered at night and during periods of poor visibility as directed by Airport Authority Airfield Operation Units or Airport Authority Facilities, Design, and Construction Division. In the event the crane is approved to remain extended during the hours from sunset to sunrise, the highest point of the crane boom will be lit with a red obstruction light in accordance with AC 70/7460-1

16.0 Job Hazard Analysis

Construction is a dynamic process; it is ever changing in physical and environmental forms until the construction process is completed. At the Airport Authority all contractors who are required to perform construction work shall use the job hazards analysis process. Its purpose is to develop a preliminary hazard analysis on proposed work operations as to their potential for injury, property damage, or both. Once potential hazards have been identified, identify procedures to eliminate or mitigate their potential for occurrence.

Direct benefits of the hazard analysis are as follows:

- Potential for injury or property damaged is eliminated;
- Identifies hazards;
- Improves safety;
- Defines job procedures;
- Provides training;
- Increases awareness;
- Provides improvement in job methods;

During the planning stages of the project work, activities are flagged that require job hazard analysis. Further work operations requiring job hazards analysis may be identified by weekly workarounds, responses to accidents or audits, safety meetings, or committee meetings, or if work activities change that have not been addressed.

17.0 Housekeeping

Leads, hoses, and extension cords shall be hung up with a nonconductive material, off all floors, stairways, and walkways. Trash such as drinking cups, cans, and scraps from lunch are not to be thrown down, but disposed of properly in marked containers.

Available material, equipment, concrete forms, pipe, etc., are to be orderly, stacked out of walkways, and from in front of doors, stairways, and ladders.

Oil, grease, and other such liquid spills shall be cleaned up at the time of spill and are not to be left unattended. Each craft is responsible for housekeeping in its respective work areas.

Where such items as protruding rebar and anchor bolts create an impalement hazard or tripping hazard they shall be properly protected and conspicuously marked.

Trash barrels and 55 gallon drums shall not be hoisted by holes cut in the sides; adequate means of support shall be used.

Site/FOD walks will be required on a weekly basis

18.0 Site Sanitation

Potable water:

- An adequate supply of potable water shall be provided on all projects.
- Portable containers used to dispense drinking water shall be capable of being tightly closed, and equipped with a tap. Water shall not be dipped from containers.
- Any container used to distribute drinking water shall be clearly marked as to the nature of its contents and not be used for any other purpose.
- The common drinking cup is prohibited.
- Where single service cups (to be used but once) are supplied, both a sanitary container for the unused cups and a receptacle for disposing of the used cups shall be provided.

Non potable water:

- Outlets for nonpotable water, such as for industrial or firefighting purposes only, shall be identified by signs to clearly indicate that the water is unsafe and is not to be used for drinking or washing purposes.
- There shall be no cross-connection, open or potential, between a system furnishing potable water and a system furnishing non-potable water. Toilets at construction sites:

Washing facilities:

Employees can use public restroom expects when projects engaged in activities where employees are exposed to sludge, chemicals, or other harmful contaminants must provide adequate washing facilities onsite. This can be accomplished by receiving permission to use the WCAA non-public facilities or constructing a temporary wash station. This would require a potable water holding tank, soap, towels, and weather protection. Such facilities should be mobile and maintained in close proximity to work locations and along the route of travel to the parking area.

Food consumption:

—In an attempt to prevent the possibility of contracting a hand-to-mouth disease, all employees must wash their hands before breaks, lunch, and going home. All consumables such as food, candy, dip tobacco, cigarettes, coffee, sodas, etc., should be not stored or consumed in the work area. These items should not leave the personal car parking lot unless sanitary lunch areas are designated for that purpose.

19.0 Compressed Gasses

Care shall be exercised in handling all compressed gas cylinders. They shall not be dropped, jarred, or exposed to temperature extremes. Cylinders shall have the valve cap or valve protection device in place at all times, except when in actual use or connected to a welding set. Cylinders shall not be rolled and shall not be lifted by the valve or valve cap; a suitable cradle or other device shall be used. Cylinders shall have their contents properly identified.

Cylinders not having fixed hand wheels shall have keys, handles, or non-adjustable wrenches on the valve stems while the cylinders are in service. Compressed gas cylinders, whether full or empty, shall be stored and transported in an upright position and chained or otherwise secured so they cannot fall or be upset. Oxygen cylinders in storage shall be separated from fuel-gas cylinders or combustible materials (especially oil or grease) a minimum distance of 20 feet or by a 5 foot high noncombustible barrier. Cylinders shall not be placed where they might become part of an electric circuit or within 5 feet of an electrical outlet. Employees shall never force connections which do not fit nor shall they tamper with the safety relief devices of cylinder valves.

Before the regulator is removed from a cylinder, the valve shall be closed and all pressure released from the regulator. A leaking cylinder shall not be used. Such cylinders shall be taken outdoors away from sources of ignition. The supervisor shall be notified. A flame shall never be used to detect gas leaks. The recessed top of cylinders shall not be used as a place for tools.

Oxygen--Oil, grease, or similar materials shall not be allowed to come in contact with any valve, fitting, regulator, or gauge of oxygen cylinders:

- Oxygen shall never be used as a substitute for compressed air.
- When an oxygen cylinder is in use, the valve should be opened fully in order to prevent leakage around the valve stem.

Acetylene--Acetylene cylinders shall be properly secured and always used, transported, or stored in a vertical position. Cylinders shall be protected from sparks, flames, and contact with energized electrical equipment:

- An acetylene cylinder valve shall not be opened more than one and one-half turns of the spindle and preferably no more than three-fourths of a turn.
- Employees shall not use acetylene in a free state at pressures higher than 15 psi.

20.0 Safe Supports and Scaffolds

Each project shall designate at least one competent person to oversee scaffold operations. Operations shall include scaffold erection, moving, dismantling, altering, repairing, etc. The competent person shall ensure the proper type of scaffolding is selected for each task, and that only experienced and properly trained employees erect, move, dismantle, alter, or repair scaffolds. The competent person shall ensure each scaffold is erected in accordance with Contractors requirements, manufacturer's specifications, and applicable OSHA standards.

The competent person shall inspect each scaffold prior to employees accessing the scaffold, before each shift, and after any occurrence that could affect the scaffold's structural integrity and shall tag or ensure the scaffold is tagged according to the scaffold tagging procedure.

Scaffold Tagging Procedure

Prior to assigning an employee or crew to work off of a scaffold, the foreman responsible for the employee or crew shall interface with a scaffold competent person to ensure the scaffold is complete and properly tagged. The foreman shall then visually inspect the scaffold to verify it is complete and properly tagged according to the following:

- Green Tag--This scaffold was built to meet OSHA regulations; it is safe to use.
- Yellow Tag--This scaffold does not meet OSHA regulations; personal fall arrest systems are required. This tag is used in situations where guardrails cannot be constructed on all open sides of a scaffold platform because of an interference or obstruction.
- Red Tag--This scaffold is not complete; DO NOT USE.

Employees shall not use a scaffold unless it is properly tagged according to the scaffold tagging procedure.

Scaffold Requirements No employee, or any material or equipment, shall be supported or permitted to be supported on any portion of a pole structure, scaffold, ladder, walkway, or other elevated structure, crane or derrick, etc., without it first being determined that such support is adequately strong and properly secured in place.

Employees shall check all scaffolding prior to use to ensure it is of sufficient strength and rigidity to safely support the weight of persons and material to which it will be subjected. Employees shall not use a scaffold from 4 to 6 feet in height, having a minimum horizontal dimension of less than 45 inches, unless there is present a standard guardrail, with midrail and toe board, to provide adequate employee protection. Employees shall not use a scaffold over 6 feet in height unless there is present a standard guardrail, with midrail and toeboard to provide adequate employee protection.

Scaffold planks shall be secured in place and extend over their end supports by not less than 6 inches (unless cleated) nor more than 12 inches. Scaffolds shall not be moved without first removing all loose tools, materials, and equipment resting on the scaffold deck.

The footing or anchorage points for scaffolds shall be sound, rigid, and capable of carrying the maximum intended load without settling or displacement. Unstable objects such as barrels, boxes, loose brick, or concrete blocks shall not be used to support scaffolds or planks. Scaffolds shall be erected level and plum and rigidly braced to prevent swaying and displacement. Scaffolds shall not be altered or moved horizontally while being used or occupied except when specifically designed for such use. Movable scaffolds shall have the casters or wheels locked to prevent movement.

The width of all scaffolds, ramps, and platforms shall be sufficient to prevent congestion of persons, materials, or equipment, and in no case shall they be less than 18 inches wide. They shall extend to the entire platform depth.

Synthetic or natural fiber rope shall not be used as guardrails. Employees working on suspended scaffolds shall be protected by an independent lifeline, body harness, and lanyard. Safe access shall be provided for all scaffolds. Structural members should not be used as a means of access.

Employees shall not use a scaffold unless it is properly tagged according to the project scaffold tagging procedure. Only competent persons may erect, dismantle, or alter a scaffold. Only scaffold grade planks (appropriately marked) shall be used for work platforms. Scaffold planks shall not be painted with non-transparent materials which could obscure defects. Scaffold foundations shall utilize the manufacturer's footer plates.

Mobile Scaffolds:

Platforms shall be tightly planked for the full width of the scaffold except for necessary entrance opening. Platforms shall be secured in place. All tools and equipment shall be removed from the scaffold before it is moved or repositioned. The rollers shall be secured to the scaffold frame by using a manufactured pin or by No. 9 wire to prevent displacement. Standard guardrail systems shall be used on all open sides of the scaffold.

Swinging Scaffolds:

On suspension scaffolds designed for working load of 500 pounds, no more than two workers shall be permitted to work at one time. On suspension scaffolds with a working load of 750 pounds, no more than three workers shall be permitted to work at one time. Each employee shall be protected by an approved full body harness attached to an independent lifeline.

The lifeline shall be securely attached to substantial members of the structure (not scaffold), or to securely rigged lines, which will safely suspend the employee in case of a fall.

21.0 Ladders

Wooden ladders shall not be painted so as to obscure a defect in the wood; only a clear, nonconductive finish shall be used. All ladders shall be inspected frequently and regularly. Ladders with weakened, broken, or missing steps; broken side rails; or other defects shall be tagged and removed from service.

Ladders and scaffolds shall be sufficiently strong for their intended use. Portable metal ladders shall not be used in the vicinity of energized electrical circuits. (Exception: Such ladders may be used in specialized work, such as high voltage substations, where nonconductive ladders might present a greater hazard. These ladders shall be properly marked.)

Ladders shall not be placed in front of a door that opens toward the ladder, unless the door is open, locked, or guarded. When ascending or descending ladders, employees shall have both hands free and shall face the ladder. Only one employee shall work from a ladder at one time (except for hook type ladders). If two employees are required, a second ladder shall be used. Ladders shall not be used as scaffold platforms. Boxes, chairs, etc., shall not be used as ladders. Employees shall not use a ladder until they have been properly trained in its use.

Each quarter, a formal ladder inspection must be performed by a competent person designated by the Project Manager. If the ladder passes the inspection, it shall be tagged with a colored tie-wrap for the respective quarter. If it fails, it shall be tagged with a "DO NOT OPERATE" tag until repairs are made. When the ladders are being inspected, the Quarterly Ladder Inspection form must be filled out by the competent person.

Straight Ladders:

Portable straight ladders shall not be used without nonskid bases. The ladder shall be placed so that the distance between the bottom of the ladder and the supporting point is approximately one fourth of the ladder length between supports. Straight ladders shall not be climbed beyond the third step from the top.

When working from a portable ladder, the ladder must be securely placed, held, tied, or otherwise made secure to prevent slipping or falling. When dismounting from a ladder at an elevated position (as at a roof), the employee shall ensure that the ladder side rails extend at least 3 feet above the dismount position, or that grab bars are present.

Two Step Ladders:

Employees shall wear a body harness and lanyard, and tie off to a secure anchor whenever both hands must be used for the job or are exposed to a fall in excess of 6 feet. Ladders shall not be spliced together to form a longer ladder. A ladder shall not be placed against an unsafe support. Employees climbing a ladder with a fall exposure greater than 24 feet shall be protected by an approved cage, ladder climbing device, or by the use of a body harness, lanyard, lifeline step Ladders. The top two steps shall not be used. Step ladder legs shall be fully spread and the spreading bars locked in place. Step ladders shall not be used as straight ladders. When an employee is working on a step ladder over 6 feet high, the employee shall use a body harness and lanyard attached to a substantial anchor.

22.0 Material Handling/Ergonomics

An employee shall obtain assistance in lifting heavy objects or power equipment shall be used.

When two or more persons carry a heavy object that is to be lowered or dropped, there shall be a prearranged signal for releasing the load.

When two or more persons are carrying an object, each employee, if possible, should face the direction in which the object is being carried (The right way to lift is easiest and safest. Crouch or squat with the feet close to the object to be lifted, secure good footing, take a firm grip, bend the knees, keep the back vertical, and lift by bending at the knees and using the leg and thigh muscles. Employees shall not attempt to lift beyond their capacity. Caution shall be taken when lifting or pulling in an awkward position.)

Employees should avoid twisting or excessive bending when lifting or setting down loads. When moving a load horizontally, employees should push the load rather than pull it. When performing a task that requires repetitive lifting, the load should be positioned to limit bending and twisting. The use of lift tables, pallets, and mechanical devices shall be used in these instances. When using such tools as screwdrivers and wrenches, employees should avoid using their wrists in a bent (flexed), extended, or twisted position for long periods of time. Employees should maintain their wrists in a neutral (straight) position.

When gripping, grasping, or lifting an object such as a pipe or board, the whole hand and all the fingers should be used. Gripping, grasping, and lifting with just the thumb and index finger should be avoided.

23.0 Hand Tools

All tools, regardless of ownership, shall be of an approved type and maintained in good condition. (Tools are subject to inspection at any time. A foreman has the authority and responsibility to condemn unserviceable tools, regardless of ownership.) Defective tools shall be tagged to prevent their use and shall be returned to the tool room for repairs.

Employees shall always use the proper tool for the job performed. Hammers with metal handles, screwdrivers, knives with metal continuing through the handle, and metallic measuring tapes shall not be used on or near energized electrical circuits or equipment. Tools shall not be thrown from place to place or from person to person; tools that must be raised or lowered from one elevation to another shall be placed in tool buckets or firmly attached to hand lines.

Tools shall never be placed unsecured on elevated places. As impact tools such as chisels, punches, drift pins, etc., become mush-roomed or cracked, they shall be dressed, repaired, or replaced before further use. Chisels, drills, punches, ground rods, and pipes shall be held with suitable holders or tongs (not with the hands) while being struck by another employee. Shims shall not be used to make a wrench fit. Wrenches with sprung or damaged jaws shall not be used.

Pipe shall not be used to extend a wrench handle for added leverage unless the wrench was designed for such use. Tools with sharp edges shall be stored and handled so that they will not cause injury or damage. They shall not be carried in pockets. Wooden handles that are loose, cracked, or splintered shall be replaced. The handle shall not be taped or lashed with wire. All cutting tools such as saws, wood chisels, drawknives, or axes shall be kept in suitable guards or in special compartments.

Tools shall not be left lying around where they may cause a person to trip or stumble. When working on or above open grating, a canvas or other suitable covering shall be used to cover the grating to prevent tools or parts from dropping to a lower level where others are present or the danger area shall be barricaded or guarded. The insulation on hand tools shall not be depended upon to protect users from shock.

The noncurrent carrying metal parts of portable electric tools such as drills, saws, and grinders shall be effectively grounded when connected to a power source unless:

- The tool is an approved double-insulated type.
- The tool is connected to the power supply by means of an isolating transformer or other isolated power supply, such as a 24 volt dc system.

All powered tools shall be examined prior to use to ensure general service-ability and the presence of all applicable safety devices. The electric cord and electric components shall be given an especially thorough examination and documented on the Electrical Inspection Log. Powered tools shall be used only within their capability and shall be operated in accordance with the instructions of the manufacturer.

All tools shall be kept in good repair and shall be disconnected from the power source while repairs are being made. Electrical tools shall not be used where there is a hazard of flammable vapors, gases, or dusts.

All power tools and cord sets shall be protected by ground fault circuit interrupters.

25.0 Heaters

UL approved salamanders, radiant heaters, and space heaters are the only approved heaters on the jobsite. Heaters shall be kept at least 20 feet from buildings and other combustible items or as approved by the WCAA Fire Marshal. Job-made heaters, solid fuel salamanders, and open fires are prohibited on the jobsite.

26.0 Barrier Tape Identification System

In order to uniformly identify particular hazards, a barrier tape identification system has been developed for use by all employees working on the construction site. It has been developed so that any employee working on the jobsite, regardless of craft, can recognize and avoid a hazard when properly marked.

The following barrier tape identification system shall be used:

- General Purpose--Yellow tape (may have black in it). Used for low hazard areas where access should be monitored and controlled. "Use caution when crossing."
- General--Red tape (may have black in it). "Do not cross."
- Logistics--Multicolor triangular flagging. Used to identify lay-down areas, walkways, etc. "Use caution when crossing."
- Radiation--Yellow with magenta (purple) tape. Possible radiation hazard, X-ray, etc. "Do not cross."

The supervisor erecting the barrier tape shall hang a tag on the tape that indicates the hazard, name of employer, name of person erecting the tape, and date erected. The barriers shall be erected far enough back from the hazard to allow for adequate warning and protection from the hazard. The barrier shall be constructed so that it will stand against adverse weather conditions and construction traffic. If the hazard is of a magnitude which requires additional protection (hard barricades, lights, etc.), the supervisor responsible for erecting the barricade shall see that additional protection is provided. It is the erector's responsibility to maintain the barrier as long as the hazard is present. It is also his or her responsibility to take down and dispose of the barrier when its use is not needed.

27.0 General Electric Rules

All electrical work shall be in compliance with the latest edition of the National Electrical Code, unless otherwise provided by MIOSHA regulations. Electrical installations and maintenance shall be performed only by qualified electricians.

Extension cords used with portable electric tools shall be the 3 wire type, shall be protected from damage, and shall not be fastened with staples, hung from nails, or suspended from wires. Splices shall have soldered wire connections with insulation equal to the cable. Worn or frayed cables shall not be used.

Except where bulbs are deeply recessed in the reflector, bulbs on temporary lights shall be equipped with guards. Temporary lights shall not be suspended by their electric cords unless designed for suspension.

Receptacles for attachment plugs shall be of approved, concealed contact type. Where different voltages, frequencies, or types of current are supplied, receptacles shall be of such designs that attachment plugs are not interchangeable.

Cable passing through work areas shall be covered or elevated at least 7 feet off the ground. Unprotected cables, extension cords, etc., shall NEVER be run over by vehicles, heavy equipment, JLGs, scissor lifts, etc.

Boxes for disconnecting means shall be securely and rigidly fastened to the surface upon which they are mounted and fitted with covers.

Employees shall not be permitted to work in such proximity to any part of an electric power circuit that he or she may contact the same in the course of his or her work unless the employee is protected against electric shock by de-energizing the circuit and grounding it or by guarding it by effective insulation or other means. In work areas where the exact location of underground electric power lines is not known, workers using jackhammers, bars, or other hand tools which may contact an energized line shall be provided with insulated protective gloves.

For 15 and 20 ampere receptacle outlets on single-phase, 120 volt circuits for construction sites which are not a part of the permanent wiring of the building or structure, ground fault circuit interrupters (GFCIs) shall be used for employee protection.

Employees shall test GFCIs before they are used on a daily basis and, if there is a malfunction, shall report the problem to his or her supervisor. Each month, the Project Manager shall designate a competent person in electrical safety to test the GFCIs and report the results on the GFCI Monthly Test form. All outlets, "bang-boxes," etc., shall be tagged with a monthly inspection tag. When the competent person checks the GFCIs, the month and day they were tested shall be marked.

All electrical circuits, equipment, and conductor enclosures shall have a grounding system, which is permanent and continuous, of such capacity to conduct safely any fault current likely to be imposed on it, and has sufficiently low resistance to limit the voltage to ground and facilitate the operation of the circuit breaker in the circuit. The continuity and resistance of grounding systems shall be tested immediately after installation, after any repair or modification, and annually thereafter.

All electrical powered equipment and machinery shall be de-energized and locked out before any maintenance or repair work is performed. Where overhead power lines are encountered on a jobsite and equipment has the potential to contact the lines, the safe minimum clearances shall be followed. The lines shall be either relocated in cooperation with the utility or Owner, or safe clearance barriers shall be erected. Example: install poles and wire rope that has reflective tape or other acceptable attention getting markings at safe minimum clearance locations away from the power lines. Ground level signs shall also be constructed to make operators aware of the hazard above. The safe minimum clearances are as follows.

All employees who face the risk of electric shock from working on, near, or with electrical circuits, which is not reduced to a safe level by electrical installation, shall be trained in the recognition, avoidance, and control of specific electrical hazards.

28.0 Disposal Chutes

Whenever materials are dropped more than 20 feet to any exterior point of a building, an enclosed chute shall be used. When debris is dropped through holes in the floor without the use of chutes, that area where the material is dropped shall be enclosed with barricades not less than 42 inches high and not less than 6 feet back from the projected edges of the opening above. Warning signs of the hazards of falling material shall be posted at each level.

29.0 Non-AOA Motor Vehicles and Mechanized Equipment

All vehicles in use shall be checked at the beginning of each shift to assure that all parts, equipment, and accessories that affect safe operation are in proper operating condition and free from defects. All defects shall be corrected before the vehicle is placed in service.

No person shall use any motor vehicle, earth moving, or compacting equipment having an obstructed view to the rear unless:

- The vehicle has a reverse signal alarm distinguishable from the surrounding noise level, or
- The vehicle is backed up only when an observer signals that it is safe to do so.

Heavy machinery, equipment, or parts thereof which are suspended or held aloft shall be substantially blocked to prevent falling or shifting before employees are permitted to work under or between them.

30.0 Silica Exposure Plan

The contractor must have a written silica exposure plan for the work to be performed. The plan shall be a separate document or part of the Site Specific or Corporate plans.

31.0 Investigation, Documentation, and Reporting of Incident and Accidents

Accidents are an unexpected event which results in injury, illness of a person, or property damage. Incidents do not result in serious injury but could include a near-miss or result in a violation (unsafe act) of a procedure or policy.

Incidents and Accidents include, but are not limited to:

- Property Damage;
- Environmental Release;
- FAA Excursion;
- Illness/Exposures;
- FAA Incursion;
- Injury;
- FAA Surface Incident; and
- Unsafe Act or Condition.

All Accidents and Incidents must be reported immediately to the WCAA. When the Contractor or sub-contract is aware of an incident or accident., they must notify the Wayne County Airport Authority as soon as possible. Names of those WCAA employees who need to be notified will be agreed during the pre-construction meeting. Notification contacts can only be changed as approved by the WCAA. As required by law, other entities such as Michigan EGLE, EPA, FAA, and MIOSHA shall be notified of the incident as necessary by the Contractor.

In the event there is immediate danger to life and health contact 911. The Contractor shall assure they identify that they work on behalf of the Airport Authority. Explain what happened and give details as to the location of the events.

If an incident occurs on the AOA the incident /accident shall be reported immediately to the DTW or YIP Airfield Operation Units by contacting (734) 942-3685.

After every incident or accident, regardless of the location, an investigation shall be conducted by a qualified person. Contractor shall be responsible for all sub-contractors and will work with the sub-contractor as it relates to the investigation, documentation, and reporting of accidents and incidents. The Contractor shall have a system for investigation which shall include the following:

- The primary purpose of the investigation is to determine the cause of the incident/accident and eliminate them so that the same or similar events will not occur. Corrective actions shall be produced and implemented. In some cases this may involve interaction with the WCAA, other Contractors, Tenants, and Airlines.
- The WCAA, WCAA designees, FAA, MIOSHA, Michigan EGLE, and other entities may be involved in the investigation process.
- Protect and preserve the incident/accident scene so as not to alter information of evidence while the investigation is conducted.
- Collect photos;
- Collect statement from witnesses; and
- Documentation of the investigation.

A draft incident report shall be submitted to the WCAA with 24 hours. A final report shall be submitted to the WCAA within five days. Documentation of the investigation can be completed on Attachment 11 - Incident Documentation Template Supplemental reports and documentation can be submitted with the Incident Template.

If the incident/accident includes a Utility Strike, the the Contractor shall submit Attachment #12 – Supplemental Utility Strike Incident Form.

The contractor shall keep documentation of the following:

- Final incident/accident report with all supplemental documents.
- Copies of any citation or violation as a result of the events.
- Any notification to MIOSHA, FAA, Michigan EGLE, DOT, and EPA that involve results, and and remedial action schedules.

ATTACHMENT IX – SITE LOGISTIC PLAN

STRUCTURAL STEEL ERECTION TRAVEL PATH:

30' WIDE PRECAST CRANE TRAVEL PATH (starts 10' off edge of foundations)

ACCESS ROADS TO BE MAINTAINED DURING CONSTRUCTION

CONNECT TEMPORARY
CONSTRUCTION FENCE TO
EXISTING AOA FENCE

TEMPORARY
CONSTRUCTION FENCE

Concrete Batch Plant Area
March 2023 - December 2023

Underground Storm Detention System

Material/Equipment Laydown Areas (Typ.)

Temp. AOA
Perimeter Fencing

**Temp. Water
Line to feed
trailers.**

**Main Entrance Gate and
Christman|Brinker
Orientaiton/Check-in
Trailer**

**Christman|Brinker
Jobsite Offices**

Temporary Power Setup

**GREEN SHADED AREA IS
WETLANDS (Typ.) - STAY OUT!**

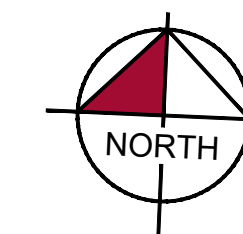
Trade Parking after paving work is completed

**EMERGENCY
MUSTER AREA**

Install New Guard Post 27, Swing Gates and Lock Prior To Removal of Existing Guard Post 27 Swing Gates

PFAS Stockpile Berm Area

Permanent AOA Perimeter Fencing



GRAPHIC SCALE IN FEET

0 75 150 300

A horizontal scale bar with four segments. The first segment (0 to 75) is white. The second segment (75 to 150) is black. The third segment (150 to 225) is white. The fourth segment (225 to 300) is black. The segments are labeled 0, 75, 150, and 300 at their right boundaries.

ATTACHMENT X – JOB HAZARD ANALYSIS (JHA)

Pre-Task / Hazard Analysis

Project Name:	Date:
Contractor:	Anticipated number of workers:
Activity / Task:	

Job Step	Hazards	Actions to Eliminate or Minimize Hazards
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

Disclaimer: This form is intended to assist the subcontractor with their safety planning efforts. The Christman Company, in receipt of this completed document, accepts no responsibility for the review, planning or execution of the work described within. It is the responsibility of the subcontractor to provide a safe work environment, free of recognized hazards, for their employees and for safety on the site as it relates to their operations and employees. The submission of this form to The Christman Company is for record purposes only.

ATTACHMENT XI –WAYNE COUNTY – LOSS MANAGEMENT MONTHLY REVIEW



Wayne County Airport Authority Loss Management Monthly Summary

Contact Name: _____

Month/Year: _____

Contract # _____

Number of
Contractors: _____

Statistics:

	Current Period	Calendar YTD	Project to Date
Total Work Hours:	_____	_____	_____
Contractors	_____	_____	_____
Total	_____	_____	_____

First Aid Cases:	_____	_____	_____
Contractors	_____	_____	_____
Total	_____	_____	_____

Near Miss Incidents:	_____	_____	_____
Contractors	_____	_____	_____
Total	_____	_____	_____

All Skilled Medical Treatment Cases:

	#	Rate	#	Rate	#	Rate
Contractors	_____	_____	_____	_____	_____	_____
Total	_____	_____	_____	_____	_____	_____

All Skilled Medical Treatment Cases Which Require Days Off from Work:

	#	Rate	#	Rate	#	Rate
Contractors	_____	_____	_____	_____	_____	_____
Total	_____	_____	_____	_____	_____	_____

Total Number of Days Away from Work:

	#	#	#
Contractors	_____	_____	_____
Total	_____	_____	_____



Wayne County Airport Authority Loss Management Monthly Summary

Description of Skilled Medical Treatment Injuries/Illnesses:
(Specify Contractor name. List body part and one-line description of incident.)

(Insert text here)

Near Miss Incidents (Specify Contractor name.)

(Insert text here)

Property Damage Incidents (Specify Contractor name.)

(Insert text here)

First Aid Only Cases.

(Insert text here)

ATTACHMENT XII – WAYNE COUNTY – CONTRACTOR EMPLOYEE FORM

**Attachment 2 – Contractor Employee Review
Contractor Safety Guidelines**

**THIS DOCUMENT MUST BE KEPT ONSITE AND
UP TO DATE**

Date: _____

Project Number: _____

From: _____

Prime Contractor

Sub-Contractor #1

Sub-Contractor #3

Sub-Contractor #2

Sub-Contractor #4

After reviewing general safety, emergency response, and specific area safety requirements with all of your on-site assigned employees and sub-contractors, please have them sign and indicate their badge ID number or the last five numbers of their social security number on this sheet. If the employee or sub-contractor refuses to sign, the superintendent shall print the employee's name and write "employee has been read the requirements but refuses to sign."

"My Superintendent/Foreman/Supervisor has been explained to me the Wayne County Airport Authority Contractor Safety Guidelines. I understand the requirements. I understand that my deviation from these requirements could be cause for my dismissal from this project."

Signature	Date	Badge ID or Last 5 SSN:	Company Name:

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WAYNE COUNTY AIRPORT AUTHORITY[illegible]



DETROIT METRO • WILLOW RUN
WAYNE COUNTY AIRPORT AUTHORITY

[illegible]

ATTACHMENT XIII – WAYNE COUNTY – CONTRACTOR INSURANCE TOOL KIT

Contractor Insurance Tool Kit

Sub-Contractor Name	Insurance Required	Amount	Policy Number	Policy Effect Date	Policy Expiration Date
1.					
2.					
3,					
4.					
5					
6					
7					

ATTACHMENT XIV – WAYNE COUNTY – REQUEST FOR MARKING UTILITIES



Request For Marking Utilities

Company Name or WCAA

Requesting the Marking: _____
Company/WCAA Division Name Project/Permit/Contract Number

Date and Time Submitted: _____
Print Date Print Time

Name/Contact of the Individual Requesting Marks: _____
Print Name Phone Number

Name of the WCAA Representative/Engineer Submitted to: _____
Print Name Phone Number

Drawing Numbers Reviewed: _____

Utilities will not be marked unless a drawing showing the limits of the marking are attached. Markings will only be valid for Twenty-one days. Underground work shall not be carried out beyond the 21 days of this request unless the Contractor documents the locations of the marking via survey at their own expense. The locations of the markings must be submitted back to the WCAA Representative.

Responsible Party (b)

Contact Numbers

☐ Electrical	WCAA and/or DTE	Call WCAA Project Rep. / 811 (a)
☐ Natural Gas	WCAA and/or Michcon	Call WCAA Project Rep. / 811 (a)
☐ Water	WCAA	Call WCAA Project Rep. / 811 (a)
☐ Sanitary/Storm	WCAA	Call WCAA Project Rep. / 811 (a)
☐ FAA Electrical/Fiber	FAA	(734) 955-5100
☐ Telephone	ATT	(734) 778-1215 or (734) 731-3270
☐ Jet Fuel Systems	Swissport	(734) 732-5748 or (734) 941-5510
☐ Glycol	WCAA and or Miss Dig	Call WCAA Project Rep. / 811 (a)
☐ All Other (list below)	Varies – See Drawings	Call WCAA Project Rep. / 811

Notes:

- (a) WCAA Project representative may vary from project to project. The Project Representative might also be a company that represents the WCAA.
- (b) It is the Contractors responsibly to mark and locate other utilities not listed on this form.

How will you document the locations? _____

I understand that I am asking WCAA and outside agencies to conduct onsite markings if they have the resources to do so and if not I will be contacted by WCAA and must coordinate to perform the markings as outlined in the WCAA Safety Standards. I understand that I must also attach a sketch to indicate the limits of the area I am requesting to be swept. If I require additional markings outside the limits or require other utilities to be marked I understand a new form must be completed. The Contractor is not to proceed without an executed Field Acknowledgments form.

Signature of the Individual Requesting the Markings

Date

ATTACHMENT XV – WAYNE COUNTY – HOT WORK PERMIT

IN CASE OF EMERGENCY CALL 911

Survey Requested By: (Company Name, Tenant, Airline, or Department):	Contact Person:	Contact Number:
---	------------------------	------------------------

AFTER INSPECTION THIS WILL PERMIT

TO _____ Name of Company Doing The Work

Identify the Ignition Source and the Job to Be Done

FROM _____ : _____ **AM/PM** **TO** _____ : _____ **AM/PM** _____ / _____ / _____ (day/month/year)

Extend TO: _____ : _____ **AM/PM** **TO** _____ : _____ **AM/PM** **BY** _____ / _____ / _____ (day/month/year)

Extend TO: _____ : _____ **AM/PM** **TO** _____ : _____ **AM/PM** **BY** _____ / _____ / _____ (day/month/year)

The operator shall take the necessary precaution to eliminate all possible fire hazards and to prevent damage to any construction work, building materials, equipment and all other property, both public and personal involved in connection with their work to be done. The provisions of the following shall apply at all times. If a portable **WELDING CART** is being utilized, then at least **ONE PORATBLE EXTINGUSIHER** MUST be attached to the cart with a **MINIUM RATING of 2A-10BC. THESE ARE THE MINIMUM REQUIREMENTS.** If it is determined that the hazard warrants additional extinguishers and/or other life safety precautions, then the permit holder will be **REQUIRED** to meet all the conditions before beginning their operations.

BY ORDER OF THE FIRE MARSHALL
THE FOLLOWING CONDITIONS SHALL BE INSPECTED PRIOR TO APPROVAL

C	N/C	Working On Walls or Ceilings	C	N/C	Within 35 feet of Work
		Combustible Coverings shall NOT be used			Fixed site combustibles and flammable liquids shall be protected with covers, guards, or metal shields
		Combustibles shall be moved away, from the opposite side of the wall being worked on			Combustible floors shall be wet down, covered with damp sand, metal or other shields
		Covers suspended beneath work to collect sparks shall be non-combustible			Movable combustible materials and flammable liquids shall be relocated outside the work area.
C	N/C	Work on Enclosed Equipment			The floors shall be swept clean of combustibles
		The Equipment will be cleaned of All combustibles			All Wall and floor openings shall be covered
		The container will be purged of all vapors and verified with a gas detection meter			
		Equipment has been safely taken out of service and properly locked and tagged out.	C	N/C	Tar Kettle Operations
C	N/C	Work in Enclosed Equipment (Confined Spaces)			A minimum of two (2) portable extinguishers with a minimum rating of 2A-20BC shall be maintained in a proper working order not closer than ten (10) feet nor more than fifty (50) feet from the TAR KETLE operations area.
		Contractor shall properly ventilate the space and have a gas meter onsite full time to monitor the space. Contractor must have their own recue equipment onsite and be Confined Space Entry Trained			Propane fuel tanks shall be secured and maintained at all times. Also, all supply hoses and regulators shall be maintained in the appropriate manner (i.e. free of leaks, tears, cuts
C	N/C	Fire Watch	C	N/C	Miscellaneous
		No ignition source within 100 feet of aircraft			If working in a fuel farm or fueling station a verify that the contractor has a written Safety Plan
		Operator shall supply extinguishers of appropriate size and type. They shall be located not more than fifty (50) feet from the operations			While working in ANY terminal contractor must use a ventilator or smoke eater to remove the welding hot work fumes
		Operator shall be trained in use of extinguishing equipment and in sounding of fire alarm and notifications. It is the responsibility of the Company/Tenant to ensure that operating personnel have received such training.			
		Provided during operations, with FINAL inspections thirty (30) minutes after operations cease			

AT THE TIME OF THE SURVEY THE ABOVE CONDITONS WERE FOUND. ANY NON-COMPLAINT ITEMS SHALL BE CORRECTED BEFORE OPERATIONS BEGIN

ISSUED BY: _____ **DATE:** _____ **TIME:** _____ **INC.#** _____

REMARKS: _____

ATTACHMENT XVI – WAYNE COUNTY – UTILITY FIELD AWARENESS FORM



FIELD ACKNOWLEDGEMENT FORM

	Utility	Date Req.	Date Completed
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			
11.			
12.			

Project/Contract #: _____

Name of Contractor : _____

Limits of Work Included:

☐

Yes

or

☐

No

PLEASE ATTACH SUBSEQUENT FORMS THAT HAVE BEEN COMPLETED FOR PROPER VERIFICATION

Contractor will list out those utilities that were marked, cleared, or could not be located below:

Marked

Cleared

Not in Scoped Area or
could not locate

Notes

(a) WORK SHALL NOT COMMENCE IF THE UTILITY NOT LOCATED. CONTRACTOR SHALL REQUEST A MEETING WITH THE WCAA TO CLAIRIFY PRIOR TO PROCEEDING.

Marking documentation attached?

☐

Yes

or

☐

No

The above data has been checked with the Contract Drawings. When close clearances are indicated, hand excavation must be used to determine the exact location, existing lines, and interferences in the vicinity of work must be marked indicating locations and depth prior to excavation.

Printed Name

Date

Contractor Signature

ATTACHMENT XVII – WAYNE COUNTY – UTILITY INTERRUPTION FORM



Utility Interruption Form

Permit # or Contract # _____

Circle Airport: DTW or YIP

Utility System – List System(s) to be taken out of Service (Gas, Water, Electrical) and General Location	Buildings/Systems/Equipment to be effected	Maximo Work Order #	Shut Down Date and Time	Start Up Date and Time	Contractor Rep Initials	WCAA Rep Initials
1.						
2.						
3,						
4.						
5						
6						
7						

Contractor Representative Signature and Date

WCAA Representative Signature and Date

This document is to be forwarded to the WCAA Project Representative and the WCAA Facilities & Infrastructure Divisions at least five (5) business days prior to work. The shutdowns will not be authorized unless this form signed and received in advance as specified. Please use additional forms as needed. WCAA Project Representative will be required to notify those affected by the shut down. In some cases a back-up utility source may be necessary – PLEASE SEE PROJECT REPRESENTATIVE TO DETERMINE IF BACK-UP UTILITY SOURCES ARE REQUIRED. Upon authorization of the document a copy will must forwarded to the Maintenance Call Center located at Building 703. The Call Center Representative will assure it gets forwarded to the correct departments.

ATTACHMENT XVIII – WAYNE COUNTY – UTILITY SYSTEM CUSTODY FORM



Utility System Custody Form

This form serves as authorization for the Contractor to take full custody of a Utility System during their Project Work. All WCAA Safety Rules and Regulations still apply while the system is under the custody by the undersigned (Contractor). Contractors will be in full control of the entire system while under their custody. The Contractor will be required to coordinate with WCAA for this request. If custody times need to be extended than a new form must be completed and approved.

Permit # or Contract # _____

Circle Airport: DTW or YIP

Requested System	Maximo Work Order #	Area of Isolation (i.e. list panel#, breaker # , transformer #, regulator # of where custody begins and ends)	Requested Day and Time to be turned over	Time and Day System to be turned back to WCAA
1.				
2.				
3,				
4.				
5				

Contractor Representative Signature and Date

WCAA Representative Signature and Date

This document is to be forwarded to the all the WCAA Department affected listed below five (5) business days prior to work. Upon authorization of the document a copy will must forwarded to the Maintenance Call Center located at Building 703. The Call Center Representative will assure it gets forwarded to the correct departments that are effected. Call Center Fax:

ATTACHMENT XIX – DAILY CONSTRUCTION CHECKLIST

Daily Construction Checklist – AOA Construction

	Item	Compliant (Y/N)	Action Required
1	Excavation adjacent to runways, taxiways, and aprons improperly backfilled.		
2	Mounds of earth, construction materials, temporary structures, and other obstacles near any open runway, taxiway, or taxi lane; in the related Object Free area and aircraft approach or departure areas/zones; or obstructing any sign or marking		
3	Runway resurfacing projects resulting in lips exceeding 3 in (7.6 cm) from pavement edges and ends.		
4	Heavy equipment (stationary or mobile) operating or idle near AOA, in runway approaches and departures areas, or in OFZ.		
5	Equipment or material near NAVAIDs that may degrade or impair radiated signals and/or the monitoring of navigation and visual aids. Unauthorized or improper vehicle operations in localizer or glide slope critical areas, resulting in electronic interference and/or facility shutdown.		
6	Tall and especially relatively low visibility units (that is, equipment with slim profiles) — cranes, drills, and similar objects — located in critical areas, such as OFZ and approach zones.		
7	Improperly positioned or malfunctioning lights or unlighted airport hazards, such as holes or excavations, on any apron, open taxiway, or open taxi lane or in a related safety, approach, or departure area.		
8	Obstacles, loose pavement, trash, and other debris on or near AOA. Construction debris (gravel, sand, mud, paving materials) on airport pavements may result in aircraft propeller, turbine engine, or tire damage. Also, loose materials may blow about, potentially causing personal injury or equipment damage.		
9	Inappropriate or poorly maintained fencing during construction intended to deter human and animal intrusions into the AOA. Fencing and other markings that are inadequate to separate construction areas from open AOA create aviation hazards.		
10	Improper or inadequate marking or lighting of runways (especially thresholds that have been displaced or runways that have been closed) and taxiways that could cause pilot confusion and provide a potential for a runway incursion. Inadequate or improper methods of marking, barricading, and lighting of temporarily closed portions of AOA create aviation hazards.		
11	Wildlife attractants — such as trash (food scraps not collected from construction personnel activity), grass seeds, tall grass, or standing water — on or near airports.		
12	Obliterated or faded temporary markings on active operational areas.		
13	Misleading or malfunctioning obstruction lights. Unlighted or unmarked obstructions in the approach to any open runway pose aviation hazards.		
14	Misleading or malfunctioning obstruction lights. Unlighted or unmarked obstructions in the approach to any open runway pose aviation hazards.		
15	Failure to issue, update, or cancel NOTAMs about airport or runway closures or other construction related airport conditions.		
16	Failure to mark and identify utilities or power cables. Damage to utilities and power cables during construction activity can result in the loss of runway / taxiway lighting; loss of navigation, visual, or approach aids; disruption of weather reporting services; and/or loss of communications.		
17	Restrictions on ARFF access from fire stations to the runway / taxiway system or airport buildings.		
18	Objects, regardless of whether they are marked or flagged, or activities anywhere on or near an airport that could be distracting, confusing, or alarming to pilots during aircraft operations.		
19	Water, snow, dirt, debris, or other contaminants that temporarily obscure or derogate the visibility of runway/taxiway marking, lighting, and pavement edges. Any condition or factor that obscures or diminishes the visibility of areas under construction.		
20	Spillage from vehicles (gasoline, diesel fuel, oil) on active pavement areas, such as runways, taxiways, aprons, and airport roadways.		
21	Failure to maintain drainage system integrity during construction (for example, no temporary drainage provided when working on a drainage system).		
22	Failure to provide for proper electrical lockout and tagging procedures. At larger airports with multiple maintenance shifts/workers, construction contractors should make provisions for coordinating work on circuits.		
23	Failure to control dust. Consider limiting the amount of area from which the contractor is allowed to strip turf.		
24	Exposed wiring that creates an electrocution or fire ignition hazard. Identify and secure wiring, and place it in conduit or bury it.		
25	Site burning, which can cause possible obscuration.		
26	Construction work taking place outside of designated work areas and out of phase.		

Project: _____ Name of Contractor: _____

Date of Inspection: _____ Signature of Inspector: _____

ATTACHMENT XX – CONTRACTOR INCIDENT/ACCIDENT DOCUMENTATION



DETROIT METRO • WILLOW RUN
WAYNE COUNTY AIRPORT AUTHORITY

CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

Incident Date _____ Incident Time _____

Contract Number _____ Contract Name _____

Prime Contractor Name _____ Sub Contractor Name _____

☐ DTW ☐ YIP Work Area _____

Name of the Employee(s) Involved _____

Employee #1 _____ Employee #2 _____

Employer _____ Employer _____

Job Title _____ Job Title _____

Employee Length _____ years/months Employee Length _____ years/months

Employee #1 Employment Category

☐ Full Time ☐ Part Time ☐ Temporary ☐ Seasonal ☐ Non-Employee ☐ Other

Employee #2 Employment Category

☐ Full Time ☐ Part Time ☐ Temporary ☐ Seasonal ☐ Non-Employee ☐ Other

INCIDENT TYPE - Check one or all that applies

- ☐ Property Damage ☐ Environmental Release ☐ Injury
☐ FAA Incursion ☐ FAA Surface Incident ☐ Unsafe Act
☐ Utility Strike*if checked the Supplement Utility Strike Form must be completed

VEHICLES/EQUIPMENT INVOLVED

#1 Make _____ Model _____ Plate _____

#2 Make _____ Model _____ Plate _____

CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

Description of Damages (type in field below)

INJURY INFORMATION

Describe the Nature of the Injury and Part of the Body

Nature _____ Part of Body _____

☐ Fatality ☐ Lost Workday (away from work) ☐ Medical Treatment ☐ First Aid ☐ Other

Specific Incident Location _____

On Premise ☐ Yes ☐ No

Incident During

☐ Normal Work Period ☐ Rest Period ☐ Meal Period
☐ Overtime ☐ Entering ☐ Leaving

GENERAL DETAILS OF THE INCIDENT

- 1 Describe specifically what the employee did after reporting to the job

- 2 Describe exactly what the he or she was doing when the incident occurred

CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

3 Describe what the employee did during the previous 24 hours

Employees Supervisors Name _____ Title _____

Describe where the Supervisor was located at the time of the Incident _____

Was there a written procedure for the Job? ☐ Yes ☐ No

Was the procedure being followed? ☐ Yes ☐ No

Did the Procedure include safety Rules? ☐ Yes ☐ No

Were the Safety Rules Followed? ☐ Yes ☐ No

Are the Safety Rules enforced by Supervision routinely? ☐ Yes ☐ No

Was the employee trained concerning the Safety Rule/Procedure? ☐ Yes ☐ No

If Yes When (provide date) _____

Did the incident involve

☐ Machines ☐ Equipment ☒ Processes ☒ Hazardous Material ☐ Other

Were there Witnesses? ☐ Yes ☒ No

Name and Phone Number of Witness _____

Witness Statement obtained and attached? ☐ Yes ☐ No

If no, why not?

CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

Employee(s) Involved Statement obtained and attached? ☐ Yes ☐ No

If no. why not?

BASIC INCIDENT CAUSES

Please list all contributing factors, engineering/administrative issues, educational/training deficiencies, safety procedural/programmatic flaws, etc. *(Attach copies of all photos, sketches, drawings, witness statements, and any other information that contributed to the investigation)*

Photos obtained and attached? ☐ Yes ☐ No



CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

If no. why not?

--

RECOMMENDATIONS TO PREVENT FURTHER REOCCURRENCE

Recommendation #1_____

Assigned Responsibility_____

Completion Date_____

Recommendation #2_____

Assigned Responsibility_____

Completion Date_____

Recommendation #3_____

Assigned Responsibility_____

Completion Date_____

Recommendation #4_____

Assigned Responsibility_____

Completion Date_____

Recommendation #5_____

Assigned Responsibility_____

Completion Date_____



CONTRACTOR - INCIDENT-ACCIDENT DOCUMENTATION TEMPLATE

Cost of the incident will be submitted to the WCAA once the information has been obtained by the Contractor

Reviewed by _____

Date _____

Approved by _____

Date _____

ATTACHMENT XXI – SUPPLEMENTAL UTILITY STRIKE FORM



SUPPLEMENTAL UTILITY STRIKE FORM

Incident Date _____ Incident Time _____

Contract Number _____ Contract Name _____

Check Affected Utility - one or all that apply

- | | | | |
|-------------------------------------|--|-------------------------------------|-------------------------------------|
| ☐ Electrical | ☐ Natural Gas | ☐ Sanitary | ☐ Storm |
| ☐ Fiber | ☐ Telephone/Comms | ☐ Jet Fuel | ☐ Glycol |
| ☐ Gasoline | ☐ Diesel | ☐ Steam | ☐ Condensate |
| ☐ Water | ☐ Oil | ☐ Irrigation | ☐ Duct Only |
| ☐ Other | | | |

Did the Strike Cause an outage (immediately or for the repair)? ☐ Yes ☐ No

If Yes list those that have been or will be impacted below

Who owns the Utility Impacted _____

Utilities shown on the Contract Drawings? ☐ Yes ☐ No

WCAA Utility verification Process Completed? ☐ Yes ☐ No

Was a copy of the Clearance to Excavate attached? ☐ Yes ☐ No

If not explain why?

Reviewed by _____ Date _____

PROJECT SAFETY MANUAL
SUBCONTRACTOR AND TRADE CONTRACTOR ACKNOWLEDGEMENT

I hereby acknowledge receipt and comprehension of the policies and procedures set forth in this Manual and agree to comply with its directives.

In addition, I hereby agree to comply with any field or trade-related guidelines as they apply to the portion of work for which I have been contracted and will incorporate them into the ongoing safety effort. I also understand and agree to comply with any State and Federal OSHA and ANSI regulations, standards, codes and rules as they apply to my portion of the work.

(Signed)

(Print Name)

(Title)

(Company Name)

(Date)

**Please sign and fill out the information requested above and return to
The Christman|Brinker Company.**